检测与分割

主讲人：Hao Wang
AI Thrust，信息枢纽

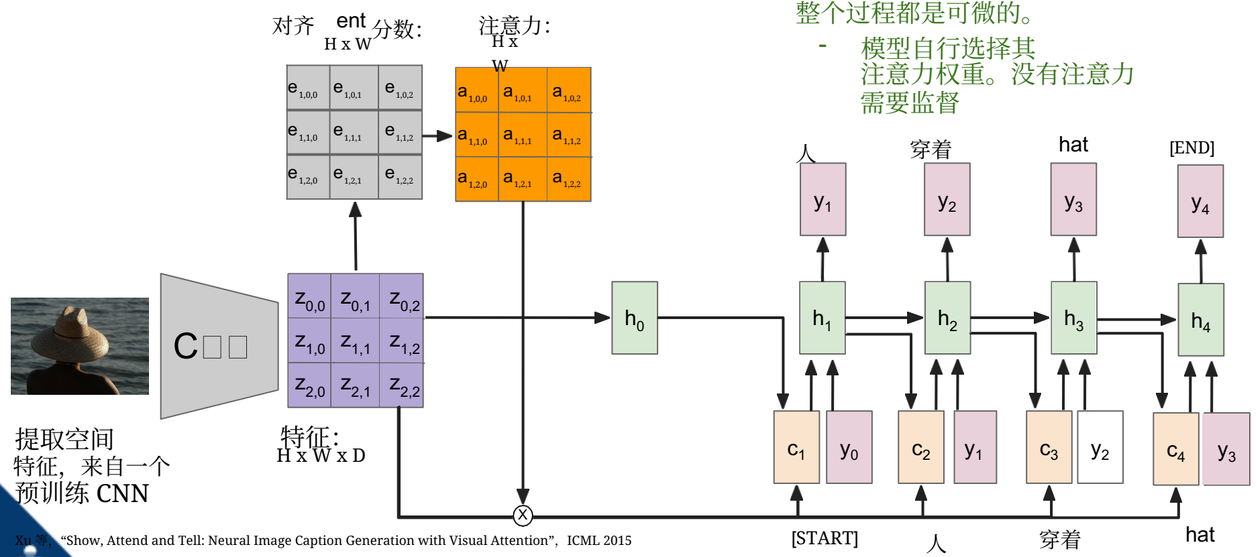
Detection & Segmentation

Instructor: Hao Wang
AI Thrust, Information hub

上周回顾

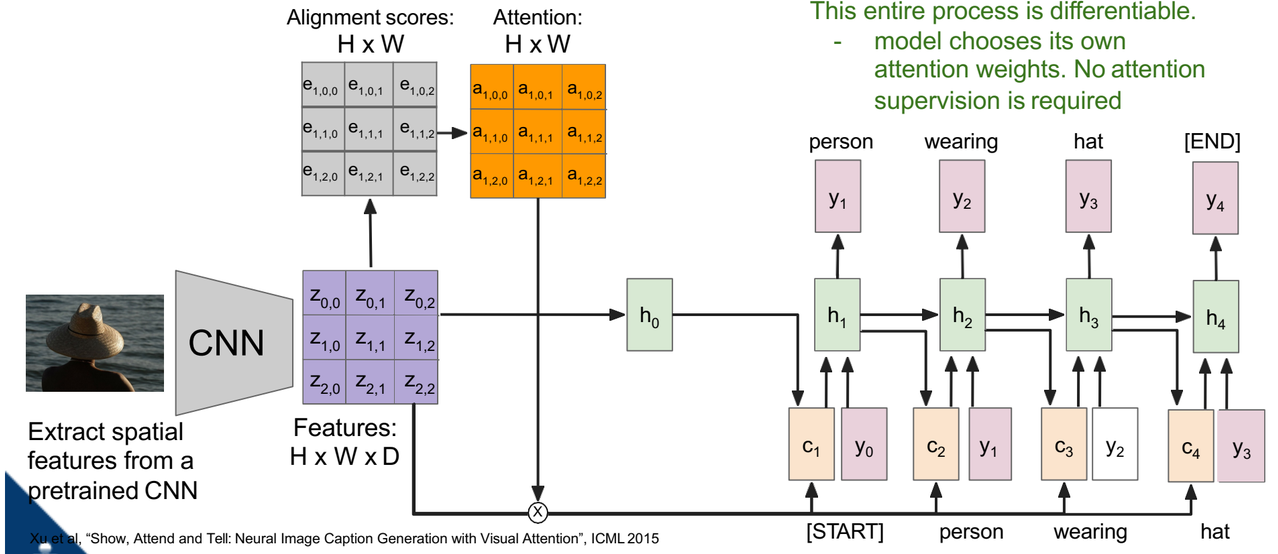
Recap of last week

使用 RNN 和 注意力的图像描述



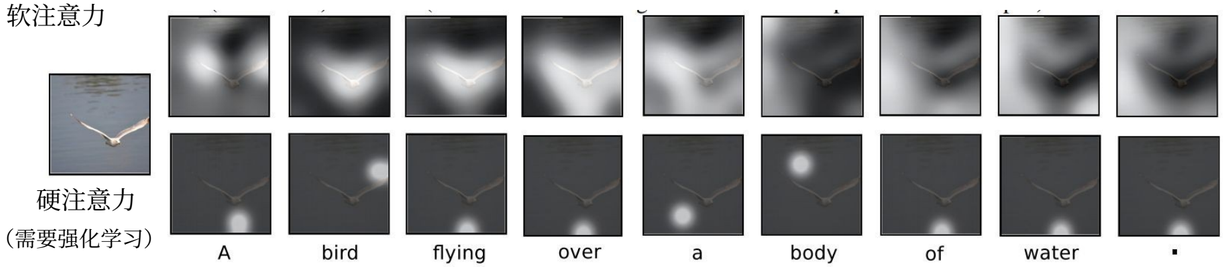
Xu 等, "Show, Attend and Tell: Neural Image Caption Generation with Visual Attention", ICML 2015

Image Captioning with RNNs & Attention



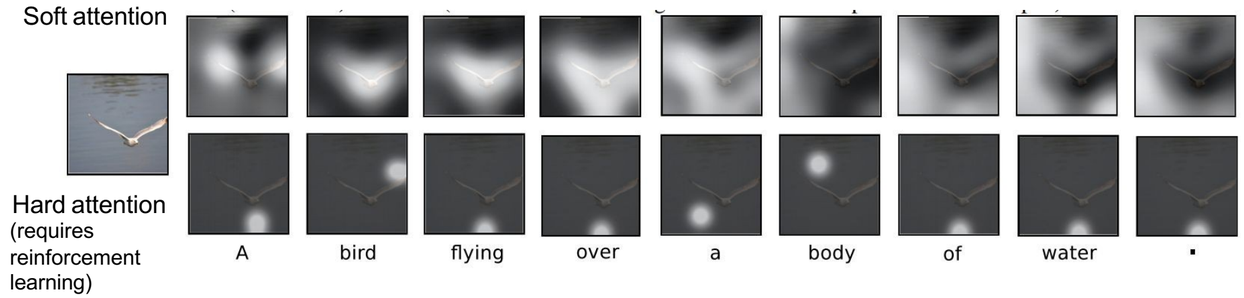
Xu et al, "Show, Attend and Tell: Neural Image Caption Generation with Visual Attention", ICML 2015

基于注意力的图像描述



Xu 等人，“Show, Attend, and Tell：基于视觉注意力的神经图像描述生成”，ICML 2015 图片版权归 Kelvin Xu、Jimmy Lei Ba、Jamie Kiros、Kyunghyun Cho、Aaron Courville、Ruslan Salakhutdinov、Richard S. Zemel 和 Yoshua Benchio 所有，2015。经许可转载。

Image Captioning with Attention



Xu et al, “Show, Attend, and Tell: Neural Image Caption Generation with Visual Attention”, ICML 2015
Figure copyright Kelvin Xu, Jimmy Lei Ba, Jamie Kiros, Kyunghyun Cho, Aaron Courville, Ruslan Salakhutdinov, Richard S. Zemel, and Yoshua Benchio, 2015. Reproduced with permission.

基于注意力的图像描述



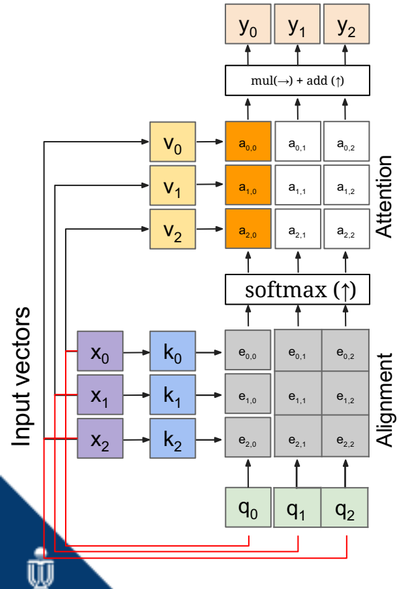
Xu 等，“Show, Attend, and Tell: Neural Image Caption Generation with Visual Attention”，ICML 2015 图片版权归 Kelvin Xu、Jimmy Lei Ba、Jamie Kiros、Kyunghyun Cho、Aaron Courville、Ruslan Salakhutdinov、Richard S. Zemel 和 Yoshua Benchio 所有，2015。经许可转载。

Image Captioning with Attention



Xu et al, “Show, Attend, and Tell: Neural Image Caption Generation with Visual Attention”, ICML 2015
Figure copyright Kelvin Xu, Jimmy Lei Ba, Jamie Kiros, Kyunghyun Cho, Aaron Courville, Ruslan Salakhutdinov, Richard S. Zemel, and Yoshua Benchio, 2015. Reproduced with permission.

通用注意力层 ➡ 自注意力层



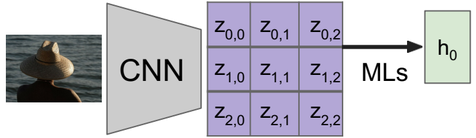
输出：
上下文向量：□（形状：D）

运算：
键向量：□ L □x q
值向量：□ L □x
查询向量：□ L □x
v 对齐：$e_{i,j}$ □ $q_i \cdot k_j$ 0 □D
注意力：**y** L softmax(□)
输出：y_j L □$_i$ $a_{i,j}$□v_i

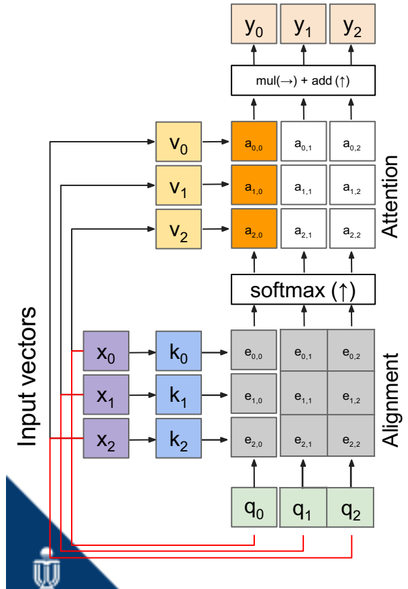
输入：
输入向量：□（形状：N x D）
查询：□（形状：M x D□）

回想一下，查询向量是输入向量的函数

编码器：h P f (□)0 w其中 □是空间 CNN 特征 **wf (.)** 是一个 MLP



General attention layer ➡ Self-attention layer



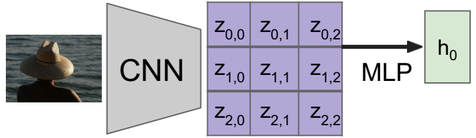
Outputs:
context vectors: **y** (shape: D_v)

Operations:
Key vectors: $\mathbf{k} = \mathbf{x}W_k$
Value vectors: $\mathbf{v} = \mathbf{x}W_v$
Query vectors: $\mathbf{q} = \mathbf{x}W_q$
Alignment: $e_{i,j} = q_j \cdot k_i / \sqrt{D}$
Attention: **a** = softmax(**e**)
Output: $y_j = \sum_i a_{i,j} v_i$

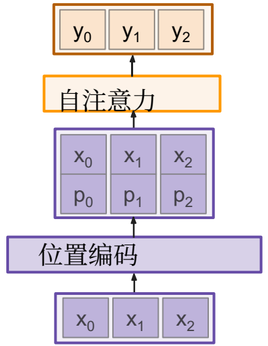
Inputs:
Input vectors: **x** (shape: N x D)
~~Queries: **q** (shape: M x D_k)~~

Recall that the query vector was a function of the input vectors

Encoder: $h_0 = f_W(\mathbf{z})$
where **z** is spatial CNN features
$f_W(.)$ is an MLP



位置编码



将特殊位置编码 p□拼接到每个输入向量 x□

我们使用一个函数 *pos*: N □ R^d 来将向量的位置 ○处理为 d 维向量

因此， p└ *pos*(○)

针对 *pos*(/) 的选项

1. 学习一个查找表：
   ○ 学习用于 *pos*(t) 的参数，其中 t □[2, T) ○ 查找表包含 T xd 个参数。

2. 设计一个满足这些期望条件的固定函数

□

直觉：

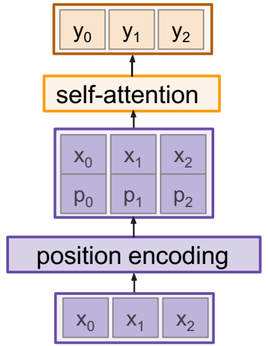
$$\mathbf{p(t)} \text{ L } \begin{bmatrix} \sin(\omega_1.t) \\ \cos(\omega_1.t) \\ \\ \sin(\omega_2.t) \\ \cos(\omega_2.t) \\ \\ \vdots \\ \\ \sin(\omega_{d/2}.t) \\ \cos(\omega_{d/2}.t) \end{bmatrix}_d$$

0：	0	0	0	0	8：	1	0	0	0
1：	0	0	0	1	9：	1	0	0	1
2：	0	0	1	0	10：	1	0	1	0
3：	0	0	1	1	11：	1	0	1	1
4：	0	1	0	0	12：	1	1	0	0
5：	0	1	0	1	13：	1	1	0	1
6：	0	1	1	0	14：	1	1	1	0
7：	0	1	1	1	15：	1	1	1	1

其中 $\omega_k = \frac{1}{10000^{2k/d}}$

Vaswani 等，“注意力就是你所需要的一切”，NeurIPS 2017

Positional encoding



Concatenate special positional encoding p_j to each input vector x_j

We use a function *pos*: N →R^d to process the position j of the vector into a d-dimensional vector

So, p_j = *pos*(j)

Options for *pos*(.)

1. Learn a lookup table:
   - Learn parameters to use for *pos*(t) for t ε [0, T)
   - Lookup table contains T x d parameters.

2. Design a fixed function with the desiderata
   -

Intuition:

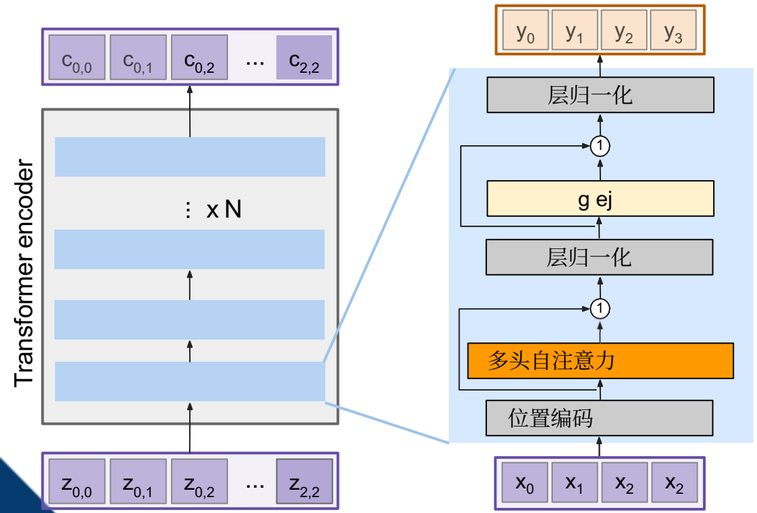
$$\mathbf{p(t)} = \begin{bmatrix} \sin(\omega_1.t) \\ \cos(\omega_1.t) \\ \\ \sin(\omega_2.t) \\ \cos(\omega_2.t) \\ \\ \vdots \\ \\ \sin(\omega_{d/2}.t) \\ \cos(\omega_{d/2}.t) \end{bmatrix}_d$$

0：	0	0	0	0	8：	1	0	0	0
1：	0	0	0	1	9：	1	0	0	1
2：	0	0	1	0	10：	1	0	1	0
3：	0	0	1	1	11：	1	0	1	1
4：	0	1	0	0	12：	1	1	0	0
5：	0	1	0	1	13：	1	1	0	1
6：	0	1	1	0	14：	1	1	1	0
7：	0	1	1	1	15：	1	1	1	1

where $\omega_k = \frac{1}{10000^{2k/d}}$

Vaswani et al, “Attention is all you need”, NeurIPS 2017

Transformer 编码器块



Transformer 编码器块：

输入：一组向量 **x** **输出**：一组向量 □

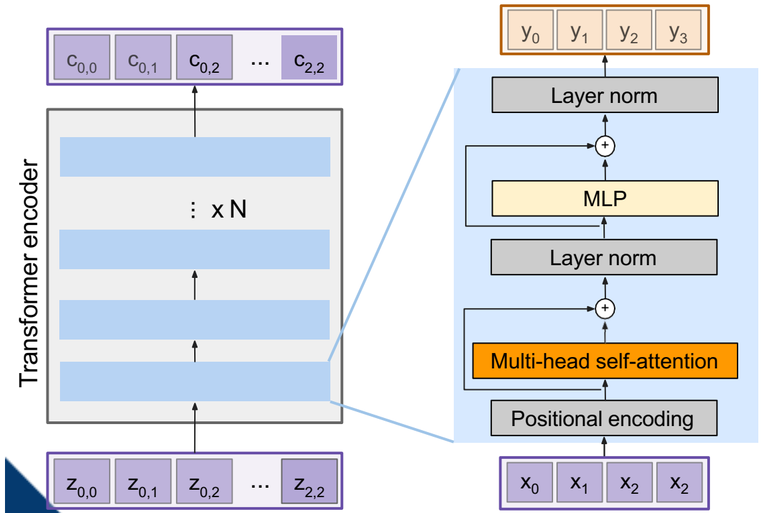
自注意力是向量之间唯一的交互。

层归一化和 MLP 对每个向量独立进行运算。

高度可扩展、高度可并行化，但内存占用高
用法。

Vaswani 等人，“Attention is all you need”，NeurIPS 2017

The Transformer encoder block



Transformer Encoder Block:

Inputs: Set of vectors **x**
Outputs: Set of vectors **y**

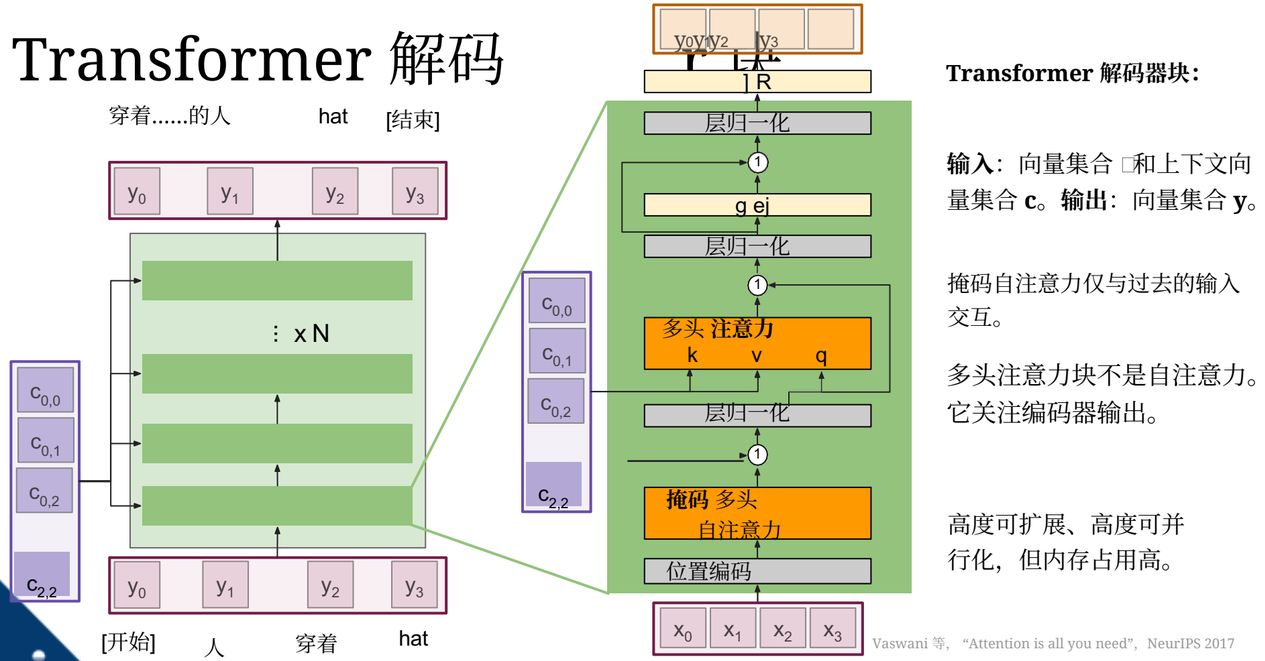
Self-attention is the only interaction between vectors.

Layer norm and MLP operate independently per vector.

Highly scalable, highly parallelizable, but high memory usage.

Vaswani et al, “Attention is all you need”, NeurIPS 2017

Transformer 解码



Vaswani 等，“Attention is all you need”，NeurIPS 2017

Transformer 解码器块：

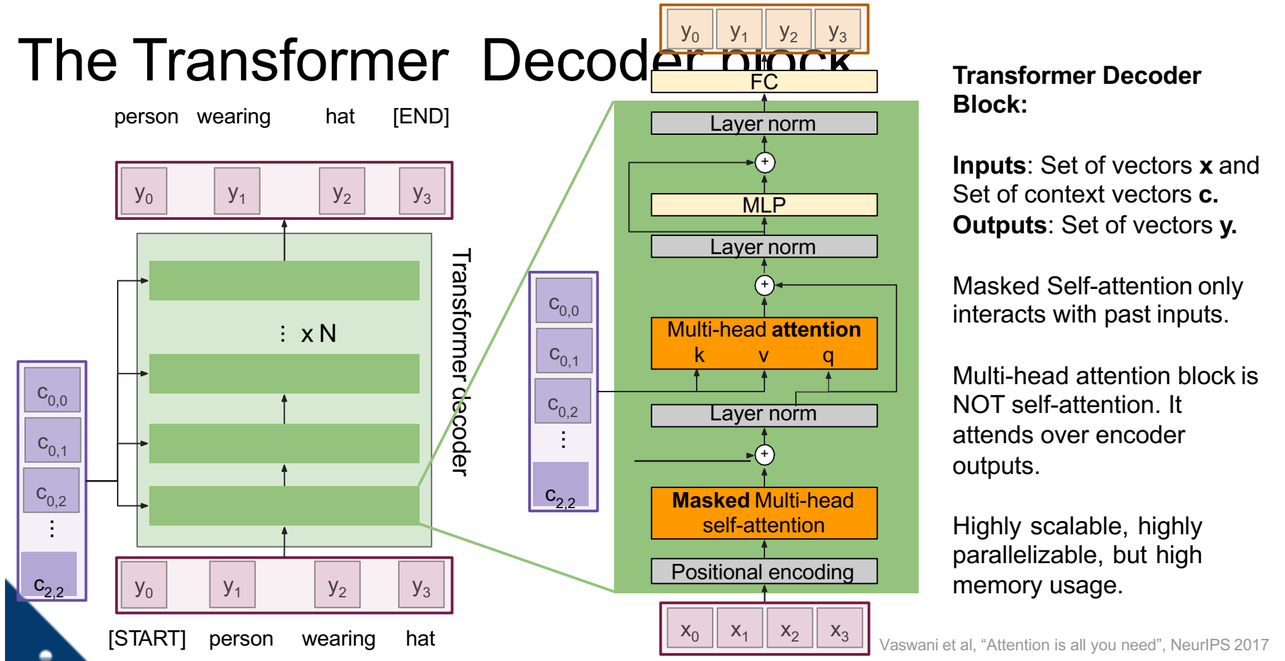
输入：向量集合 □和上下文向量集合 **c**。**输出**：向量集合 **y**。

掩码自注意力仅与过去的输入交互。

多头注意力块不是自注意力。它关注编码器输出。

高度可扩展、高度可并行化，但内存占用高。

The Transformer Decoder block



Vaswani et al, “Attention is all you need”, NeurIPS 2017

Transformer Decoder Block:

Inputs: Set of vectors **x** and Set of context vectors **c.**
Outputs: Set of vectors **y.**

Masked Self-attention only interacts with past inputs.

Multi-head attention block is NOT self-attention. It attends over encoder outputs.

Highly scalable, highly parallelizable, but high memory usage.

今日

- **计算机视觉任务**
  - **语义分割**
  - 目标检测
  - 实例分割
- 可视化与理解
  - 模型层可视化
  - 显著性图
  - CAM & Grad-CAM
- 基础模型与提示分割

Today

- **Computer Vision Tasks**
  - **Semantic Segmentation**
  - Object Detection
  - Instance Segmentation
- Visualization & Understanding
  - Model Layers Visualization
  - Saliency Maps
  - CAM & Grad-CAM
- Foundation Models and Prompt Segmentation

计算机视觉任务

分类



CAT

无空间范围

语义分割



草，CAT，树，S□□

没有对象，只有像素

目标检测



狗，狗，CAT

实例分割



狗，狗，CAT

多个对象

本图像为CC0 公共领域

Computer Vision Tasks

Classification



CAT

No spatial extent

Semantic Segmentation



GRASS, CAT, TREE, SKY

No objects, just pixels

Object Detection



DOG, DOG, CAT

Instance Segmentation



DOG, DOG, CAT

Multiple Object

This image is CC0 public domain

回顾：图像分类：计算机视觉中的核心任务



本图像由 [illegible] 提供，
依据以下许可：[illegible]

（假设给定一组可能的标签）{狗, 猫, 卡车 , 飞机 , ...}

→ t k□

Recall: Image Classification: A core task in Computer Vision



This image by Nikita is licensed under CC-BY 2.0

(assume given a set of possible labels)
{dog, cat, truck, plane, ...}

→ cat

语义分割

分类

CAT

无空间范围

语义分割



草地, CAT, 树, SKY

没有物体，只有像素

目标检测

DOG, DOG, CAT

实例分割

DOG, DOG, CAT

多目标

Semantic Segmentation

Classification

CAT

No spatial extent

Semantic Segmentation



GRASS, CAT, TREE, SKY

No objects, just pixels

Object Detection

DOG, DOG, CAT

Instance Segmentation

DOG, DOG, CAT

Multiple Object

语义分割：问题



草地、CAT、树、天空、.....

配对训练数据：对于每张训练图像， 每个像素都标注有一个语义类别。



S

在测试时，对新图像的每个像素进行分类。

Semantic Segmentation: The Problem



GRASS, CAT, TREE, SKY, ...

Paired training data: for each training image, each pixel is labeled with a semantic category.



?

At test time, classify each pixel of a new image.

语义分割思路：滑动窗口

完整图像



Semantic Segmentation Idea: Sliding Window

Full image



语义分割思路：滑动窗口

完整图像



没有上下文就无法分类

问：我们如何纳入上下文?

Semantic Segmentation Idea: Sliding Window

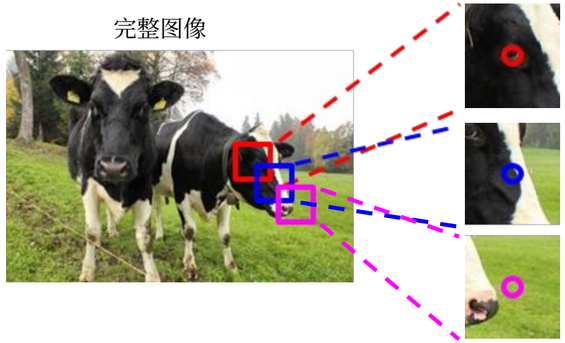
Full image



Impossible to classify without context

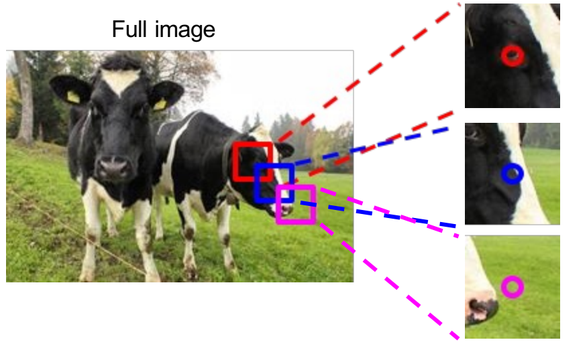
Q: how do we include context?

语义分割思路：滑动窗口



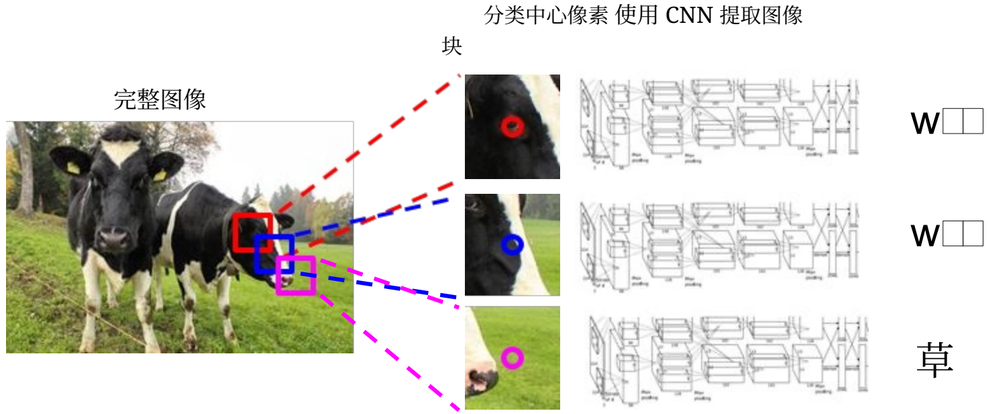
问：我们如何对此建模?

Semantic Segmentation Idea: Sliding Window



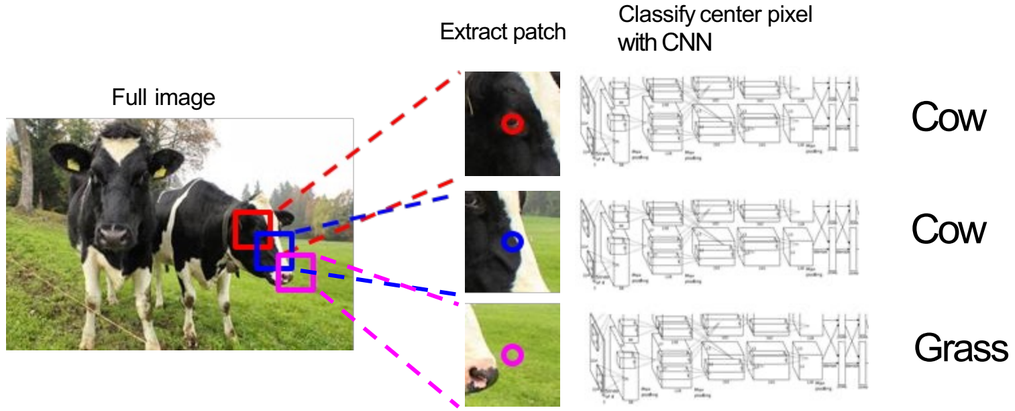
Q: how do we model this?

语义分割思路：滑动窗口



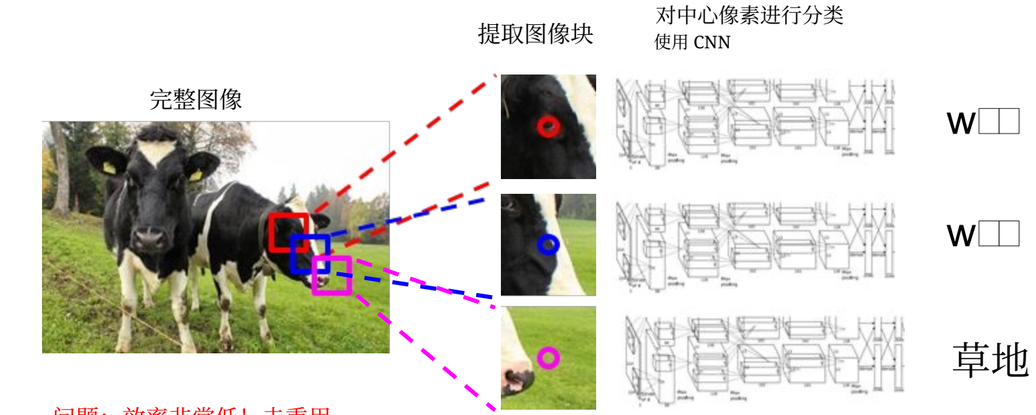
Farabet 等，“用于场景标注的层次特征学习”，TPAMI2013 Pinheiro 和 Collobert，“用于场景标注的循环卷积神经网络”，ICML 2014

Semantic Segmentation Idea: Sliding Window



Farabet et al，“Learning Hierarchical Features for Scene Labeling，” TPAMI 2013
Pinheiro and Collobert，“Recurrent Convolutional Neural Networks for Scene Labeling”，ICML 2014

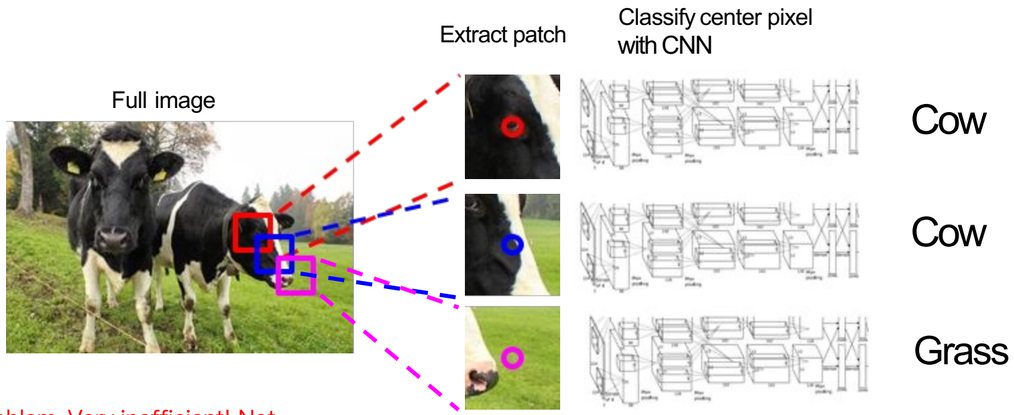
语义分割思路：滑动窗口



问题：效率非常低！未重用
重叠图像块之间的共享特征

Farabet 等，“用于场景标注的层次特征学习”，TPAMI2013 Pinheiro 和 Collobert，“用于场景标注的循环卷积神经网络”，ICML 2014

Semantic Segmentation Idea: Sliding Window



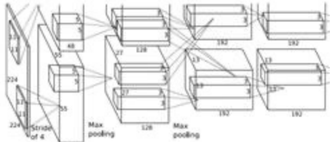
Problem: Very inefficient! Not
reusing shared features between
overlapping patches

Farabet et al, “Learning Hierarchical Features for Scene Labeling,” TPAMI 2013
Pinheiro and Collobert, “Recurrent Convolutional Neural Networks for Scene Labeling”, ICML 2014

语义分割思路：卷积

完整图像







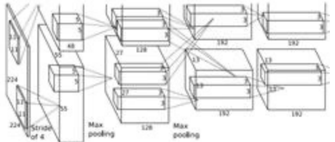
一个直观的想法：用卷积网络编码整张图像，并在其上进行语义分割。

问题：分类架构通常通过减小特征的空间尺寸来加深网络，但语义分割要求输出尺寸与输入尺寸相同。

Semantic Segmentation Idea: Convolution

Full image





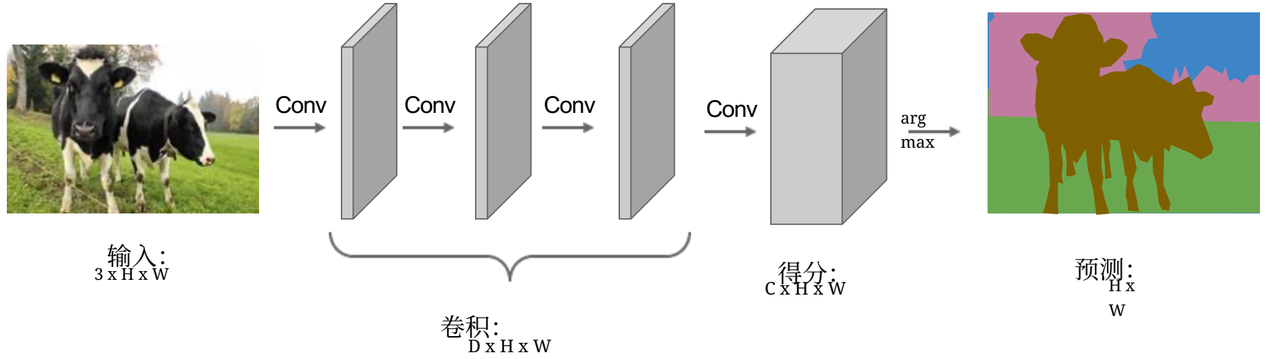


An intuitive idea: encode the entire image with conv net, and do semantic segmentation on top.

Problem: classification architectures often reduce feature spatial sizes to go deeper, but semantic segmentation requires the output size to be the same as input size.

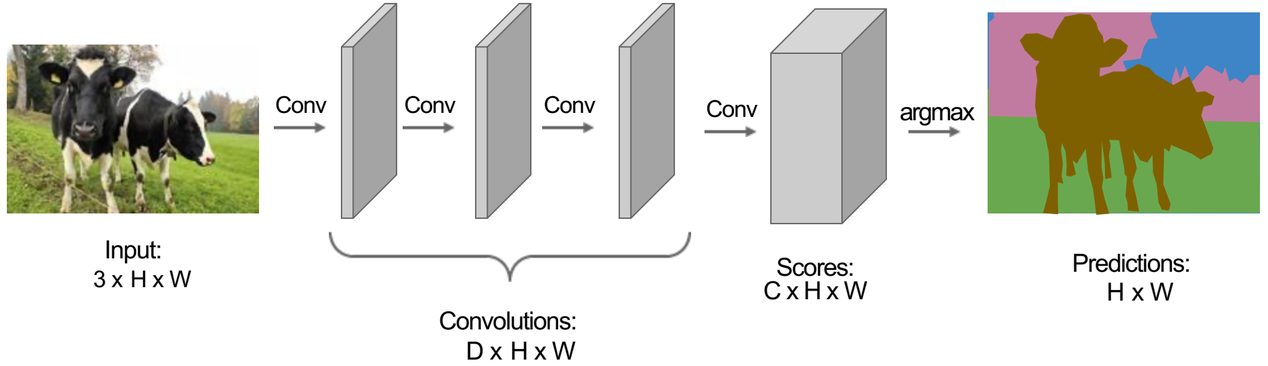
语义分割思路：全卷积

设计一个仅含卷积层、不含下采样算子的网络来预测 所有像素，一次性完成！



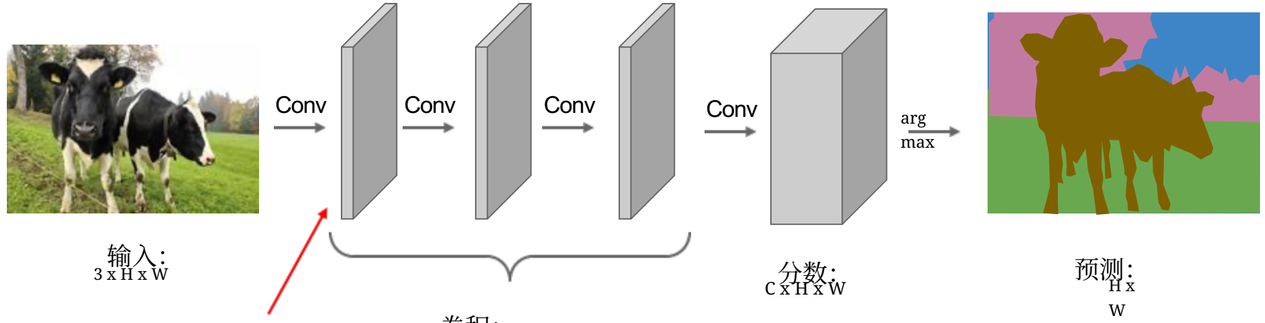
Semantic Segmentation Idea: Fully Convolutional

Design a network with only convolutional layers without downsampling operators to make predictions for pixels all at once!



语义分割思路：全卷积

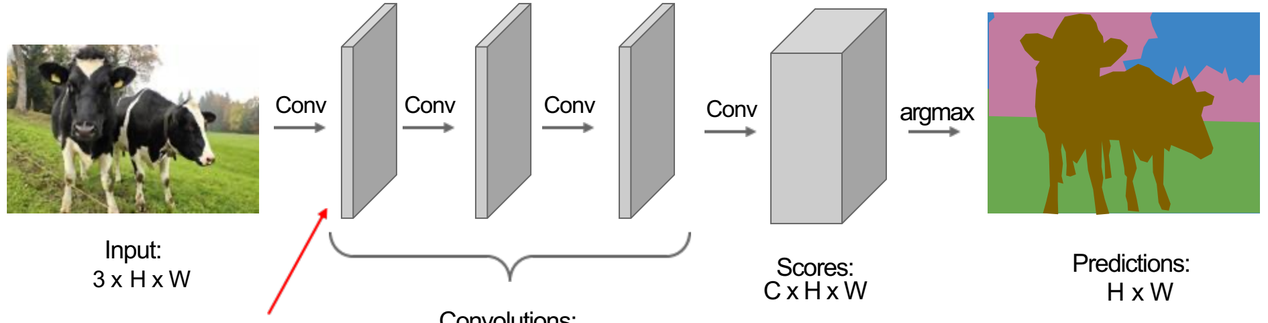
设计一个仅包含卷积层、不含下采样算子的
网络来进行预测 对所有像素一次性完成！



问题：卷积在原始图像分
辨率下将非常昂贵.....

Semantic Segmentation Idea: Fully Convolutional

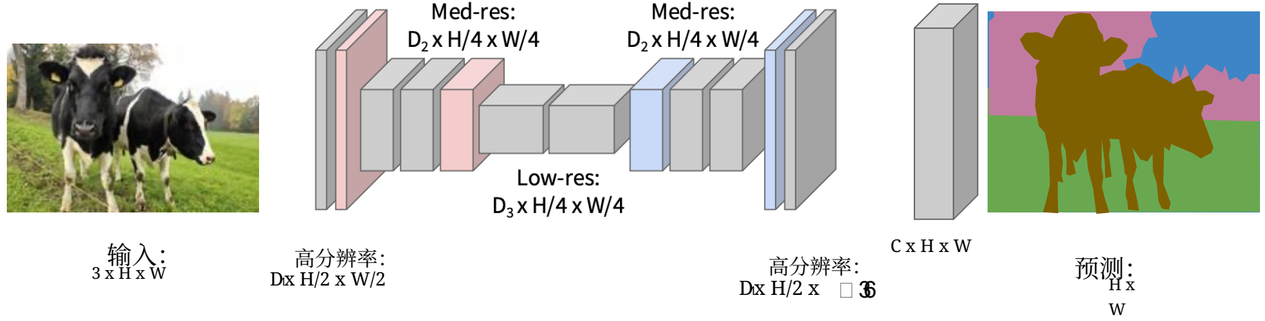
Design a network with only convolutional layers
without downsampling operators to make predictions
for pixels all at once!



Problem: convolutions at
original image resolution will be
very expensive ...

语义分割思路：全卷积

将网络设计为一系列卷积层，其中包含 下采样 和 上采样 在网络内部！



输入：
3 x H x W

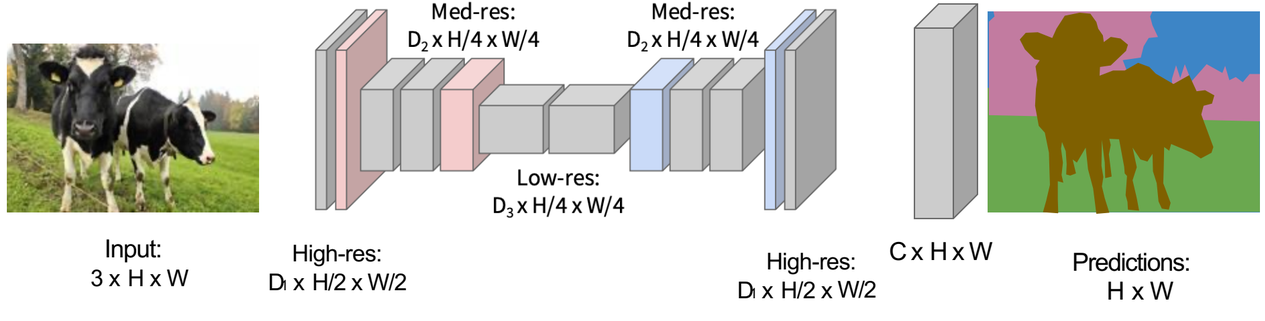
高分辨率：
D x H/2 x W/2

高分辨率：
D x H/2 x W/2

预测：
H x W

Semantic Segmentation Idea: Fully Convolutional

Design network as a bunch of convolutional layers, with downsampling and upsampling inside the network!



Input:
3 x H x W

High-res:
Di x H/2 x W/2

High-res:
Di x H/2 x W/2

Predictions:
H x W

Long、Shelhamer 和 Darrell，“用于语义分割的全卷积网络”，CVPR 2015 Noh 等，“用于语义分割的学习反卷积网络”，ICCV 2015

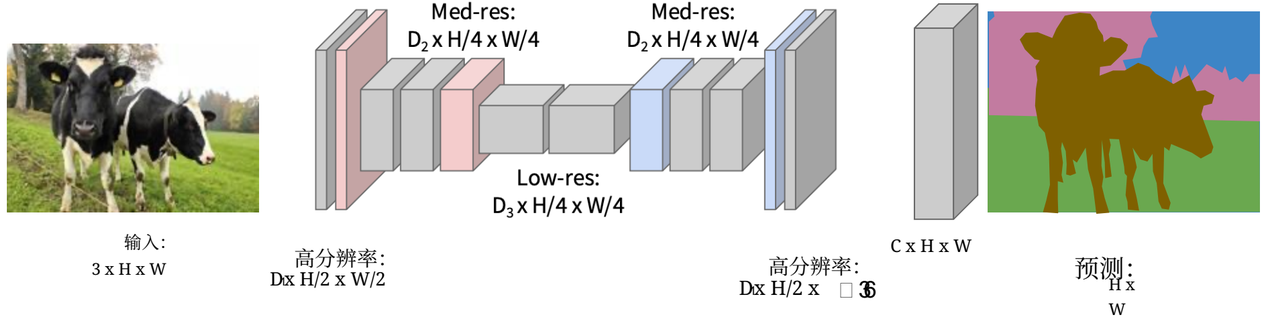
Long, Shelhamer, and Darrell, “Fully Convolutional Networks for Semantic Segmentation”, CVPR 2015
Noh et al, “Learning Deconvolution Network for Semantic Segmentation”, ICCV 2015

语义分割思路：全卷积

下采样：池化、跨步卷积

将网络设计成一系列卷积层，其中包含下采样和上采样，且这些都在网络内部完成！

上采样：???



输入：
3 x H x W

高分辨率：
D x H/2 x W/2

高分辨率：
D x H/2 x W/2

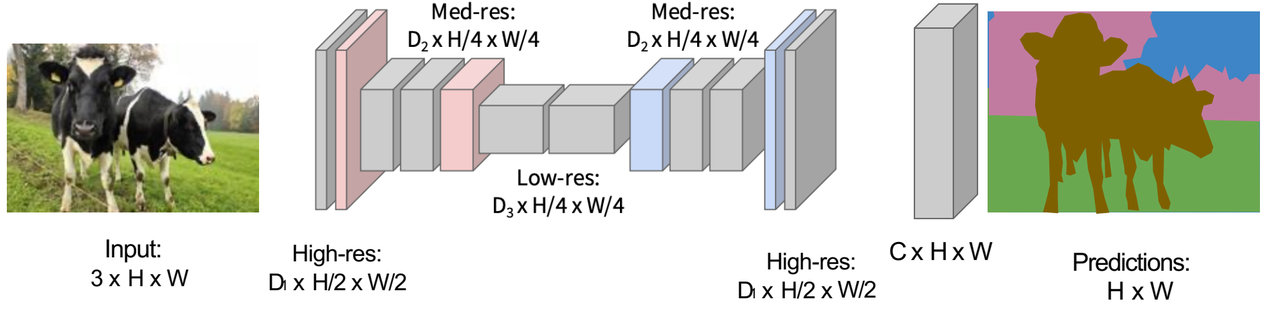
预测：
H x W

Semantic Segmentation Idea: Fully Convolutional

Downsampling:
Pooling, strided convolution

Design network as a bunch of convolutional layers, with downsampling and upsampling inside the network!

Upsampling:
???



Input:
3 x H x W

High-res:
Di x H/2 x W/2

High-res:
Di x H/2 x W/2

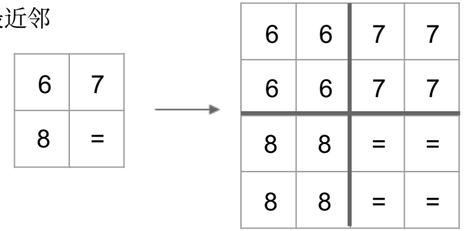
Predictions:
H x W

Long、Shelhamer 和 Darrell，《用于语义分割的全卷积网络》，CVPR 2015；Noh 等人，《学习用于语义分割的反卷积网络》，ICCV 2015

Long, Shelhamer, and Darrell, "Fully Convolutional Networks for Semantic Segmentation", CVPR 2015
Noh et al, "Learning Deconvolution Network for Semantic Segmentation", ICCV 2015

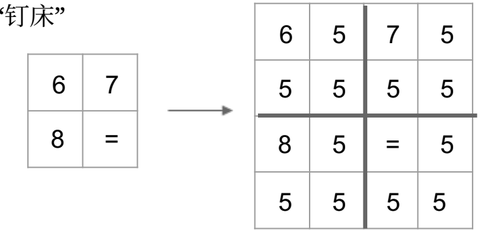
网络内上采样：“反池化”

最近邻



输入：2 x 2　　输出：4 x 4

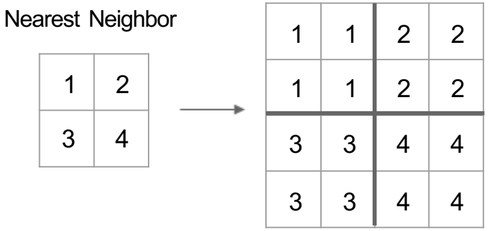
“钉床”



输入：2 x 2　　输出：4 x 4

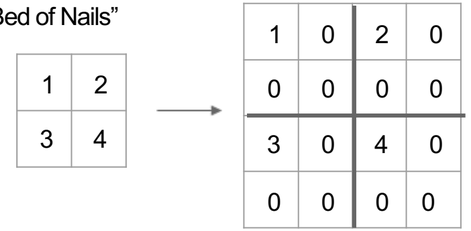
In-Network upsampling: “Unpooling”

Nearest Neighbor



Input: 2 x 2　　Output: 4 x 4

“Bed of Nails”



Input: 2 x 2　　Output: 4 x 4

网络内上采样：“最大反池化”

最大池化 记住哪个元素是最大值！

6	7	I	8
8	?	7	6
6	7	7	6
L	8	=	M

输入：4 x 4

?	I
L	M

输出：2 x 2

网络的其余部分

最大反池化：使用来自池化层的位置

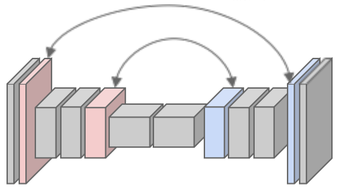
6	7
8	=

输入：2 x 2

5	5	7	5
5	6	5	5
5	5	5	5
8	5	5	=

输出：4 x 4

对应的 降采样和上采样层配对



In-Network upsampling: “Max Unpooling”

Max Pooling
Remember which element was max!

1	2	6	3
3	5	2	1
1	2	2	1
7	3	4	8

Input: 4 x 4

5	6
7	8

Output: 2 x 2

Rest of the network

Max Unpooling
Use positions from pooling layer

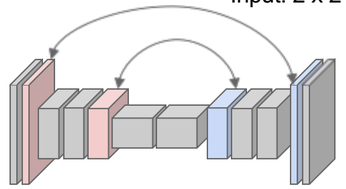
1	2
3	4

Input: 2 x 2

0	0	2	0
0	1	0	0
0	0	0	0
3	0	0	4

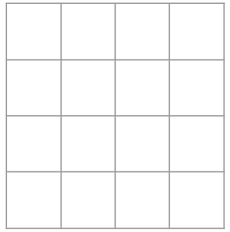
Output: 4 x 4

Corresponding pairs of downsampling and upsampling layers

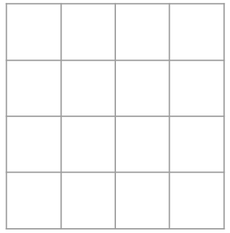


可学习的上采样

回顾：普通 3 x 3 卷积，步幅 1，填充 1



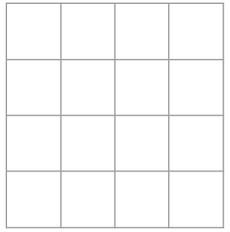
输入：4 x 4



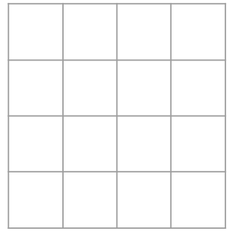
输出：4 x 4

Learnable Upsampling

Recall: Normal 3 x 3 convolution, stride 1 pad 1



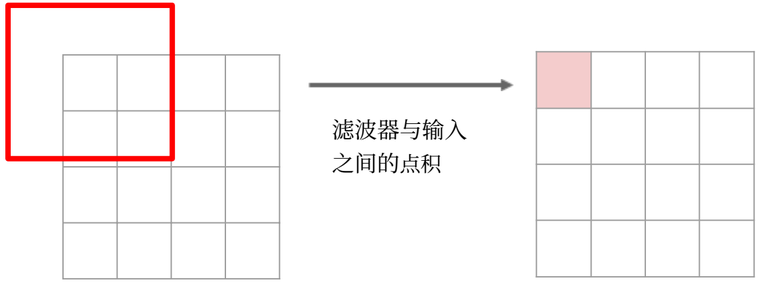
Input: 4 x 4



Output: 4 x 4

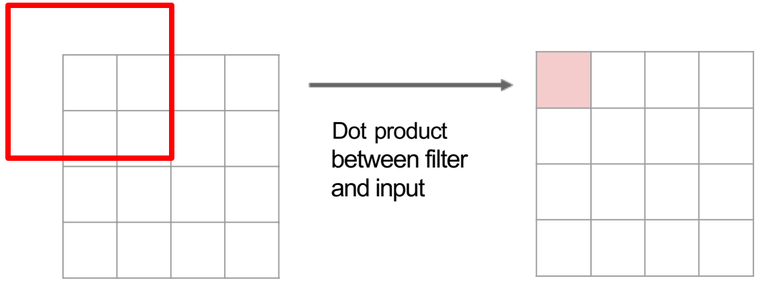
可学习上采样

回顾：普通 3 x 3 卷积，步长 1，填充 1



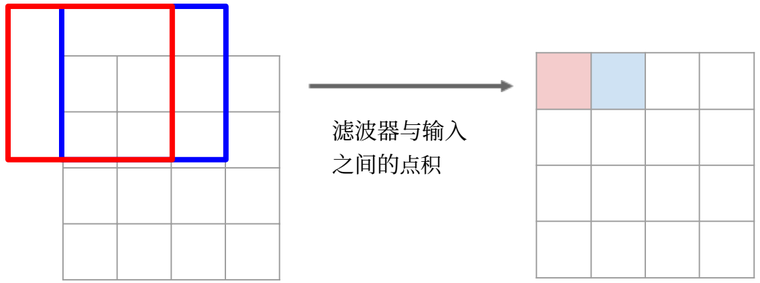
Learnable Upsampling

Recall: Normal 3 x 3 convolution, stride 1 pad 1



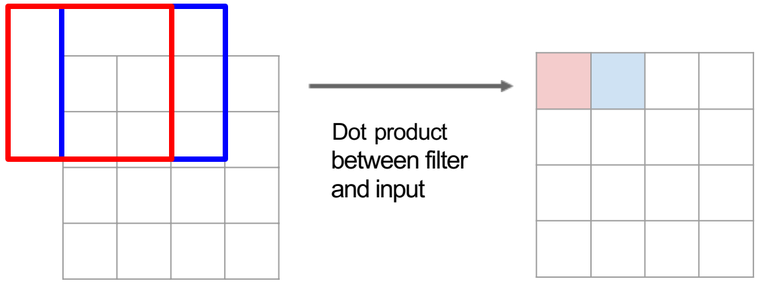
可学习上采样

回顾：普通 3 x 3 卷积，步幅 1，填充 1



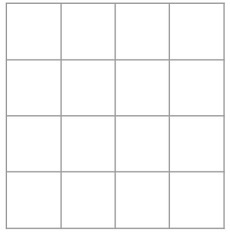
Learnable Upsampling

Recall: Normal 3 x 3 convolution, stride 1 pad 1

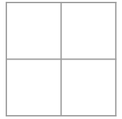


可学习上采样

回顾：普通 3 x 3 卷积，步幅 2，填充 1



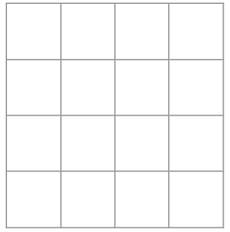
输入：4 x 4



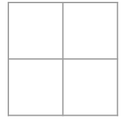
输出：2 x 2

Learnable Upsampling

Recall: Normal 3 x 3 convolution, stride 2 pad 1



Input: 4 x 4



Output: 2 x 2

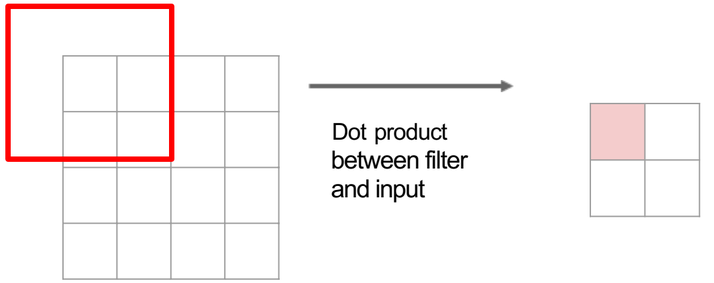
可学习上采样

回顾：常规 3 x 3 卷积，步幅 2，填充 1



Learnable Upsampling

Recall: Normal 3 x 3 convolution, stride 2 pad 1



可学习上采样

回顾：普通 3 x 3 卷积，步幅 2，填充 1



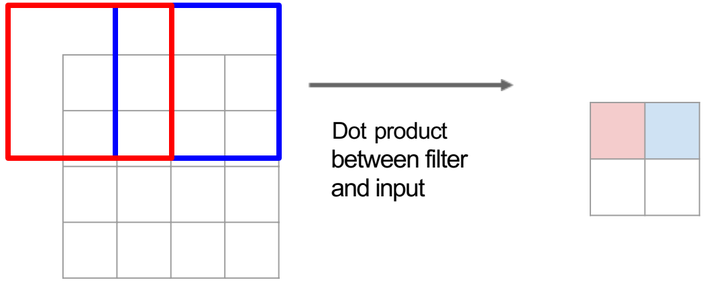
滤波器在输入中移动 2 个像素，对应输出中的每一个像素

步幅给出了 输入与输出中移动量之间的比率

我们可以将带步幅卷积解释为“可学习下采样”。

Learnable Upsampling

Recall: Normal 3 x 3 convolution, stride 2 pad 1



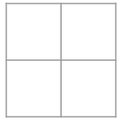
Filter moves 2 pixels in the input for every one pixel in the output

Stride gives ratio between movement in input and output

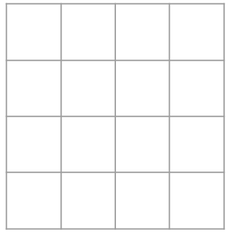
We can interpret strided convolution as “learnable downsampling”.

可学习上采样：转置卷积

3 x 3 转置卷积，步长 2，填充 1



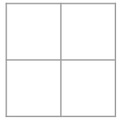
输入：2 x 2



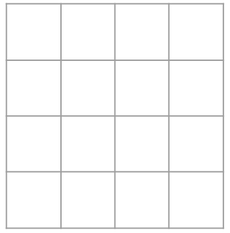
输出：4 x 4

Learnable Upsampling: Transposed Convolution

3 x 3 transposed convolution, stride 2 pad 1



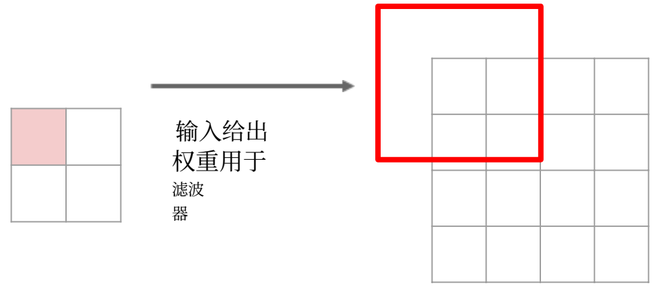
Input: 2 x 2



Output: 4 x 4

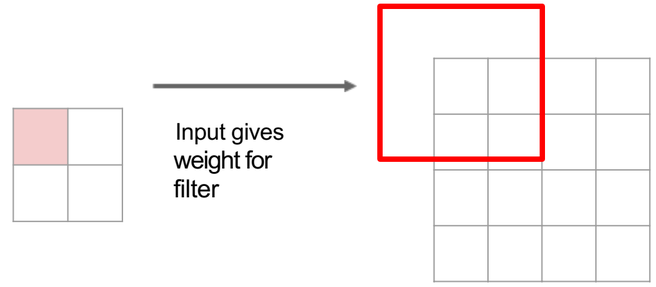
可学习上采样：转置卷积

3 x 3 转置卷积，步幅 2 填充 1



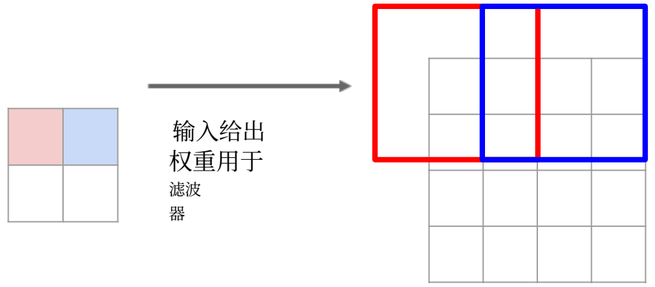
Learnable Upsampling: Transposed Convolution

3 x 3 transposed convolution, stride 2 pad 1



可学习上采样：转置卷积

3 x 3 转置卷积，步幅 2 填充 1

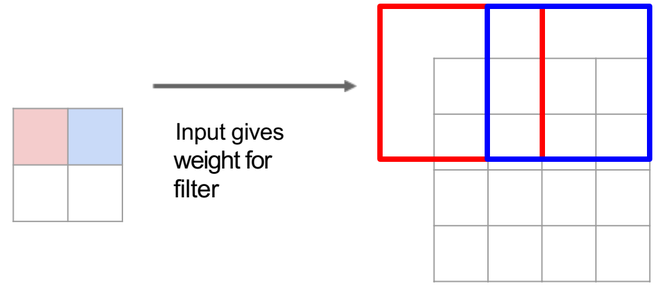


滤波器在输出中移动2个像素，对应 输入中的每1个像素

步幅给出比率，介于 输出与输入中的移动

Learnable Upsampling: Transposed Convolution

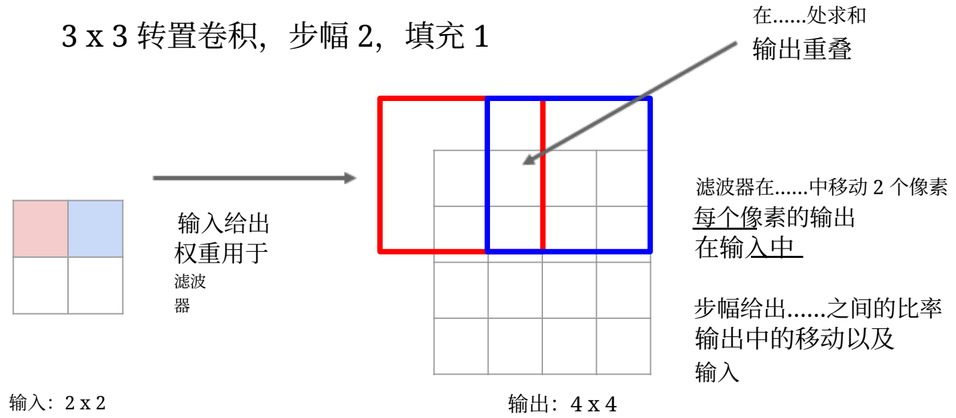
3 x 3 transposed convolution, stride 2 pad 1



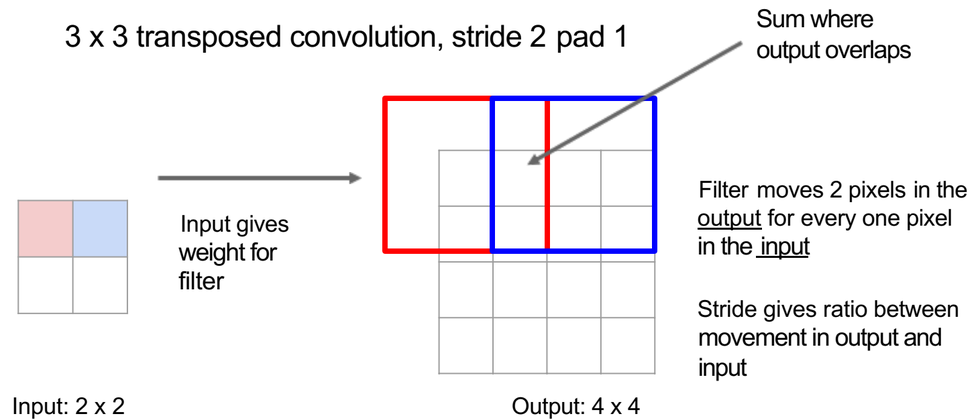
Filter moves 2 pixels in the output for every one pixel in the input

Stride gives ratio between movement in output and input

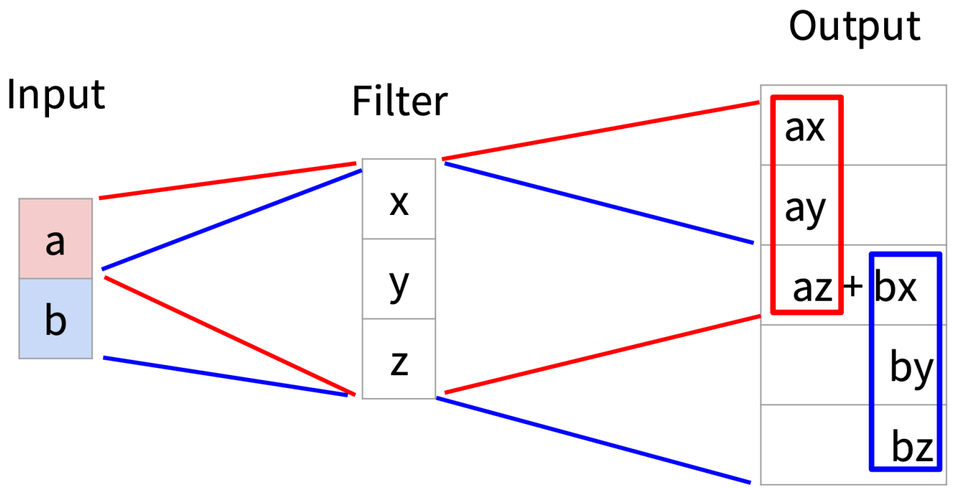
可学习上采样：转置卷积



Learnable Upsampling: Transposed Convolution

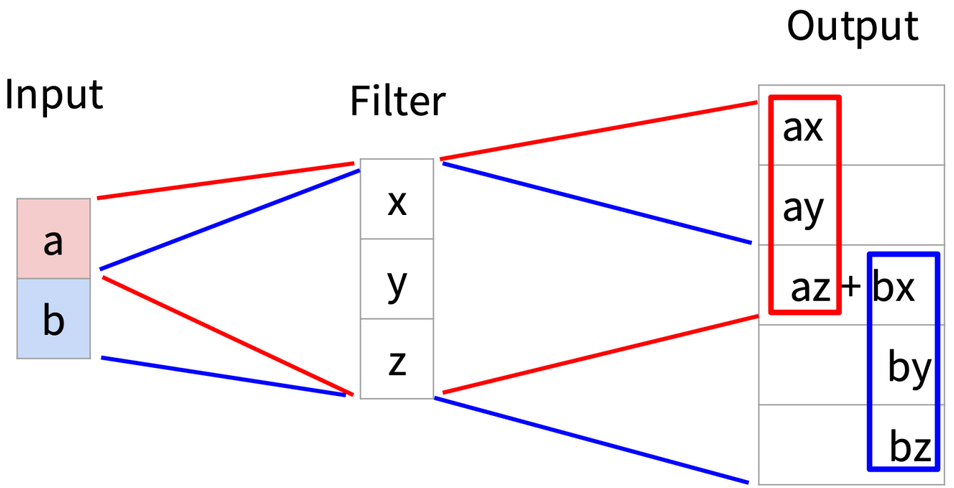


可学习上采样：一维示例



Output contains copies of the filter weighted by the input, summing at where at overlaps in the output

Learnable Upsampling: 1D Example



Output contains copies of the filter weighted by the input, summing at where at overlaps in the output

语义分割思路：全卷积

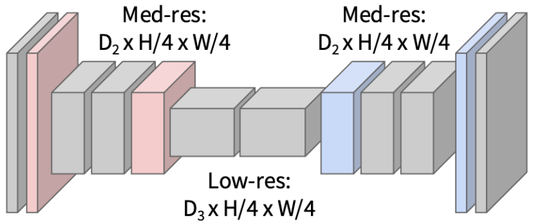
下采样：池化、步长卷积

将网络设计为一组卷积层，其中包含 下采样和上采样，并且这些操作都在网络内部完成！

上采样：反池化或步长转置卷积



输入：
3 x H x W



高分辨率：
Dx H/2 x W/2

高分辨率：
Dx H/2 x W/2



预测结果：
H x W

Semantic Segmentation Idea: Fully Convolutional

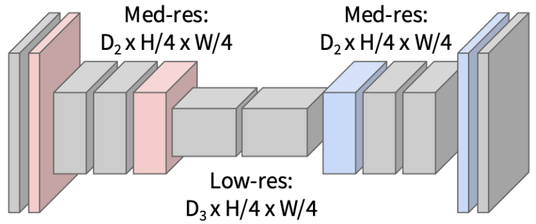
Downsampling:
Pooling, strided convolution

Design network as a bunch of convolutional layers, with downsampling and upsampling inside the network!

Upsampling:
Unpooling or strided transposed convolution



Input:
3 x H x W



High-res:
Di x H/2 x W/2

High-res:
Di x H/2 x W/2

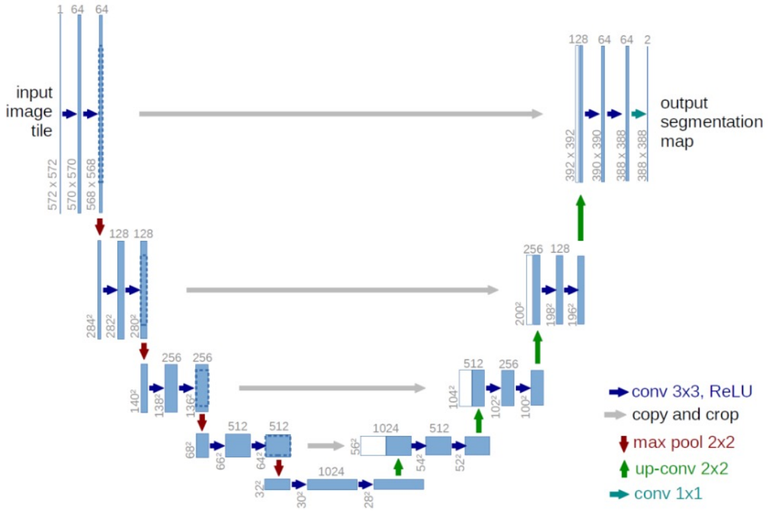


Predictions:
H x W

Long、Shelhamer 和 Darrell，“用于语义分割的全卷积网络”，CVPR 2015 Noh 等，“用于语义分割的学习反卷积网络”，ICCV 2015

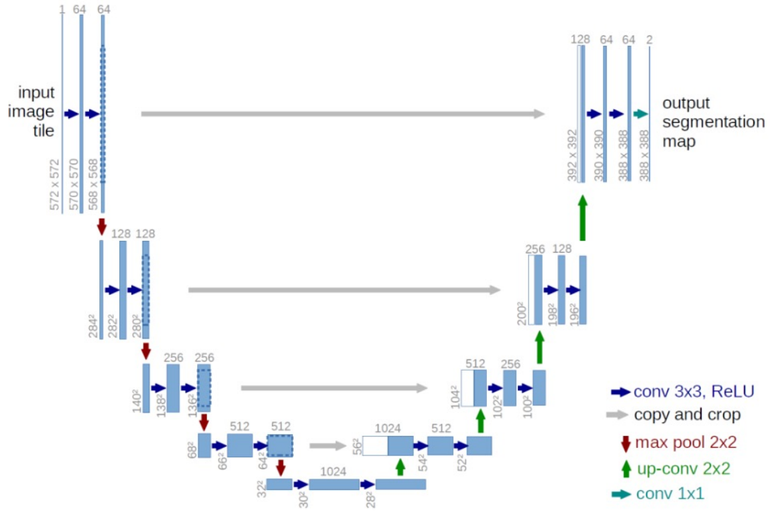
Long, Shelhamer, and Darrell, “Fully Convolutional Networks for Semantic Segmentation”, CVPR 2015
Noh et al, “Learning Deconvolution Network for Semantic Segmentation”, ICCV 2015

U-Net



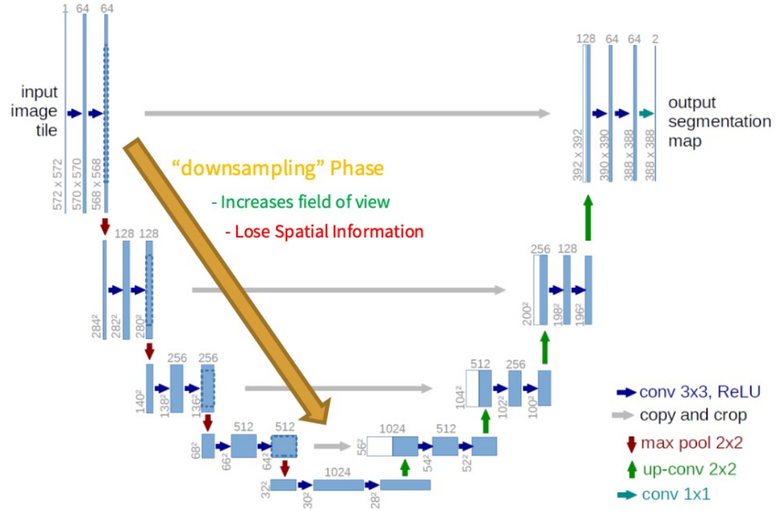
Ronneberger 等人 (2015) U-net 架构

U-Net



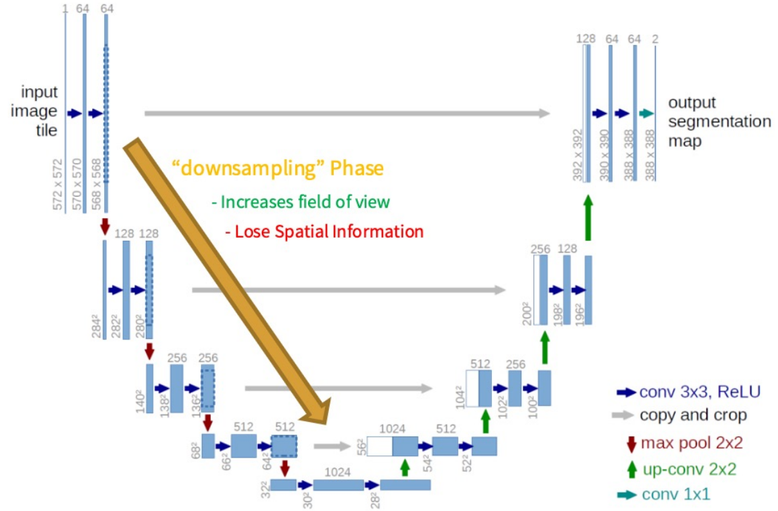
Ronneberger et al. (2015) U-net Architecture

U-Net



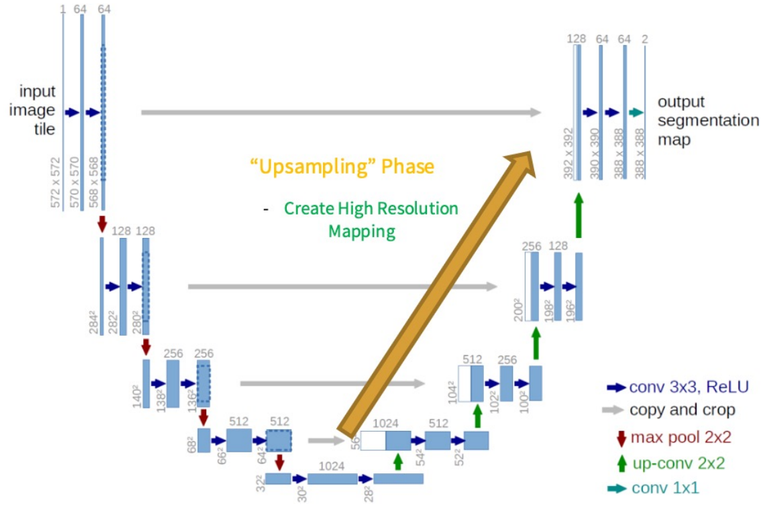
Ronneberger et al. (2015) U-net 架构

U-Net



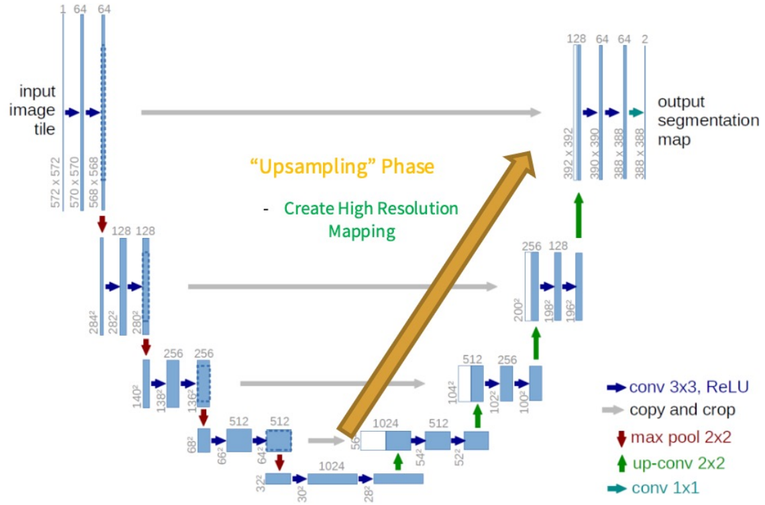
Ronneberger et al. (2015) U-net Architecture

U-Net



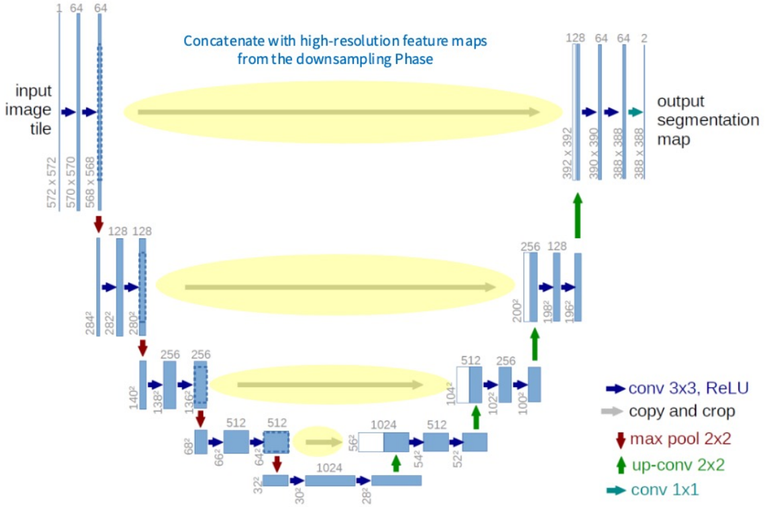
Ronneberger 等人 (2015) U-net 架构

U-Net



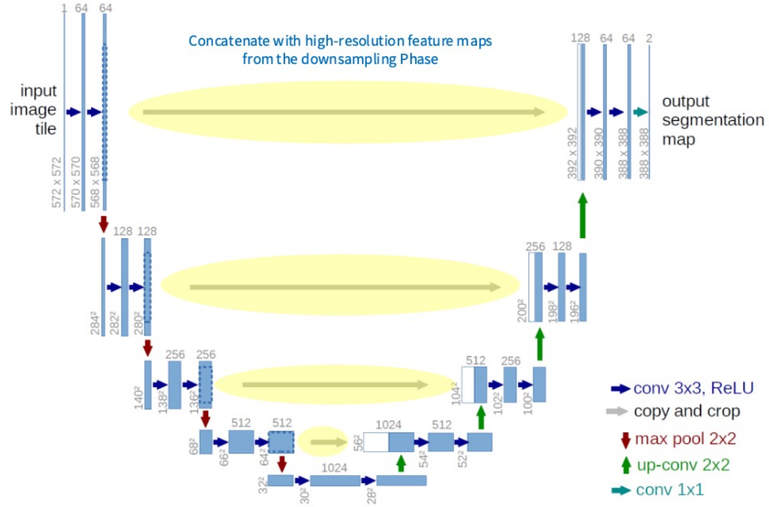
Ronneberger et al. (2015) U-net Architecture

U-Net



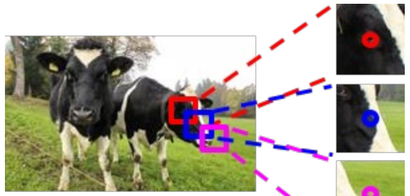
Ronneberger 等人 (2015) U-net 架构

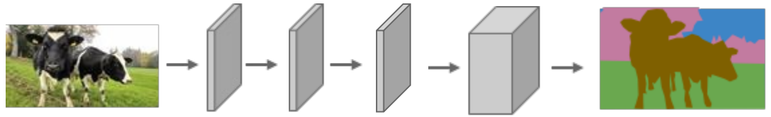
U-Net



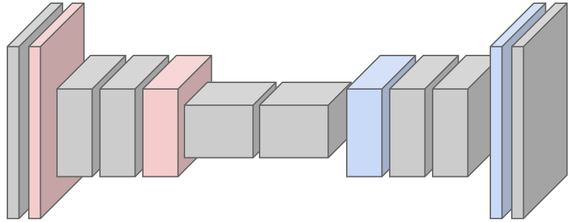
Ronneberger et al. (2015) U-net Architecture

语义分割：总结



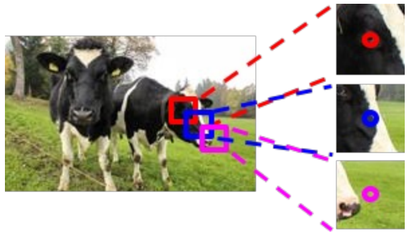


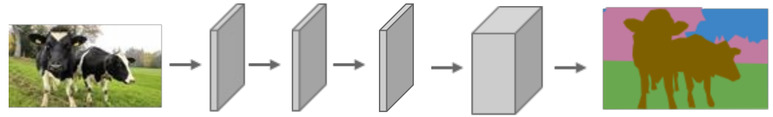




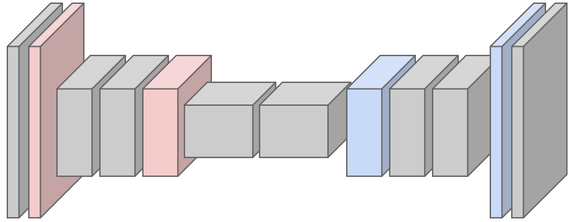


Semantic Segmentation: Summary







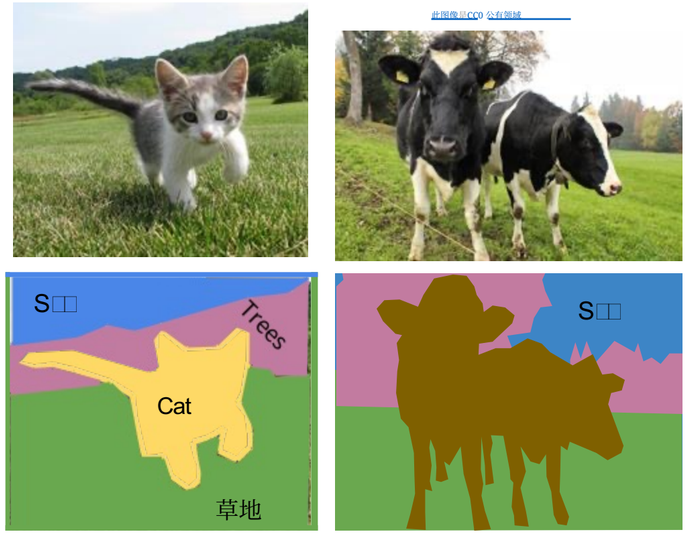




语义分割

为图像中的每个像素标注类别标签

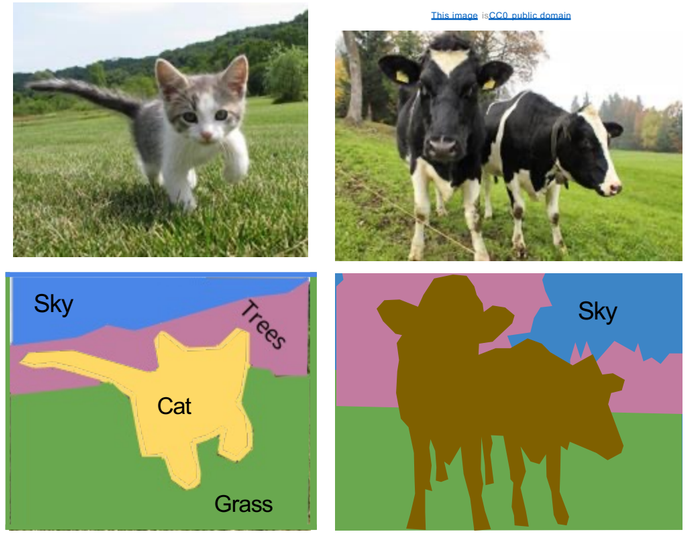
不区分实例，只关注像素



Semantic Segmentation

Label each pixel in the image with a category label

Don't differentiate instances, only care about pixels



今天

- **计算机视觉任务**
  - 语义分割 - **目标检测** - 实例分割
- 可视化与理解
  - 模型层可视化 - 显著性图 - CAM 与 Grad-CAM
- 基础模型与提示分割

Today

- **Computer Vision Tasks**
  - Semantic Segmentation
  - **Object Detection**
  - Instance Segmentation
- Visualization & Understanding
  - Model Layers Visualization
  - Saliency Maps
  - CAM & Grad-CAM
- Foundation Models and Prompt Segmentation

目标检测

分类

CAT

无空间范围

语义分割

草地，CAT，树，SK

没有对象，只有像素

目标检测

狗，狗，CAT

实例分割



狗，狗，CAT

多对象

Object Detection

Classification

CAT

No spatial extent

Semantic Segmentation

GRASS, CAT, TREE, SK

No objects, just pixels

Object Detection

DOG, DOG, CAT

Instance Segmentation



DOG, DOG, CAT

Multiple Object

目标检测

分类

CAT

无空间范围

语义分割

草地, CAT, 树, SK

没有物体，只有像素

目标检测



狗, 狗,CAT

实例分割



狗, 狗,CAT

多目标

Object Detection

Classification

CAT

No spatial extent

Semantic Segmentation

GRASS, CAT, TREE, SK

No objects, just pixels

Object Detection



DOG, DOG, CAT

Instance Segmentation

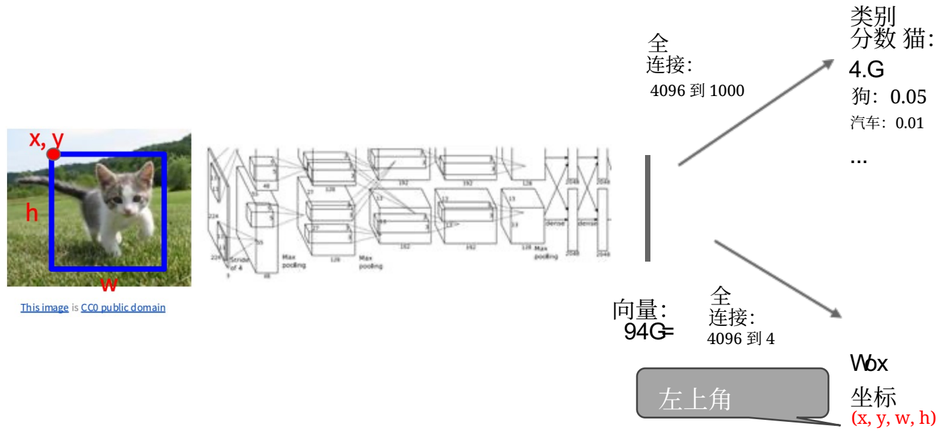


DOG, DOG, CAT

Multiple Object

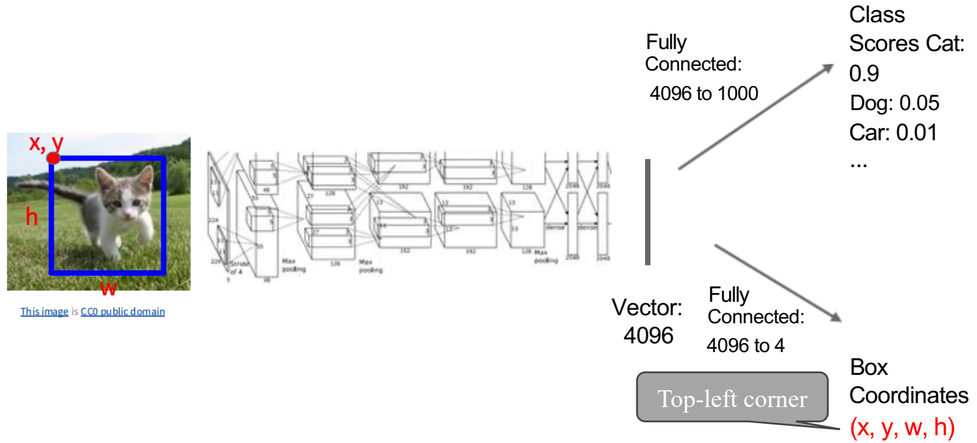
目标检测：单个对象

(分类 4 定位)



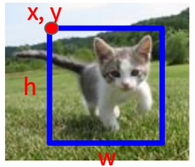
Object Detection: Single Object

(Classification + Localization)

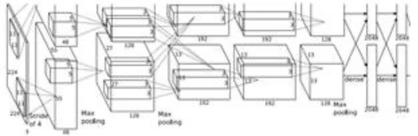


目标检测：单目标

（分类 4 定位）



This image is CC0 public domain



全
连通：
4096 到 1000

类别
分数 猫：
4.G
狗：0.05
汽车：0.01
...

Softmax
t oss

正确标签：
Cat

向量：
94G=

全
连接：
4096 到 4

Wbx
坐标
(x, y, w, h)

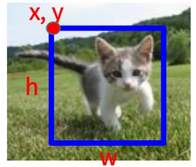
L2 损失

正确边界框：
(x', y', w', h')

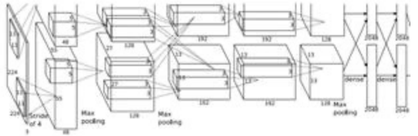
将定位视为一个
回归问题！

Object Detection: Single Object

(Classification + Localization)



This image is CC0 public domain



Fully
Connected:
4096 to 1000

Class
Scores Cat:
0.9
Dog: 0.05
Car: 0.01
...

Softmax
Loss

Correct label:
Cat

Vector:
4096

Fully
Connected:
4096 to 4

Box
Coordinates
(x, y, w, h)

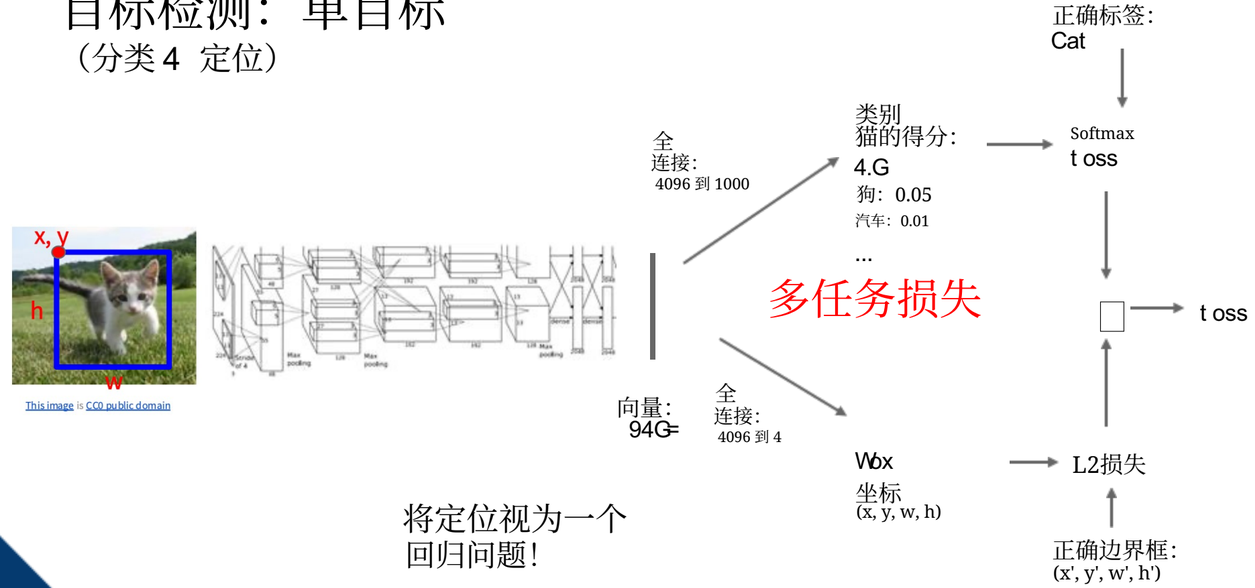
L2 Loss

Correct box:
(x', y', w', h')

Treat localization as a
regression problem!

目标检测：单目标

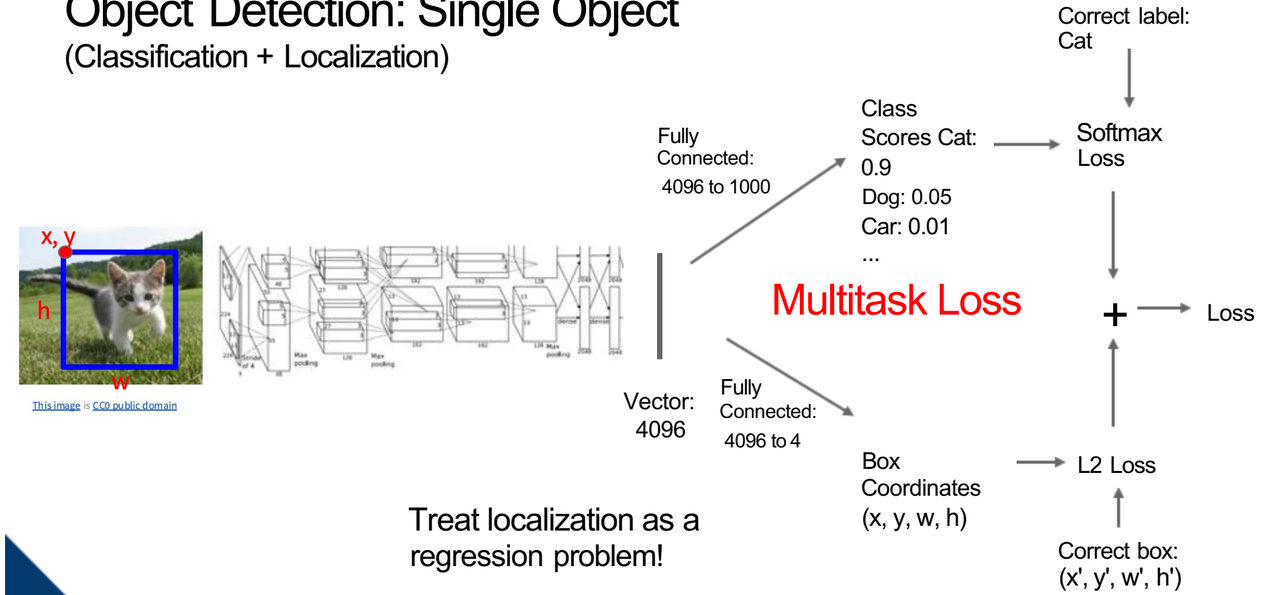
（分类 4 定位）



将定位视为一个
回归问题！

Object Detection: Single Object

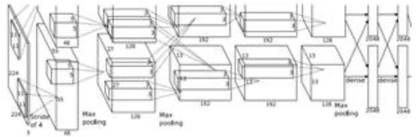
(Classification + Localization)



Treat localization as a
regression problem!

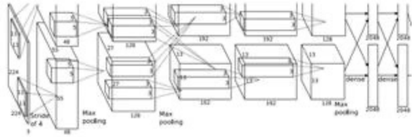
目标检测：多个对象





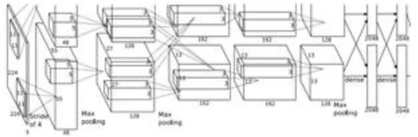
CAT：(x, y, w, h)





狗：(x, y, w, h) 狗：(x, y, w, h) CAT：(x, y, w, h)

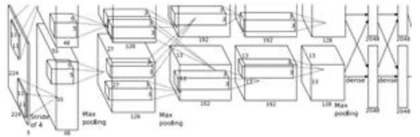




鸭子: (x, y, w, h) 鸭子: (x, y, w,h)....

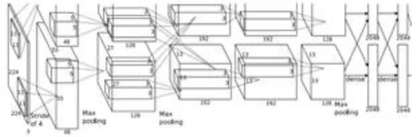
Object Detection: Multiple Objects





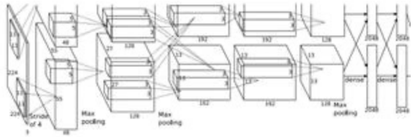
CAT: (x, y, w, h)





DOG: (x, y, w, h)
DOG: (x, y, w, h)
CAT: (x, y, w, h)



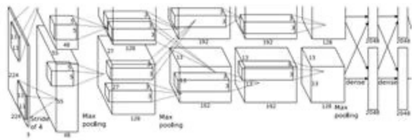


DUCK: (x, y, w, h)
DUCK: (x, y, w, h)
….

目标检测：多个目标

每张图像需要不同的输出数量！

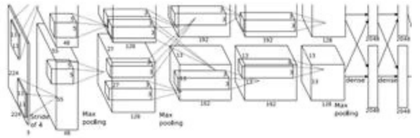




CAT: (x, y, w, h)

4个数字

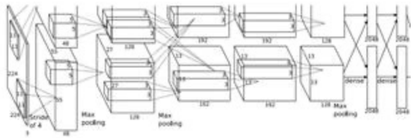




狗: (x, y, w, h) 狗: (x, y, w, h)
CAT: (x, y, w, h)

12 个数字





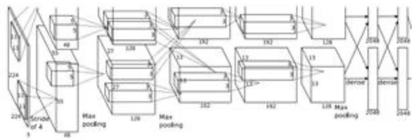
鸭子：(x, y, w, h) 鸭子：(x, y, w,h)....

许多数字！

Object Detection: Multiple Objects

Each image needs a different number of outputs!

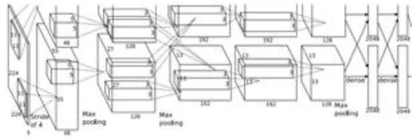




CAT: (x, y, w, h)

4 numbers

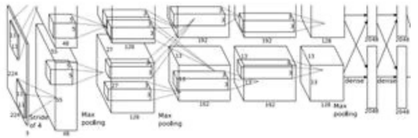




DOG: (x, y, w, h)
DOG: (x, y, w, h)
CAT: (x, y, w, h)

12 numbers





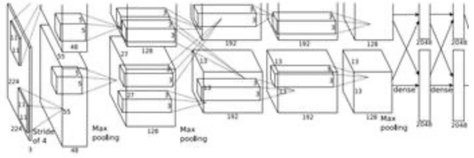
DUCK: (x, y, w, h)
DUCK: (x, y, w, h)
….

Many numbers!

目标检测：多个对象

将 CNN 应用于图像的许多不同裁剪区域，
CNN 将每个裁剪区域分类为对象或背景



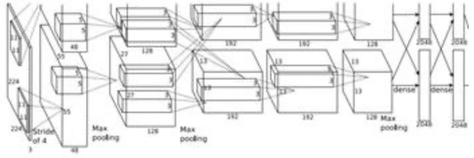


狗？否
猫？否 背景？
是

Object Detection: Multiple Objects

Apply a CNN to many different crops of the image, CNN classifies each crop as object or background



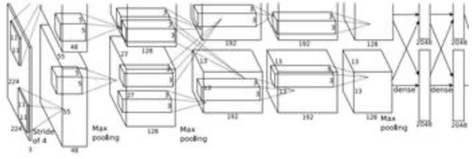


Dog? NO
Cat? NO
Background?
YES

目标检测：多个对象

将 CNN 应用于图像中许多不同的裁剪区域，CNN 将每个裁剪区域分类为目标或背景



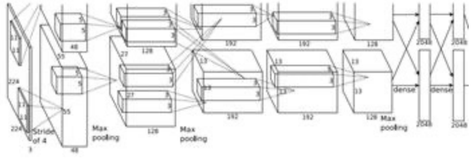


狗？ 是猫？
否 背景？ 否

Object Detection: Multiple Objects

Apply a CNN to many different crops of the image, CNN classifies each crop as object or background



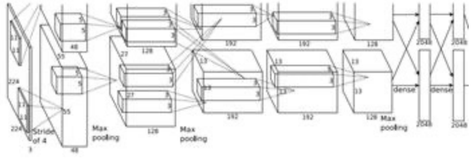


Dog? YES
Cat? NO
Background? NO

目标检测：多个对象

将 CNN 应用于图像的许多不同裁剪区域，
CNN 将每个裁剪区域分类为目标或背景



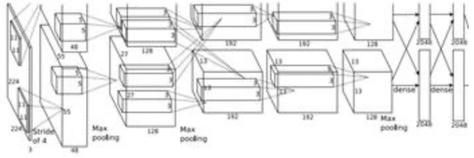


狗？ 是猫？
否 背景？ 否

Object Detection: Multiple Objects

Apply a CNN to many different crops of the image, CNN classifies each crop as object or background



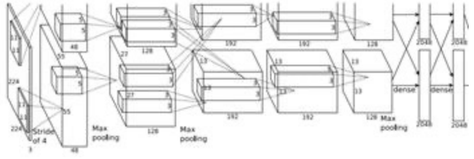


Dog? YES
Cat? NO
Background? NO

目标检测：多目标

将 CNN 应用于图像的许多不同裁剪区域，
CNN 将每个裁剪区域分类为目标或背景





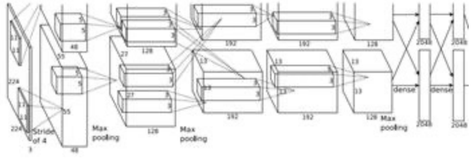
狗？否猫？
是 背景？否

问：这种方法有什么问题？

Object Detection: Multiple Objects

Apply a CNN to many different crops of the image, CNN classifies each crop as object or background



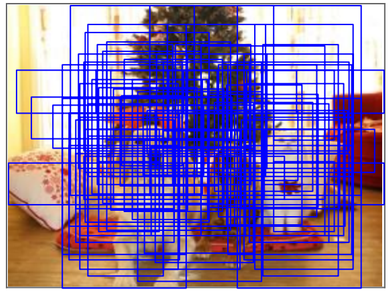


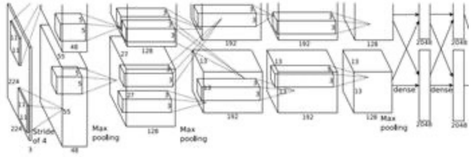
Dog? NO
Cat? YES
Background? NO

Q: What's the problem with this approach?

目标检测：多个对象

将 CNN 应用于图像的许多不同裁剪区域，
CNN 将每个裁剪区域分类为目标或背景



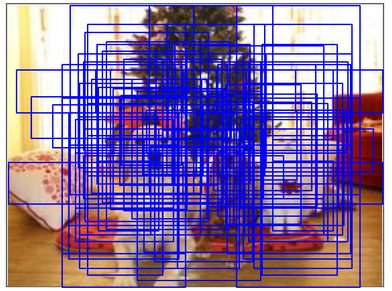


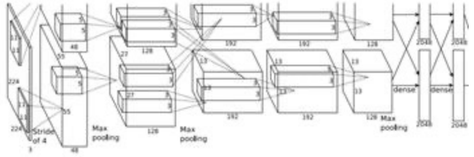
狗？否猫？
是 背景？否

问题：需要将 CNN 应用于巨大数量的
位置、尺度和纵横比，计算成本非常高！

Object Detection: Multiple Objects

Apply a CNN to many different crops of the image, CNN classifies each crop as object or background



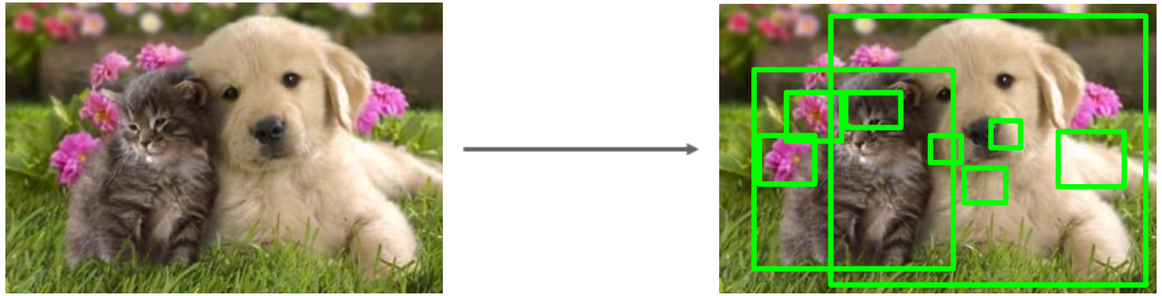


Dog? NO
Cat? YES
Background? NO

Problem: Need to apply CNN to huge number of locations, scales, and aspect ratios, very computationally expensive!

区域提议：Selective Search

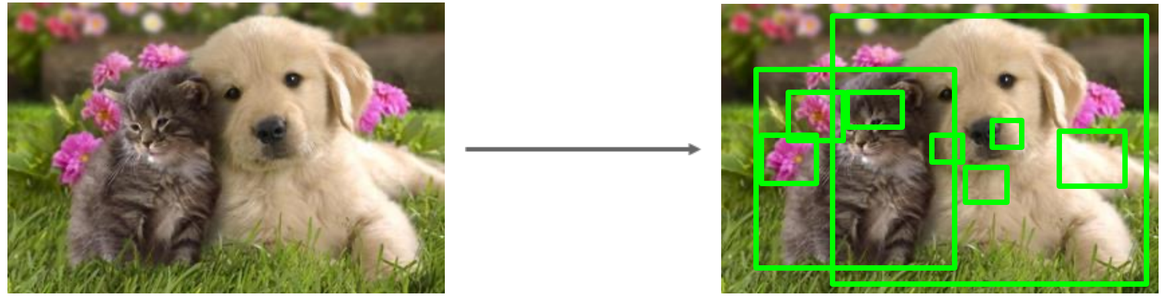
- 找出可能包含物体的“块状”图像区域
- 运行相对较快；例如，Selective Search 在CPU上几秒内即可给出2000个区域提议



Alexe 等，“测量图像窗口的目标性”，TPAMI 2012 Uijlings 等，“面向目标识别的 Selective Search”，IJCV 2013 Cheng 等，“BING：用于以300fps进行目标性估计的二值化范数梯度”，CVPR 2014 Zitnick 和 Dollar，“Edge Boxes：从边缘定位目标提议”，ECCV 2014

Region Proposals: Selective Search

- Find “blobby” image regions that are likely to contain objects
- Relatively fast to run; e.g. Selective Search gives 2000 region proposals in a few seconds on CPU



Alexe et al, “Measuring the objectness of image windows”, TPAMI 2012
Uijlings et al, “Selective Search for Object Recognition”, IJCV 2013
Cheng et al, “BING: Binarized normed gradients for objectness estimation at 300fps”, CVPR 2014
Zitnick and Dollar, “Edge boxes: Locating object proposals from edges”, ECCV 2014

R-CNN



输入图像

↑ irshic□et al，“Rich feature hierarchiesfor accurateobject detection and semanticsegmentatior□□ ⊂VPR 2014。图版权 Ross Girshick，2015；来源。经许可转载。

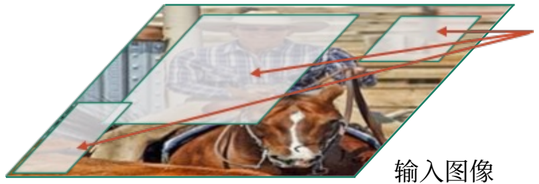
R-CNN



Input image

Girshick et al，“Rich feature hierarchies for accurate object detection and semantic segmentation”，CVPR ⊄0⁻ 4.
Figure copyright Ross Girshick, 2015;source. Reproduced with permission.

R-CNN



感兴趣区域 (RoI) 来自提议方法 (□2k)

输入图像

↑ irshic□et al，“用于精确目标检测和语义分割的丰富特征层次”，CVPR 2014。图版权 Ross Girshick，2015；来源。经许可转载。

R-CNN

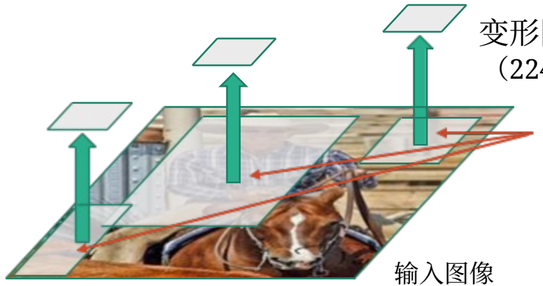


Regions of Interest (RoI) from a proposal method (~2k)

Input image

Girshick et al，“Rich feature hierarchies for accurate object detection and semantic segmentation”，CVPR 2014.
Figure copyright Ross Girshick, 2015;source. Reproduced with permission.

R-CNN



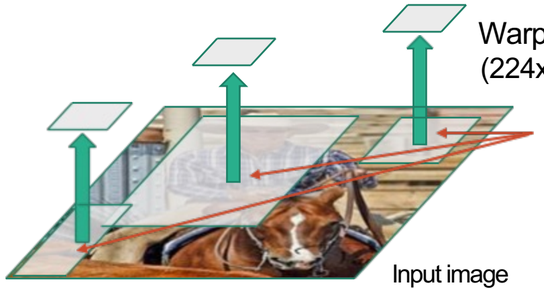
变形图像区域
（224x224 像素）

来自候选区域方法的
感兴趣区域（RoI）
（□2k）

输入图像

Girshick 等，“用于精确目标检测和语义分割的丰富特征层级”，CVPR 2014。图版权 Ross Girshick，2015；来源。经许可转载。

R-CNN



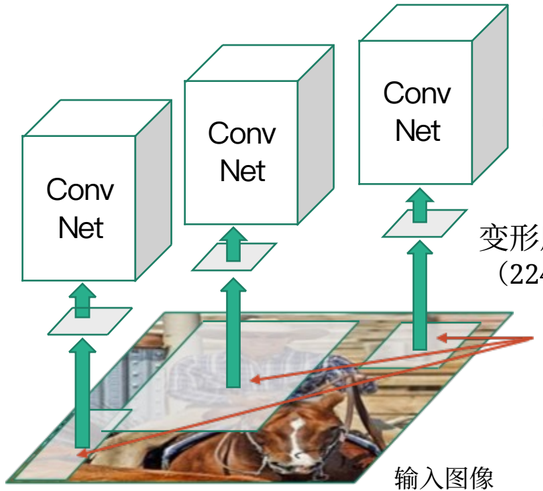
Warped image regions
(224x224 pixels)

Regions of Interest
(RoI) from a proposal
method (~2k)

Input image

Girshick et al，“Rich feature hierarchies for accurate object detection and semantic segmentation”，CVPR 2014.
Figure copyright Ross Girshick, 2015;source. Reproduced with permission.

R-CNN



将每个区域通过 ConvNet前向传播（经 ImageNet 预训练）

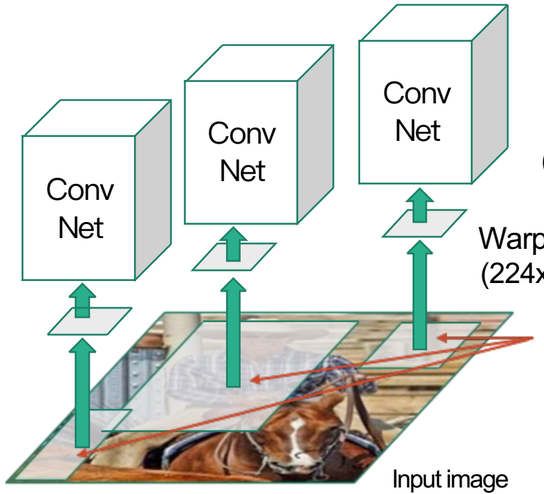
变形后的图像区域（224x224 像素）

感兴趣区域 (RoI) 来自候选提议方法（□2k）

输入图像

Girshick 等，“用于精确目标检测和语义分割的丰富特征层次结构”，CVPR 2014。
图版权 Ross Girshick，2015；来源。经许可转载。

R-CNN



Forward each region through ConvNet (ImageNet-pretranied)

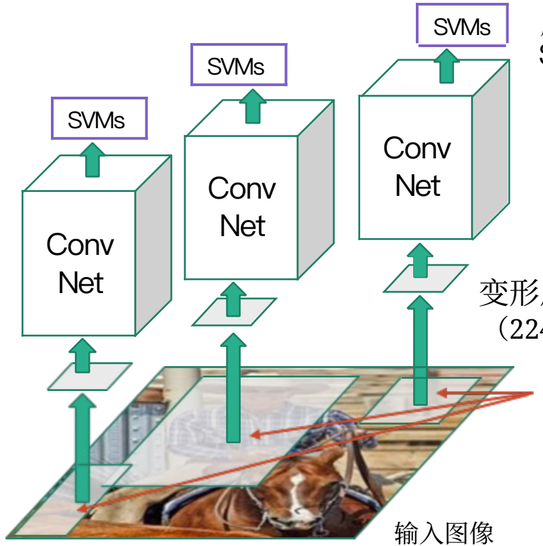
Warped image regions (224x224 pixels)

Regions of Interest (RoI) from a proposal method (~2k)

Input image

Girshick et al，“Rich feature hierarchies for accurate object detection and semantic segmentation”，CVPR 2014.
Figure copyright Ross Girshick, 2015;source. Reproduced with permission.

R-CNN



用以下内容对区域进行分类
SVMs

将每个区域前向传播
通过 ConvNet
（在 ImageNet 上预训练的）

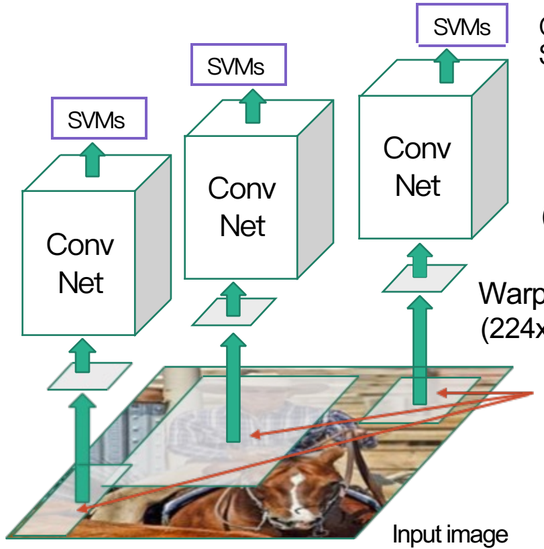
变形后的图像区域
（224x224 像素）

感兴趣区域
（RoI）来自提议
方法（□2k）

输入图像

↑ irshic□et al,"Rich feature hierarchiesfor accurateobject detection and semanticsegmentation□,CVPR 2014. 图版权 Ross Girshick，2015；来源。经许可转载。

R-CNN



Classify regions with SVMs

Forward each region through ConvNet (ImageNet-pretranied)

Warped image regions (224x224 pixels)

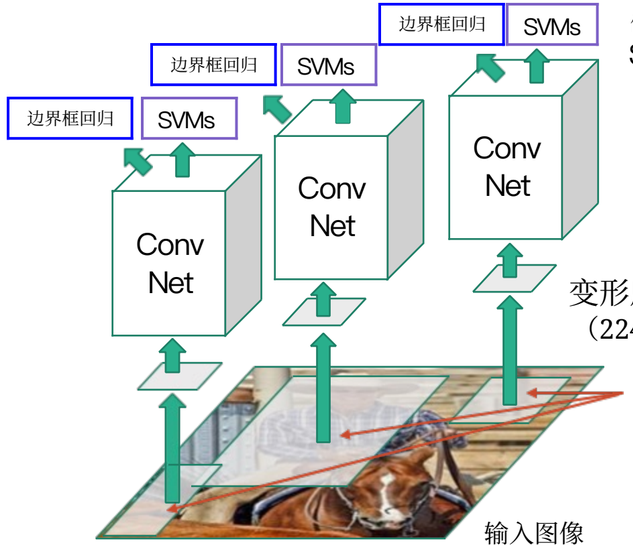
Regions of Interest (RoI) from a proposal method (~2k)

Input image

Girshick et al，"Rich feature hierarchies for accurate object detection and semantic segmentation"，CVPR 2014.
Figure copyright Ross Girshick, 2015;source. Reproduced with permission.

R-CNN

预测对 RoI 的“修正”：4 个数字：(dx, dy, dw, dh)



使用……对区域进行分类
SVMs

将每个区域前向传播
通过 ConvNet
（ImageNet 预训练的）

变形后的图像区域
（224x224 像素）

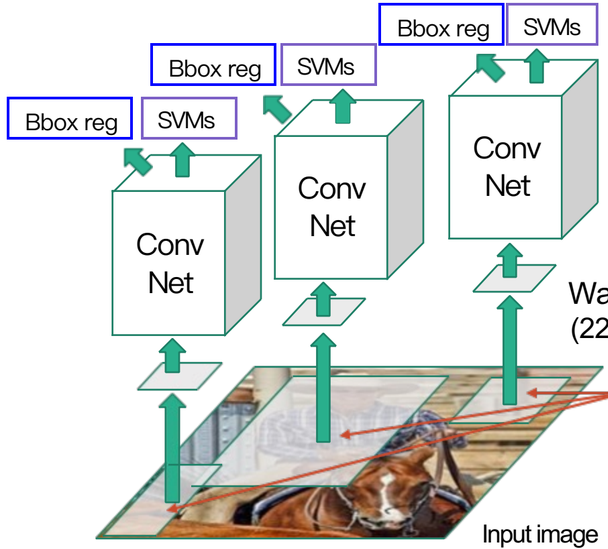
感兴趣区域
来自候选区域的 (RoI)
方法（□2k）

输入图像

Girshick 等，“用于准确目标检测和语义分割的丰富特征层次结构”，CVPR 2014。
图版权 Ross Girshick，2015；来源。经许可转载。

R-CNN

Predict“corrections”to the RoI: 4 numbers: (dx, dy, dw, dh)



Classify regions with SVMs

Forward each region through ConvNet (ImageNet-pretranied)

Warped image regions (224x224 pixels)

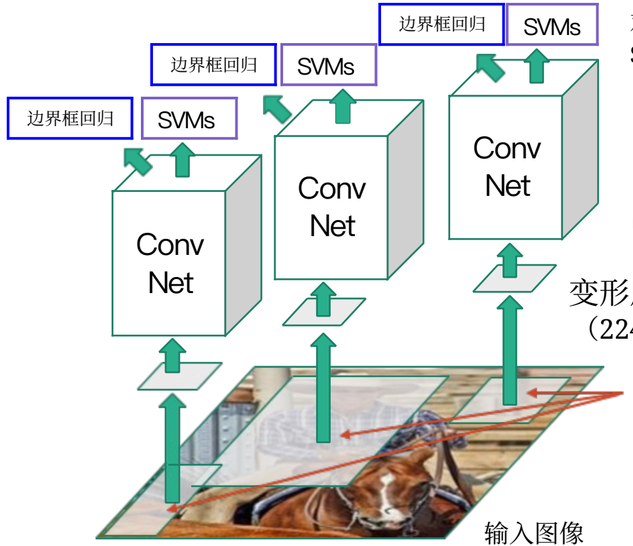
Regions of Interest (RoI) from a proposal method (~2k)

Input image

Girshick et al，“Rich feature hierarchies for accurate object detection and semantic segmentation”，CVPR 2014.
Figure copyright Ross Girshick, 2015;source. Reproduced with permission.

R-CNN

预测对 RoI 的“修正”：4 个数字：(dx, dy, dw, dh)



对区域进行分类，使用 SVMs

将每个区域前向传播通过卷积网络 (ImageNet 预训练)

变形后的图像区域（224x224 像素）

感兴趣区域（RoI）来自区域提议方法（□2k）

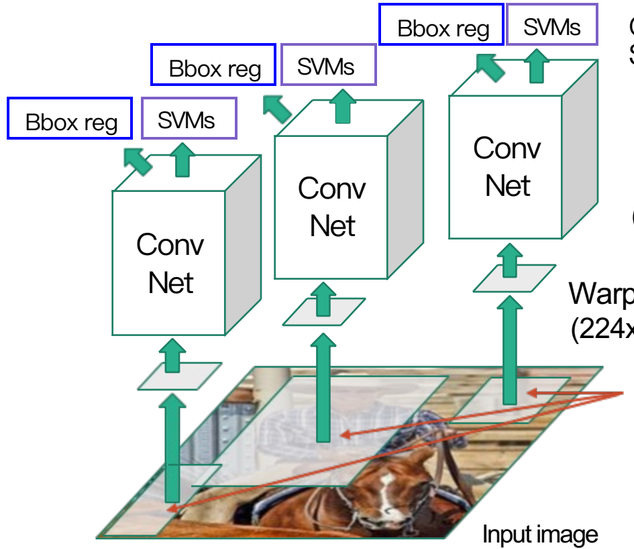
输入图像

问题：非常慢！需要对每张图像进行 □7k 次独立前向 传递！

Girshick 等，“用于精确目标检测和语义分割的丰富特征层次”，CVPR 2014。图片版权 Ross Girshick，2015；来源。经许可转载。

R-CNN

Predict“corrections”to the RoI: 4 numbers: (dx, dy, dw, dh)



Classify regions with SVMs

Forward each region through ConvNet (ImageNet-pretranied)

Warped image regions (224x224 pixels)

Regions of Interest (RoI) from a proposal method (~2k)

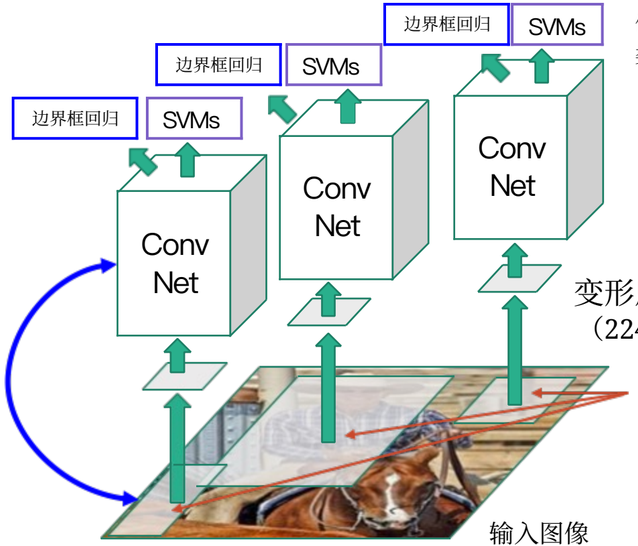
Input image

Problem: Very slow! Need to do ~2k independent forward passes for each image!

Girshick et al，“Rich feature hierarchies for accurate object detection and semantic segmentation”，CVPR 2014.
Figure copyright Ross Girshick, 2015;source. Reproduced with permission.

“慢” b-Ca a

预测对 RoI 的“修正”：4 个数值：(dx, dy, dw, dh)



使用SVMs对区域进行分类

对每个区域进行前向传播通过卷积网络（经 ImageNet 预训练）

变形后的图像区域（224x224 像素）

感兴趣区域（RoI）来自区域提议方法（□2k）

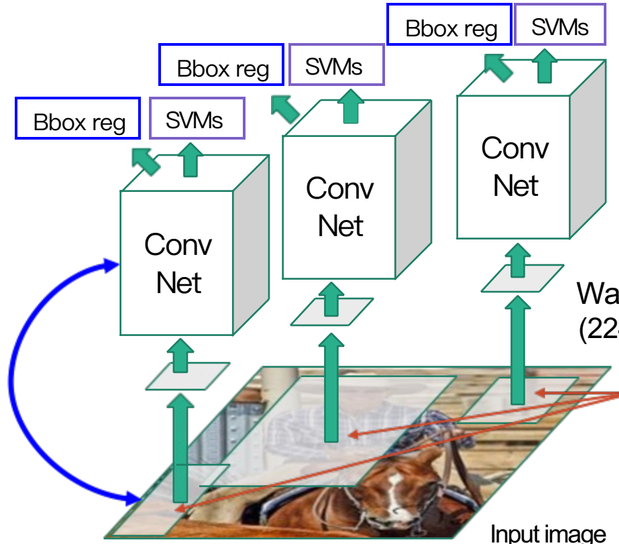
问题：非常慢！
需要执行 □7k 次
独立前向传播
每张图像都要进行多次传递！

思路：在裁剪前先让图像通过卷积网络！改为裁剪卷积特征！

Girshick 等，“用于精确目标检测和语义分割的丰富特征层级”，CVPR 2014。图片版权 Ross Girshick，2015；来源。经许可转载。

“Slow” R-CNN

Predict“corrections”to the RoI: 4 numbers: (dx, dy, dw, dh)



Classify regions with SVMs

Forward each region through ConvNet (ImageNet-pretranied)

Warped image regions (224x224 pixels)

Regions of Interest (RoI) from a proposal method (~2k)

Problem: Very slow! Need to do ~2k independent forward passes for each image!

Idea: Pass the image through convnet before cropping! Crop the conv feature instead!

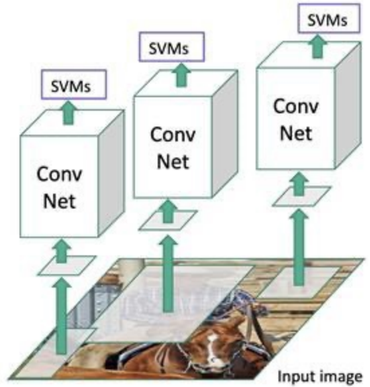
Girshick et al，“Rich feature hierarchies for accurate object detection and semantic segmentation”，CVPR 2014.
Figure copyright Ross Girshick, 2015;source. Reproduced with permission.

Fast R-CNN



输入图像

“慢速” R-CNN



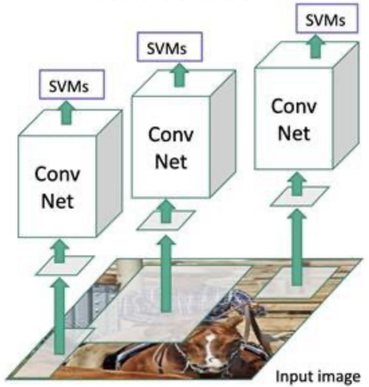
Girshick，“Fast R-CNN”，ICCV 2015。图版权 Ross Girshick，2015；来源。经许可转载

Fast R-CNN



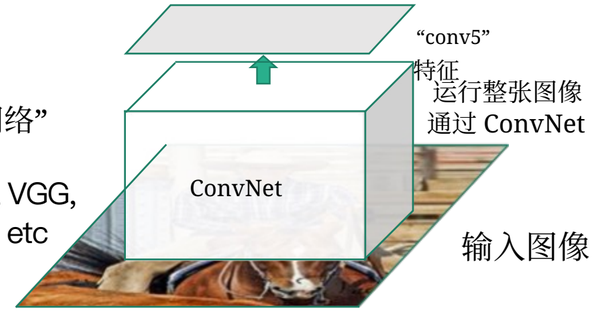
Input image

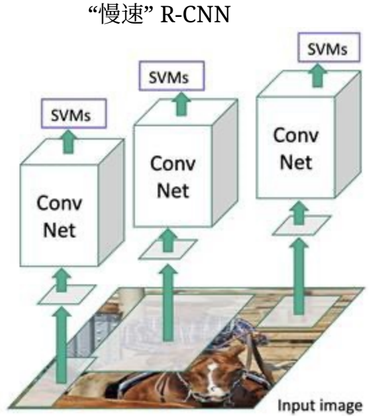
“Slow” R-CNN



Girshick，“Fast R-CNN”，ICCV 2015. Figure copyright Ross Girshick，2015; source. Reproduced with permission.

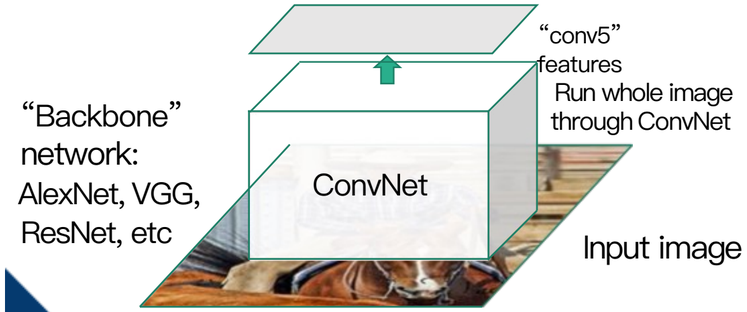
Fast R-CNN

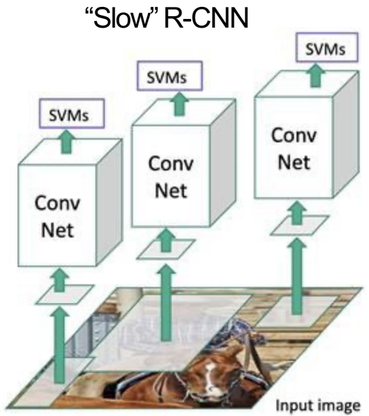




Girshick, “Fast R-CNN”, ICCV 2015。图版权 Ross Girshick，2015；来源。经许可转载.

Fast R-CNN



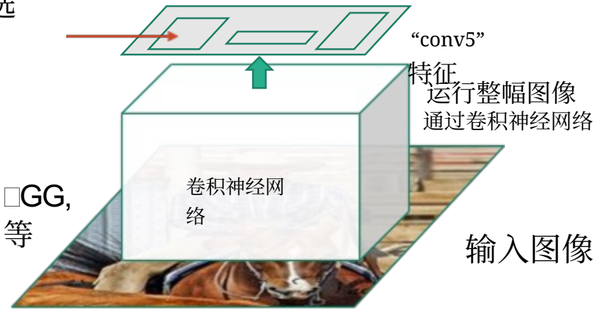


Girshick, “Fast R-CNN”, ICCV 2015. Figure copyright Ross Girshick, 2015; source. Reproduced with permission.

Fast R-CNN

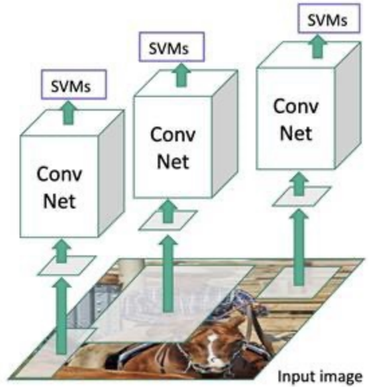
感兴趣区域
（RoIs）
来自候选
方法

“主干”
网络:
AlexNet，□GG,
ResNet 等



Girshick, “Fast R-CNN”, ICCV 2015。图片版权归 Ross Girshick 所有，2015； 来源。经许可转载.

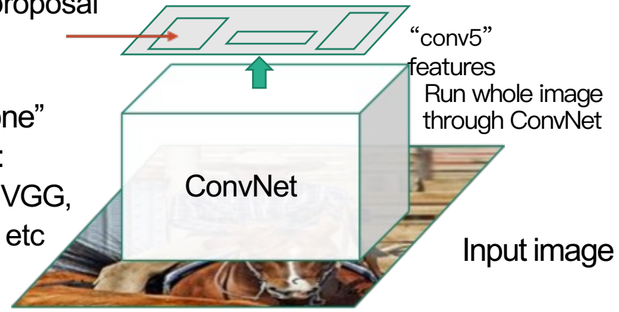
“慢速” R-CNN



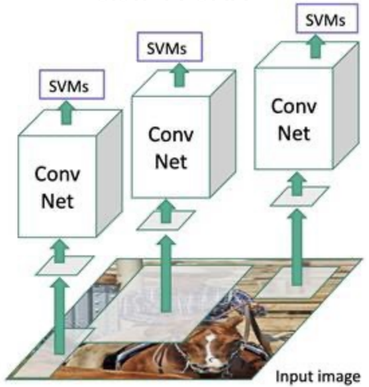
Fast R-CNN

Regions of
Interest (RoIs)
from a proposal
method

“Backbone”
network:
AlexNet, VGG,
ResNet, etc



“Slow” R-CNN

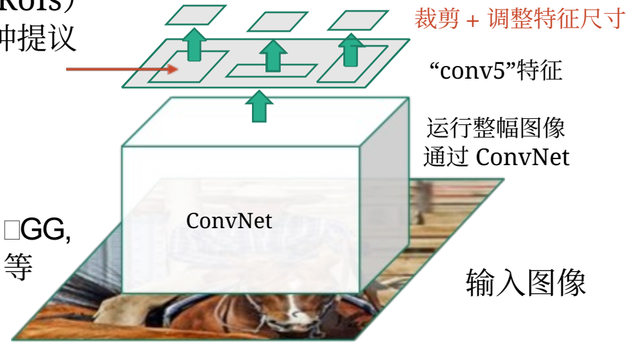


Girshick, “Fast R-CNN”, ICCV 2015. Figure copyright Ross Girshick, 2015; source. Reproduced with permission.

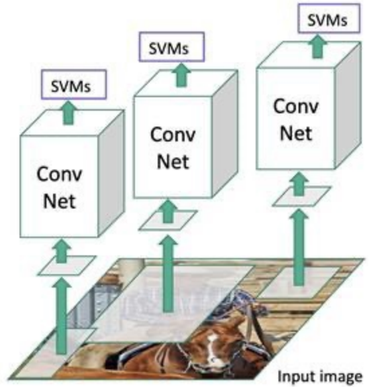
Fast R-CNN

的区域
兴趣（RoIs）
来自一种提议
方法

“主干”
网络:
AlexNet ，□GG,
ResNet 等



“慢速” R-CNN

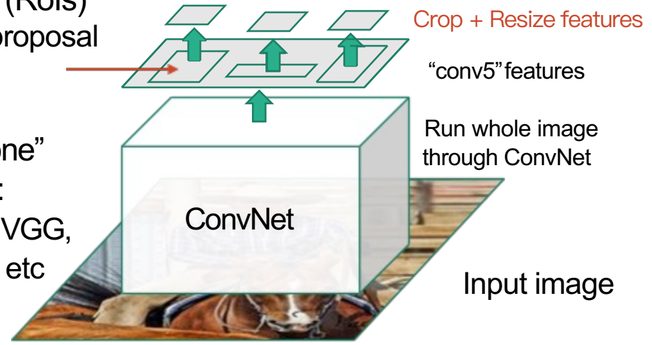


Girshick, “Fast R-CNN”, ICCV 2015。图片版权 Ross Girshick，2015；来源。经许可转载

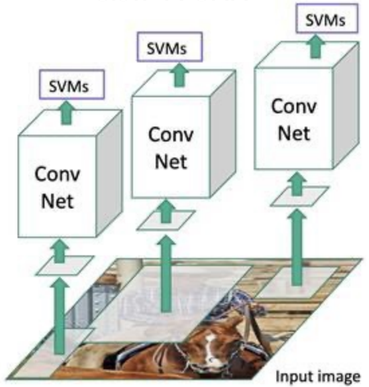
Fast R-CNN

Regions of
Interest (RoIs)
from a proposal
method

“Backbone”
network:
AlexNet, VGG,
ResNet, etc

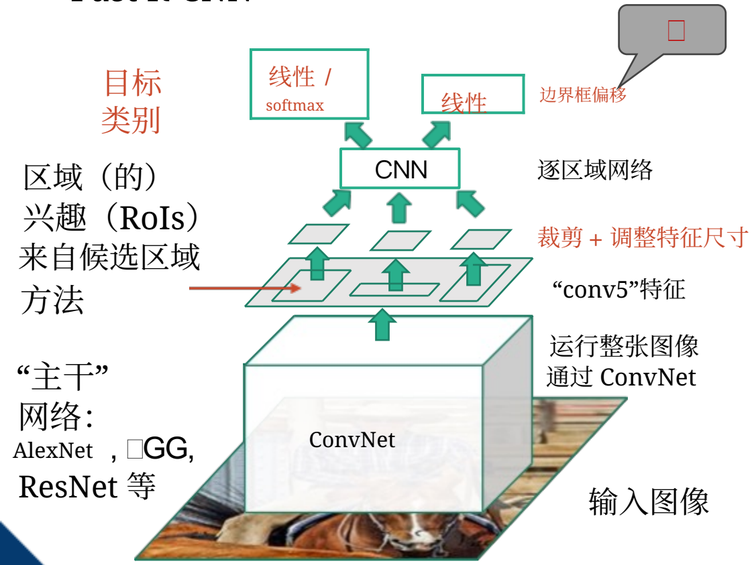


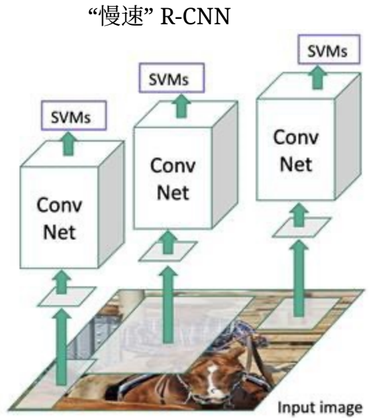
“Slow” R-CNN



Girshick, “Fast R-CNN”, ICCV 2015. Figure copyright Ross Girshick, 2015; source. Reproduced with permission.

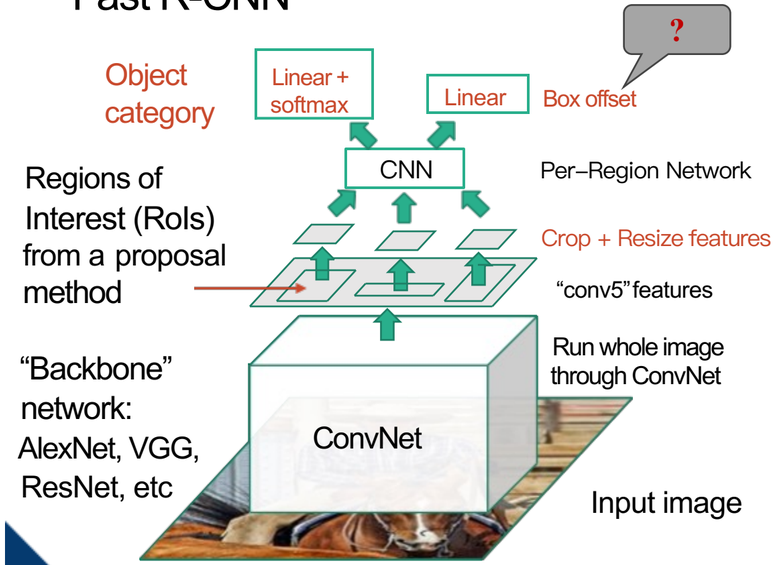
Fast R-CNN

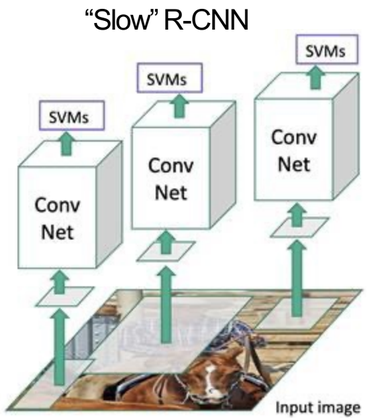




Girshick, “Fast R-CNN”, ICCV 2015。图版权归 Ross Girshick 所有，2015； 来源。经许可转载.

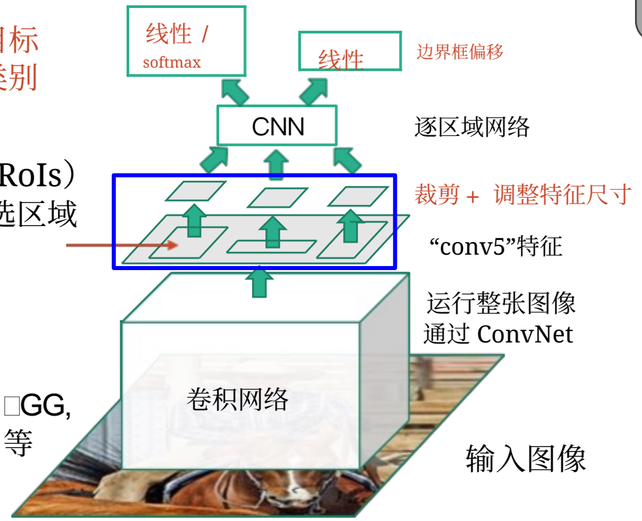
Fast R-CNN





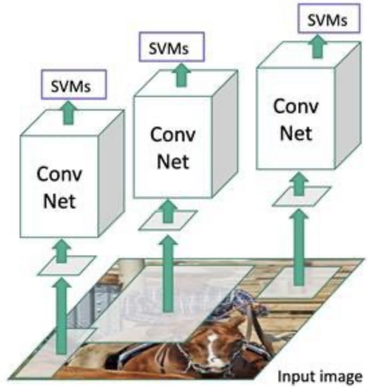
Girshick, “Fast R–CNN”, ICCV 2015. Figure copyright Ross Girshick, 2015; source. Reproduced with permission.

Fast R-CNN



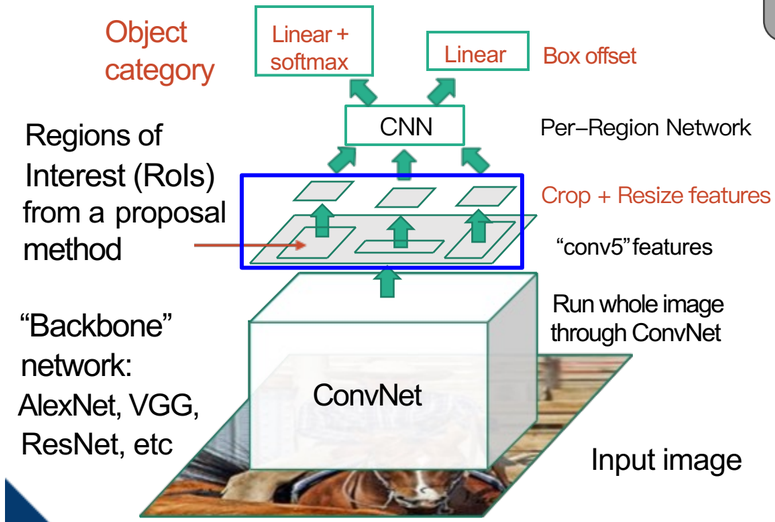
Fast 的局限性 R-CNN?

“慢速” R-CNN



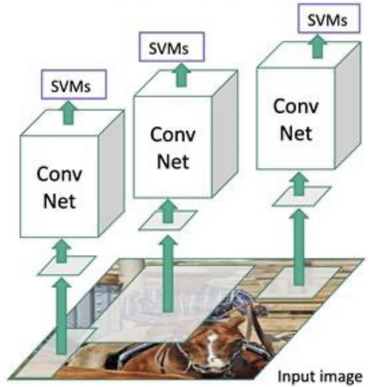
Girshick, “Fast R-CNN”, ICCV 2015。图片版权 Ross Girshick，2015； 来源。经许可转载.

Fast R-CNN



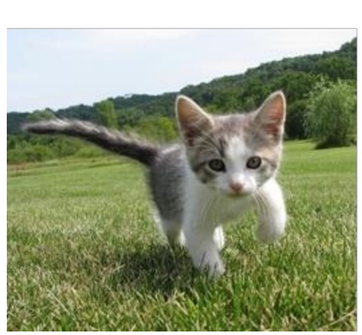
Limitations of Fast R-CNN?

“Slow” R-CNN



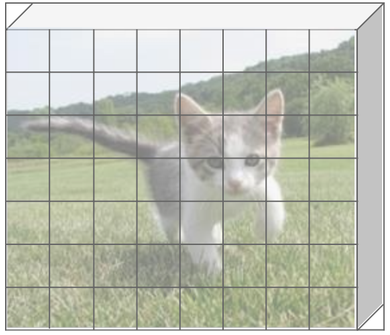
Girshick, “Fast R–CNN”, ICCV 2015. Figure copyright Ross Girshick, 2015; source. Reproduced with permission.

区域提议网络



输入图像（例如 3 x 640 x 480）
特征





图像

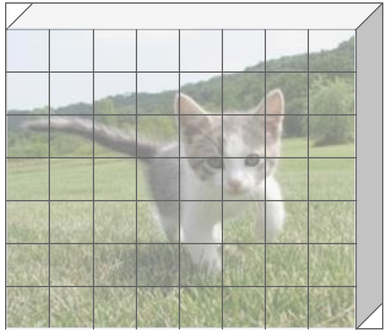
（例如 512 x 64 x 5: ）

Region Proposal Network



Input Image
(e.g. 3 x 640 x 480)
features



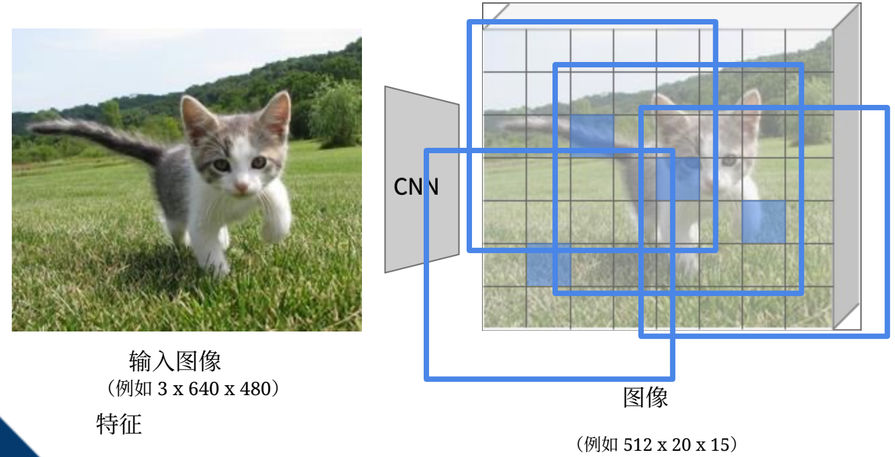


Image

(e.g. 512 x 20 x 15)

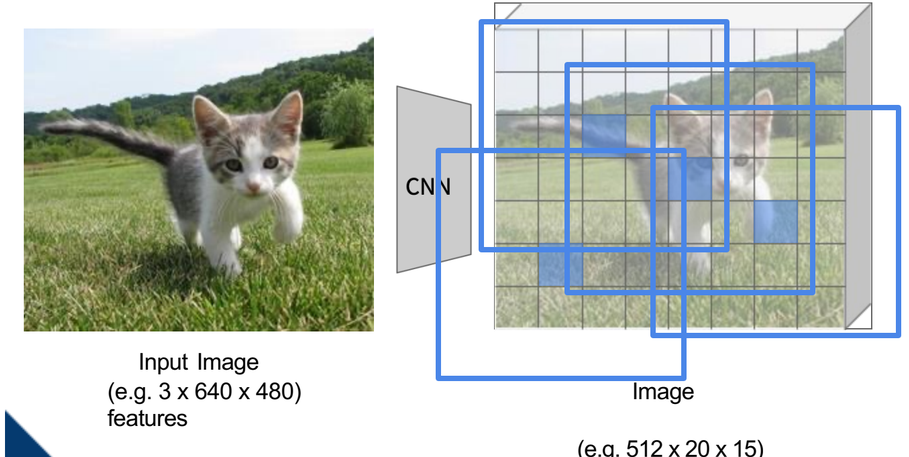
区域提议网络

想象特征图中的每个点处都有一个固定大小的锚框



Region Proposal Network

Imagine an anchor box of fixed size at each point in the feature map

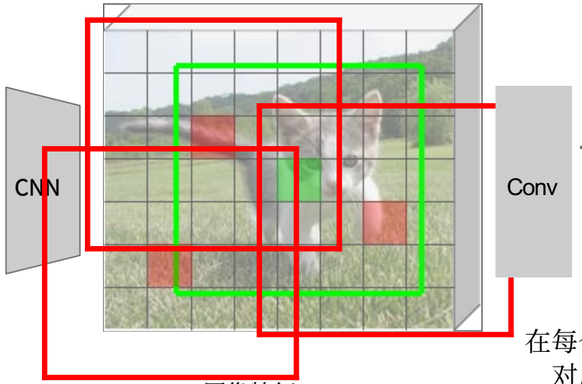


区域提议网络

想象在特征图的每个点
上都有一个固定大小的锚框



输入图像
（例如 3 x 640 x 480）



图像特征
（例如 512 x 20 x 15）

Conv

锚框是一个目标吗?
1 x 20 x 15

在每个点处，预测是否
对应的锚框
包含一个对象（二值
分类）

Region Proposal Network

Imagine an anchor box of
fixed size at each point in
the feature map



Input Image
(e.g. 3 x 640 x 480)

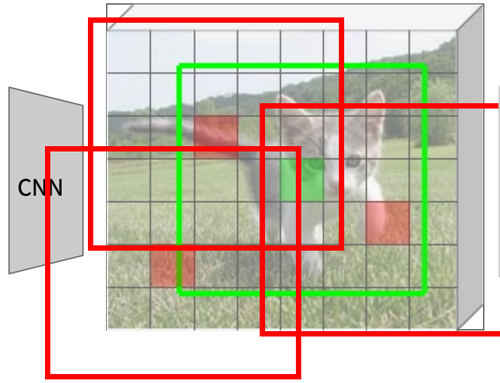


Image features
(e.g. 512 x 20 x 15)

Conv

Anchor is an object?
1 x 20 x 15

At each point, predict whether
the corresponding anchor
contains an object (binary
classification)

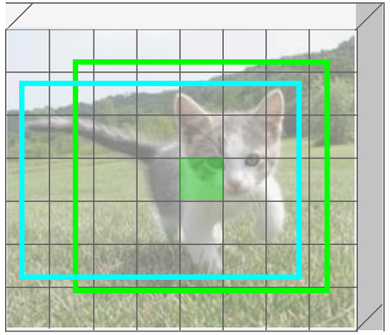
区域提议网络

想象在特征图的每个点
上都有一个固定大小的锚框



输入图像（例
如 3 x 640 x 480）

CNN



图像特征（例
如 512 x 20 x 15）

Conv

锚框是一个对象吗?
1 x 20 x 15

边界框修正
4 x 20 x 15
(w_1, w_2, w_y, w_h)

对于正样本框，还要预测从锚框到真实框的修正量（每个像素回归4个数值）

Region Proposal Network

Imagine an anchor box of
fixed size at each point in
the feature map



Input Image
(e.g. 3 x 640 x 480)

CNN

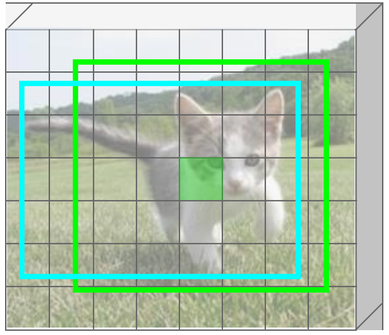


Image features
(e.g. 512 x 20 x 15)

Conv

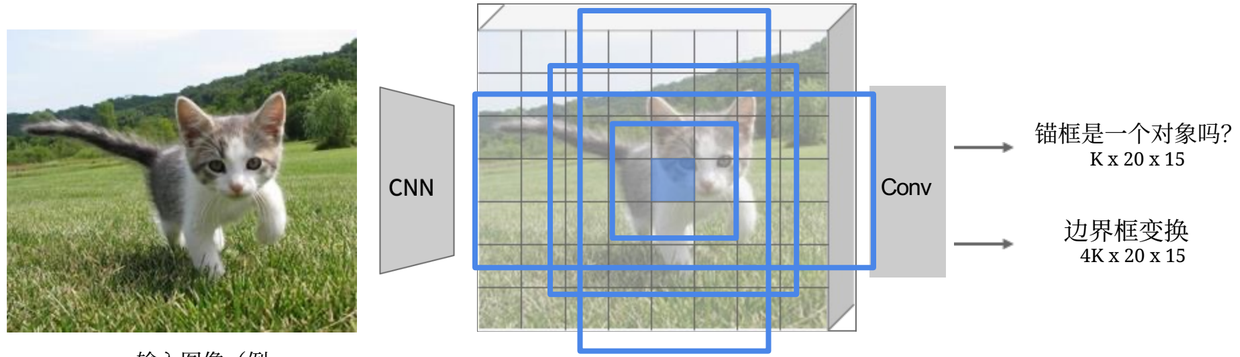
Anchor is an object?
1 x 20 x 15

Box corrections
4 x 20 x 15
(dx, dy, dw, dh)

For positive boxes, also predict a
corrections from the anchor to
the ground-truth box (regress 4
numbers per pixel)

区域提议网络

在实践中，在每个点使用
K 个不同大小/尺度的锚框

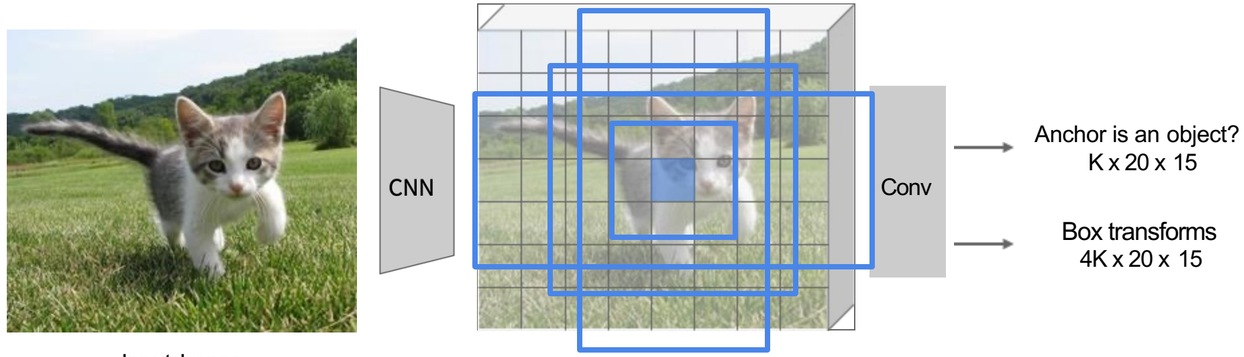


输入图像（例
如 3 x 640 x 480）

图像特征（例
如 512 x 20 x 15）

Region Proposal Network

In practice use K different
anchor boxes of different
size / scale at each point



Input Image
(e.g. 3 x 640 x 480)

Image features
(e.g. 512 x 20 x 15)

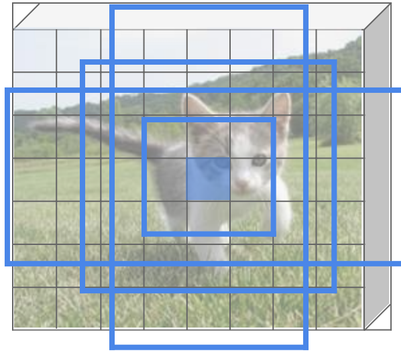
区域候选网络

在实践中，在每个点使用
K 个不同大小/尺度的锚框



输入图像（例
如 3 x 640 x 480）

CNN



图像特征
（例如 512 x 20 x 15）

Conv

锚框是一个对象吗?
K x 20 x 15

边界框变换
4K x 20 x 15

将 K*20*15 个框按
它们的“对象性”分数，取
前 □OGG 个作为我们的候选区域

Region Proposal Network

In practice use K different anchor boxes of different size / scale at each point



Input Image
(e.g. 3 x 640 x 480)

CNN

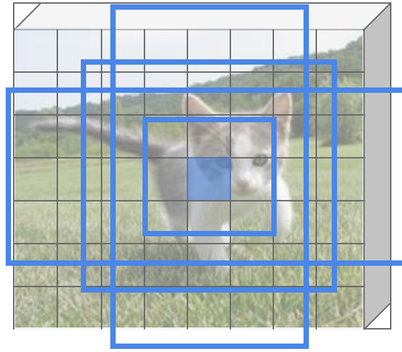


Image features
(e.g. 512 x 20 x 15)

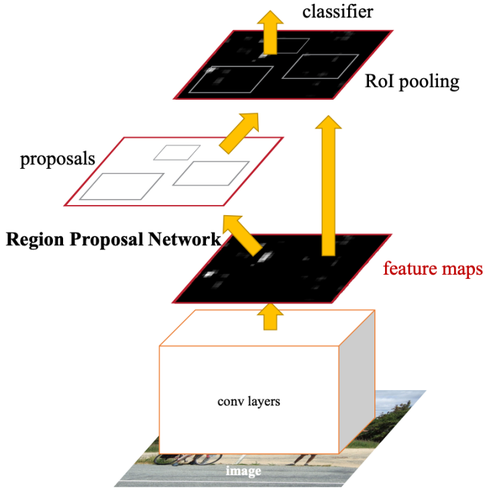
Conv

Anchor is an object?
K x 20 x 15

Box transforms
4K x 20 x 15

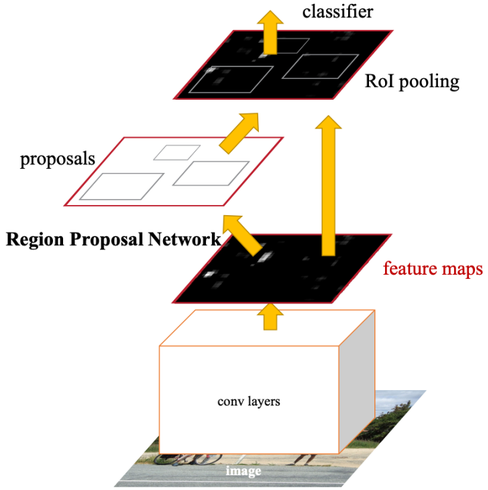
Sort the K*20*15 boxes by their “objectness” score, take top ~300 as our proposals

Faster R-CNN



- **核心创新：“共享卷积特征”**
- **问题：**先前的方法（Fast R-CNN）依赖于外部的、缓慢的 候选区域生成方法（Selective Search），这些方法在 CPU 上运行。
- **解决方案：**集成一个 **区域提议网络（RPN）**，它与 **Fast R-CNN 检测器**共享相同的主干特征。
- **结果：** 生成候选框的成本几乎为 **零** （每张图像约 10ms）。

Faster R-CNN

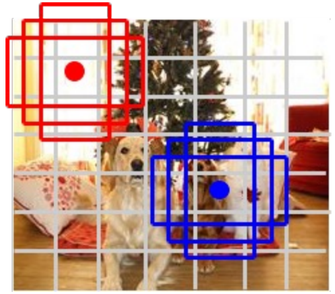


- **The Core Innovation: "Shared Convolutional Features"**
- **The Problem:** Previous methods (Fast R-CNN) relied on external, slow proposal methods (Selective Search) running on CPU.
- **The Solution:** Integrate a **Region Proposal Network (RPN)** that shares the same backbone features with the **Fast R-CNN detector**.
- **Result:** The cost of generating proposals is nearly **zero** (approx. 10ms per image).

单阶段目标检测器：YOLO / SSD / RetinaNet



输入图像 3 x H x W



将图像划分为 7 x 7 的网格

一组以每个网格单元为中心的基准框，此处 B N3

在每个网格单元内：

C 从 B 个中的每一个回归基础框到一个最终框，该框具有 5 个数字：(dx, dy, dh, dw, confidence)

C 为 C 个中的每一个预测分数类别（包括背景作为一个类别）

C 看起来很像 RPN ，but 特定类别！

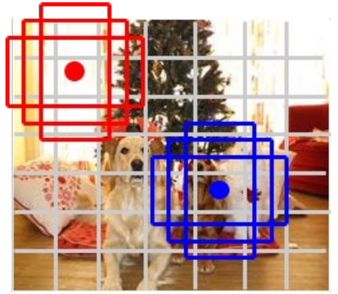
输出：7 x 7 x (5 * B 3 C)

Redmon 等，“你只看一次：统一的实时目标检测”，CVPR 2016 Liu 等，“SSD：单次多框检测器”，ECCV 2016 Lin 等，“用于密集目标检测的焦点损失”，ICCV2017

Single-Stage Object Detectors: YOLO / SSD / RetinaNet



Input image 3 x H x W



Divide image into grid 7 x 7

a set of base boxes centered at each grid cell Here B = 3

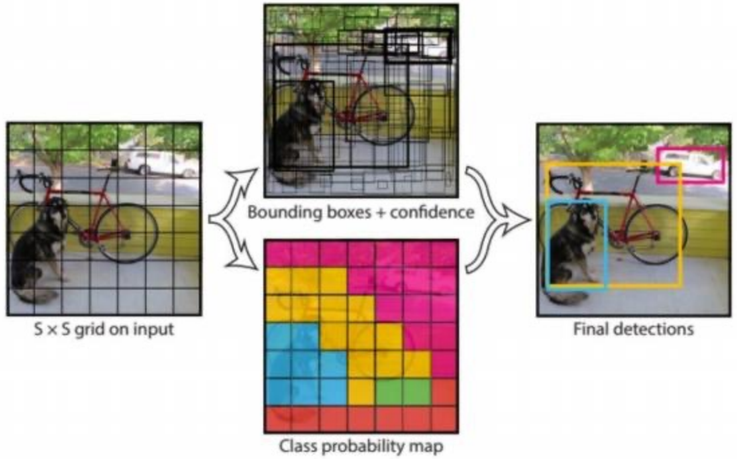
Within each grid cell:

- Regress from each of the B base boxes to a final box with 5 numbers: (dx, dy, dh, dw, confidence)
- Predict scores for each of C classes (including background as a class)
- Looks a lot like RPN, but category-specific!

Output: 7 x 7 x (5 * B + C)

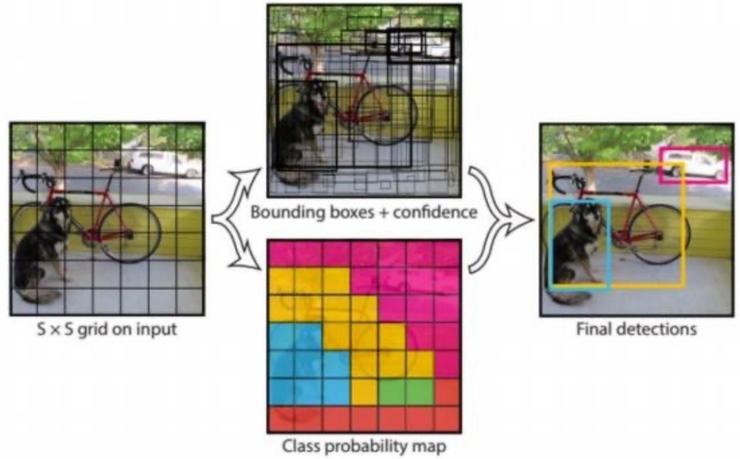
Redmon et al, “You Only Look Once: Unified, Real-Time Object Detection”, CVPR 2016
Liu et al, “SSD: Single-Shot MultiBox Detector”, ECCV 2016
Lin et al, “Focal Loss for Dense Object Detection”, ICCV 2017

k e Ye (k □□ e □□Y□□□ e □no)实时目标检测



Redmon 等，“You only look once：统一的实时目标检测（2015）。”

YOLO (You Only Look Once) real-time object detection



Redmon et al. "You only look once: unified, real-time object detection (2015)."

YOLO



SxS 网格

YOLO



SxS Grid

YOLO



对每个框输出：

- P(object)：框包含对象的概率
- B 个边界框（x, y, h, w）
- P(class)：属于某一类别的概率

□x□
n rid

Redmon 等，“You only look once：统一的实时目标检测（2015）。”

YOLO



For each box output:

- P(object): probability that the box contains an object
- B bounding boxes (x, y, h, w)
- P(class): probability of belonging to a class

SxS
Grid

Redmon et al. "You only look once: unified, real-time object detection (2015)."

YOLO



对于每个框的输出：

- P(object)：框包含物体的概率
- B 边界框 (x, y, h, w)
- P(class)：属于某一类别的概率

PL=

□x□
n rid

YOLO



For each box output:

- P(object): probability that the box contains an object
- B bounding boxes (x, y, h, w)
- P(class): probability of belonging to a class

B=2

SxS
Grid

YOLO



对于每个框的输出：

- P(object)：框包含物体的概率
- B 边界框 (x, y, h, w)
- P(class)：属于某一类别的概率

PL=

□x□
n rid

Redmon 等，“You only look once：统一的实时目标检测（2015）。”

YOLO



For each box output:

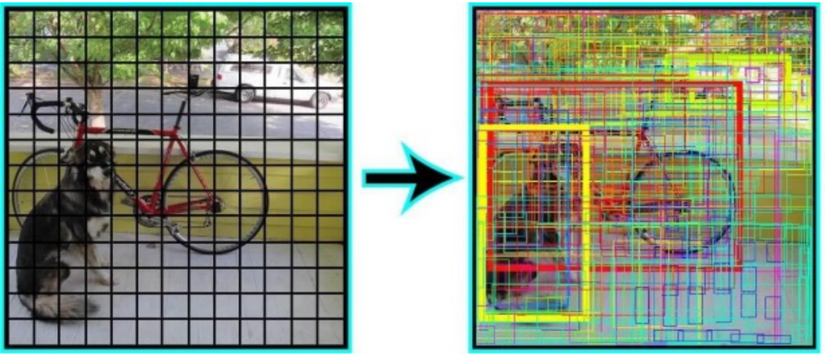
- P(object): probability that the box contains an object
- B bounding boxes (x, y, h, w)
- P(class): probability of belonging to a class

B=2

SxS
Grid

Redmon et al. "You only look once: unified, real-time object detection (2015)."

YOLO

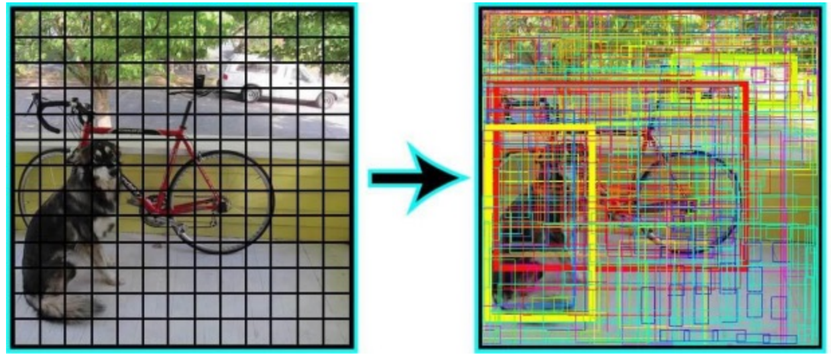


SxS 网格

许多具有不同目标概率的边界框

Redmon 等，“You only look once：统一的实时目标检测（2015）。”

YOLO

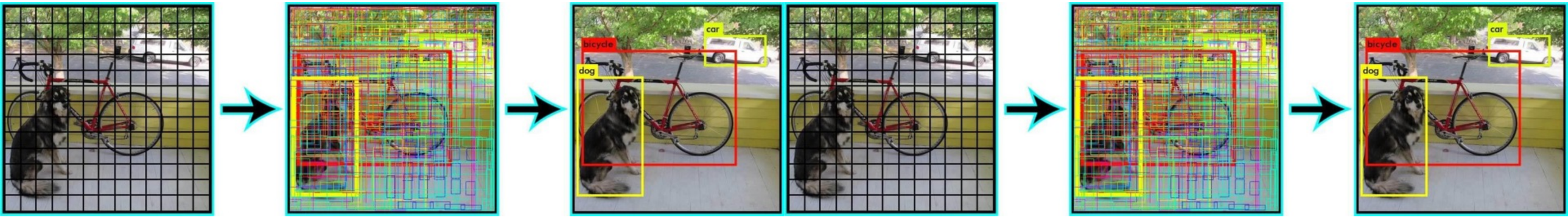


SxS Grid

Many bounding boxes with different object probabilities

Redmon et al. "You only look once: unified, real-time object detection (2015)."

YOLO



SxS 网格

Redmon 等，“You only look once：统一的实时目标检测（2015）。”

YOLO

SxS Grid

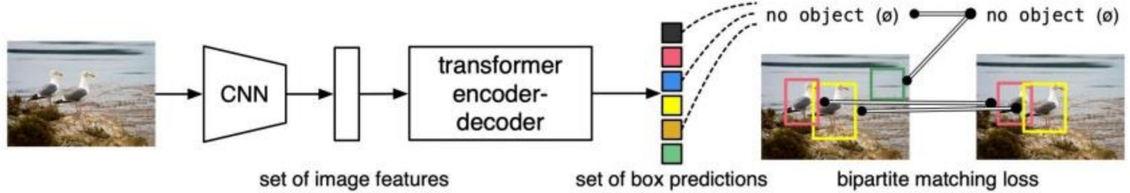
Redmon et al. "You only look once: unified, real-time object detection (2015)."

基于 Transformer 的目标检测：DETR

简单的目标检测流程：直接从 Transformer 输出一组边界框

无锚框，无边界框变换回归

通过二分匹配将预测边界框与 GT 边界框匹配；训练以回归边界框坐标



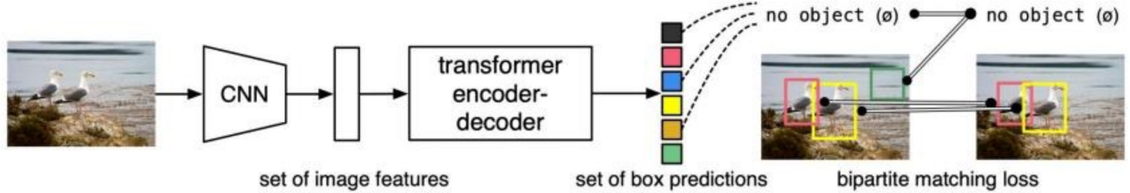
Carion 等，“基于 Transformer 的端到端目标检测”，ECCV 2020

Object Detection with Transformers: DETR

Simple object detection pipeline: directly output a set of boxes from a Transformer

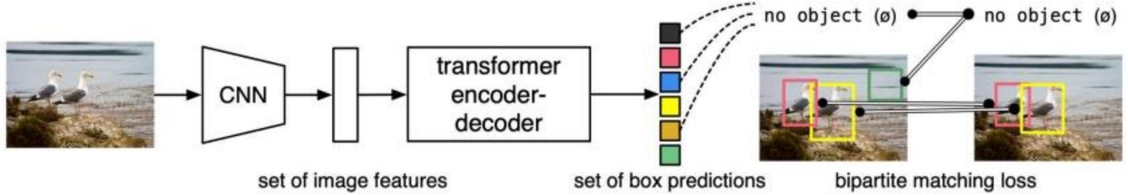
No anchors, no regression of box transforms

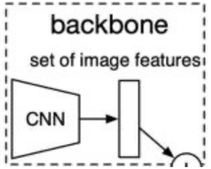
Match predicted boxes to GT boxes with bipartite matching; train to regress box coordinates



Carion et al, “End-to-End Object Detection with Transformers”, ECCV 2020

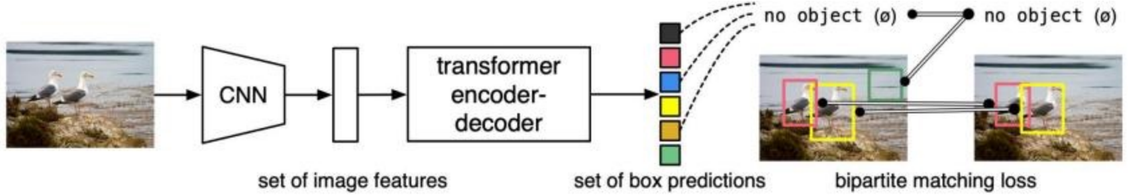
使用 Transformer 的目标检测：DETR

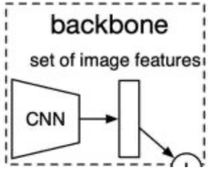




Carion 等人，“使用 Transformer 的端到端目标检测”，ECCV 2020

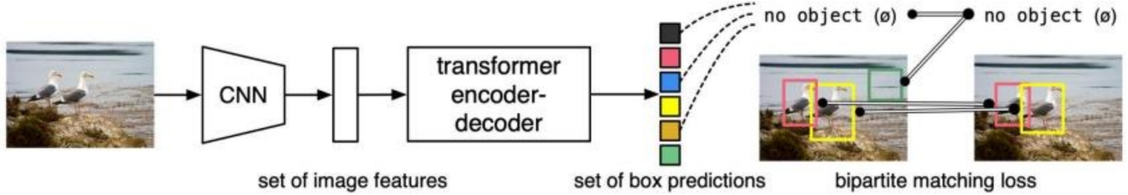
Object Detection with Transformers: DETR

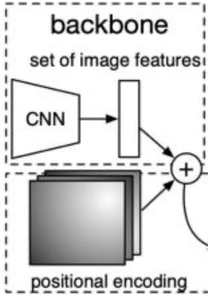




Carion et al, “End-to-End Object Detection with Transformers”，ECCV 2020

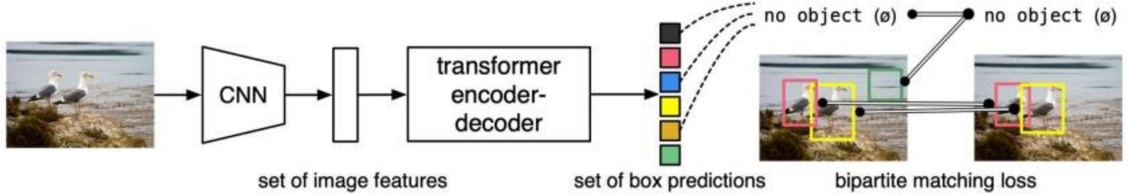
基于 Transformer 的目标检测：DETR

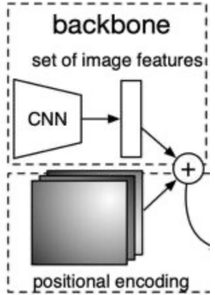




Carion 等，“基于 Transformer 的端到端目标检测”，ECCV 2020

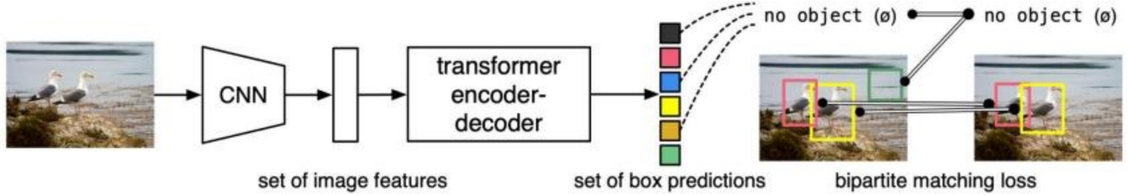
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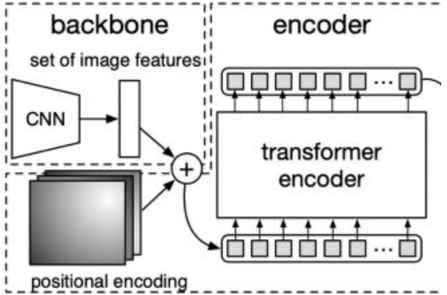




Carion et al，“End-to-End Object Detection with Transformers”，ECCV 2020

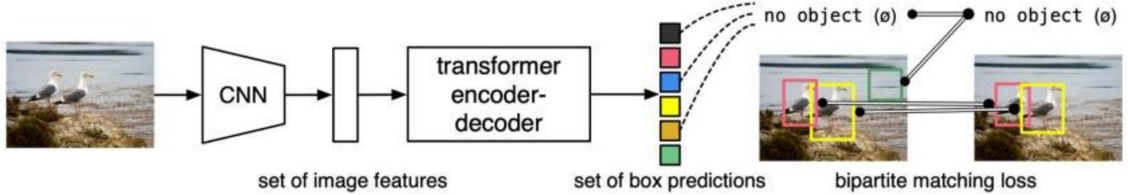
基于 Transformer 的目标检测：DETR

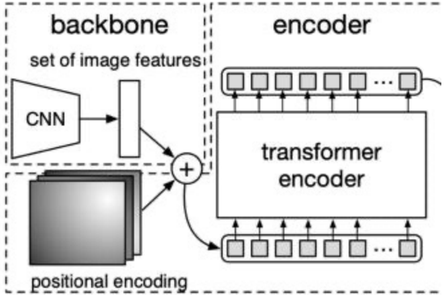




Carion 等，“基于 Transformer 的端到端目标检测”，ECCV 2020

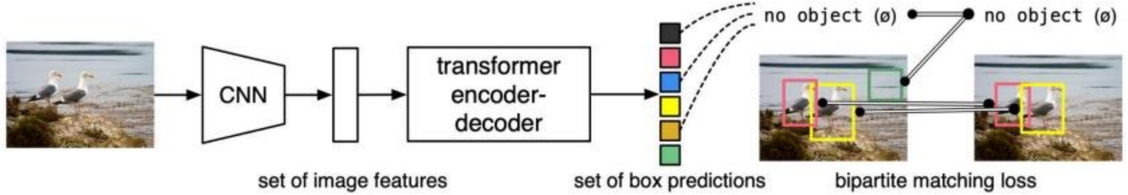
Object Detection with Transformers: DETR

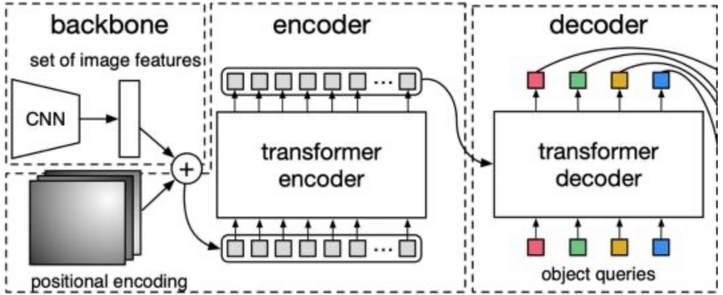




Carion et al，“End-to-End Object Detection with Transformers”，ECCV 2020

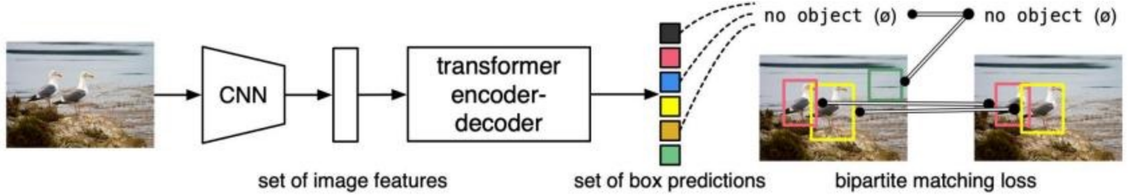
使用 Transformer 的目标检测：DETR

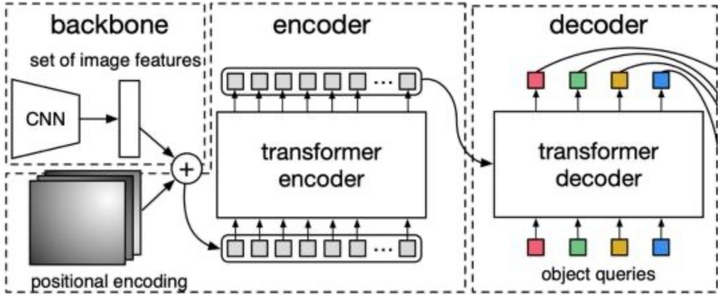




Carion 等，“基于 Transformer 的端到端目标检测”，ECCV 2020

Object Detection with Transformers: DETR

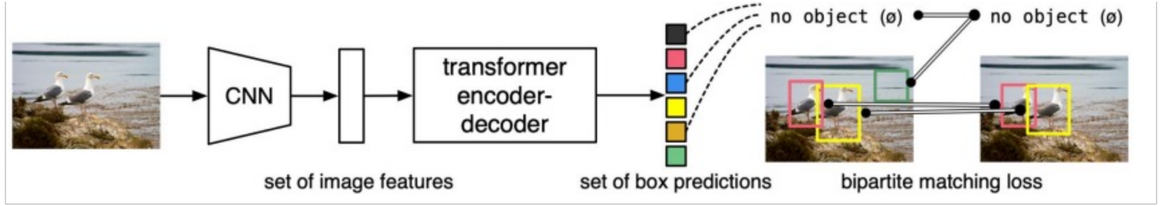


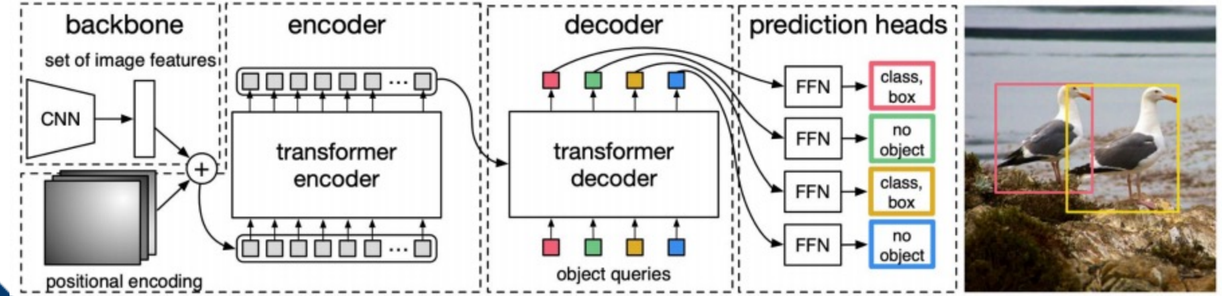


Carion et al，“End-to-End Object Detection with Transformers”，ECCV 2020

基于 Transformer 的目标检测：DETR

集合预测 □
匈牙利匹配

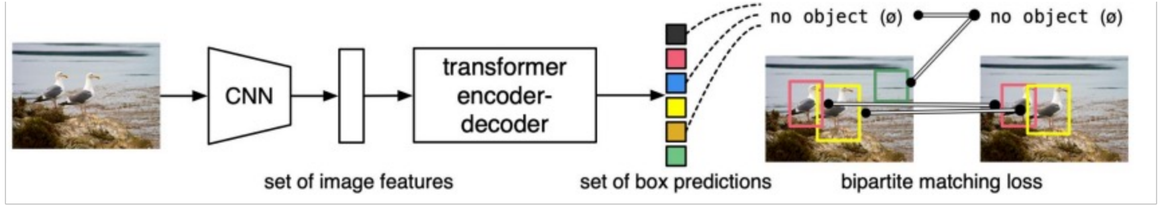


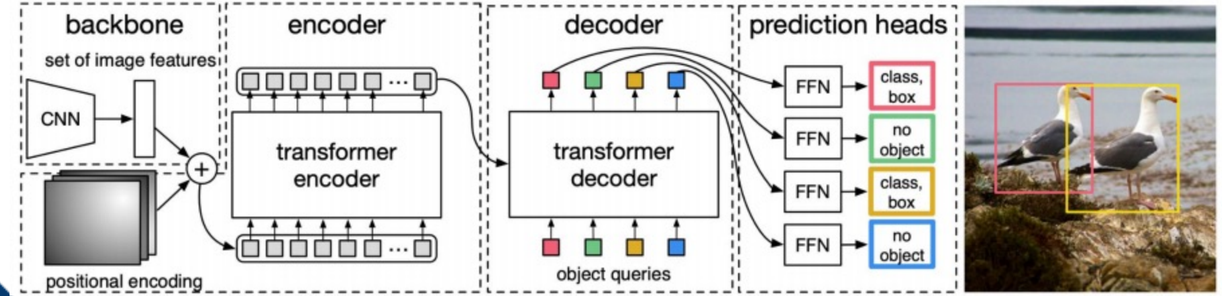


Carion 等，“基于 Transformer 的端到端目标检测”，ECCV 2020

Object Detection with Transformers: DETR

Set Prediction +
Hungarian Matching





Carion et al，“End-to-End Object Detection with Transformers”，ECCV 2020

为什么选择
DETR?

先前方法：
Ø 复杂设计（锚框、IoU 阈值、NMS） Ø 大量工程化
组件 Ø 重复预测 → 需要后处理 Ø 有限的全局推理
（大多为局部感受野）

DETR：

- 将检测重新表述为集合预测：CNN 骨干网络 → Transformer → 对象查询
- 关键特性：Ø 无锚框
  Ø 无 NMS Ø 端到端训
  练



Why DETR?

Previous Methods:

- Complex design (anchors, IoU thresholds, NMS)
- Heavily engineered components
- Duplicate predictions → post-processing needed
- Limited global reasoning (mostly local receptive fields)

DETR:

- Reformulates detection as set prediction: CNN backbone → Transformer → Object Queries
- Key features:
  - No anchors
  - No NMS
  - End-to-end training



今日

- **计算机视觉任务**
  - 语义分割 □ 目标检测 □ **实例分割**
- 可视化与理解
  - 模型层可视化 □ 显著性图 □ CAM & Grad-CAM
- 基础模型与提示分割

Today

- **Computer Vision Tasks**
  - Semantic Segmentation
  - Object Detection
  - **Instance Segmentation**
- Visualization & Understanding
  - Model Layers Visualization
  - Saliency Maps
  - CAM & Grad-CAM
- Foundation Models and Prompt Segmentation

实例分割

CAT

无空间范围

草地，CAT，树，SK

没有对象，只有像素

分类

语义分割

DOG, DOG, CAT



DOG, DOG, CAT

多对象

对象检测

实例分割

Instance Segmentation

CAT

No spatial extent

GRASS, CAT, TREE, SK

No objects, just pixels

Classification

Semantic Segmentation

DOG, DOG, CAT



DOG, DOG, CAT

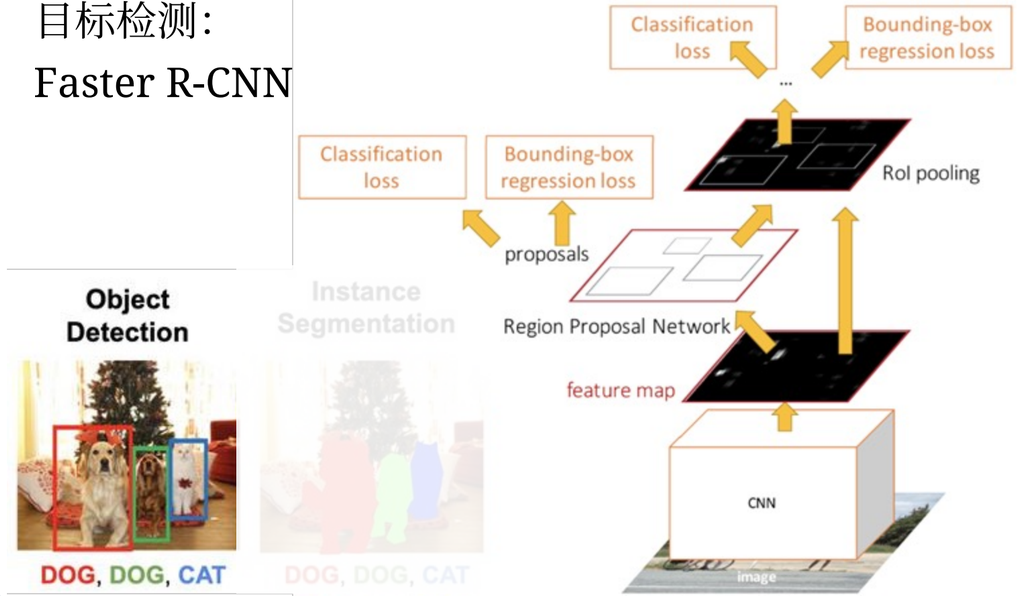
Multiple Object

Object Detection

Instance Segmentation

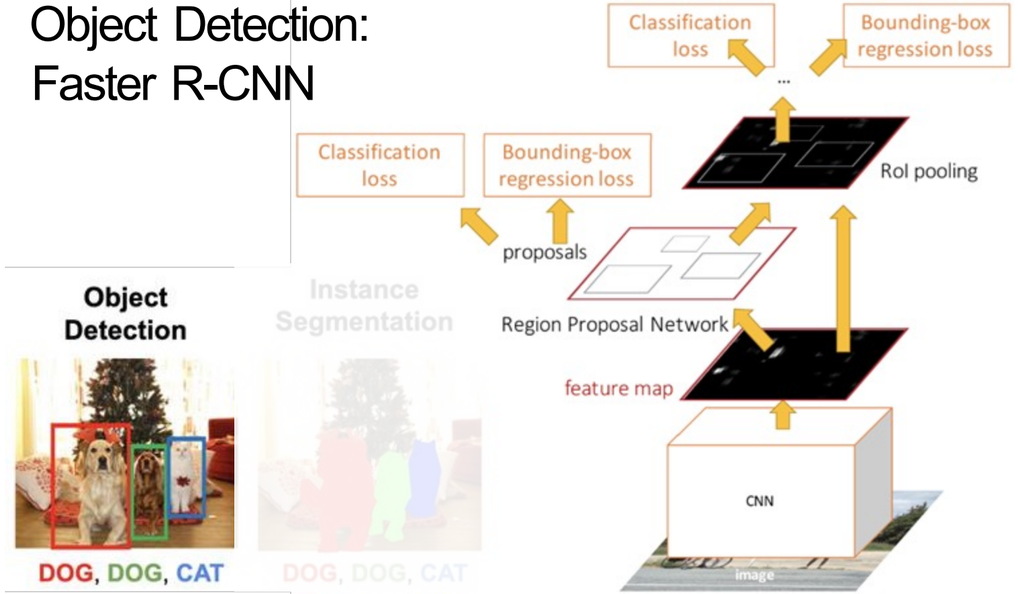
目标检测：

Faster R-CNN

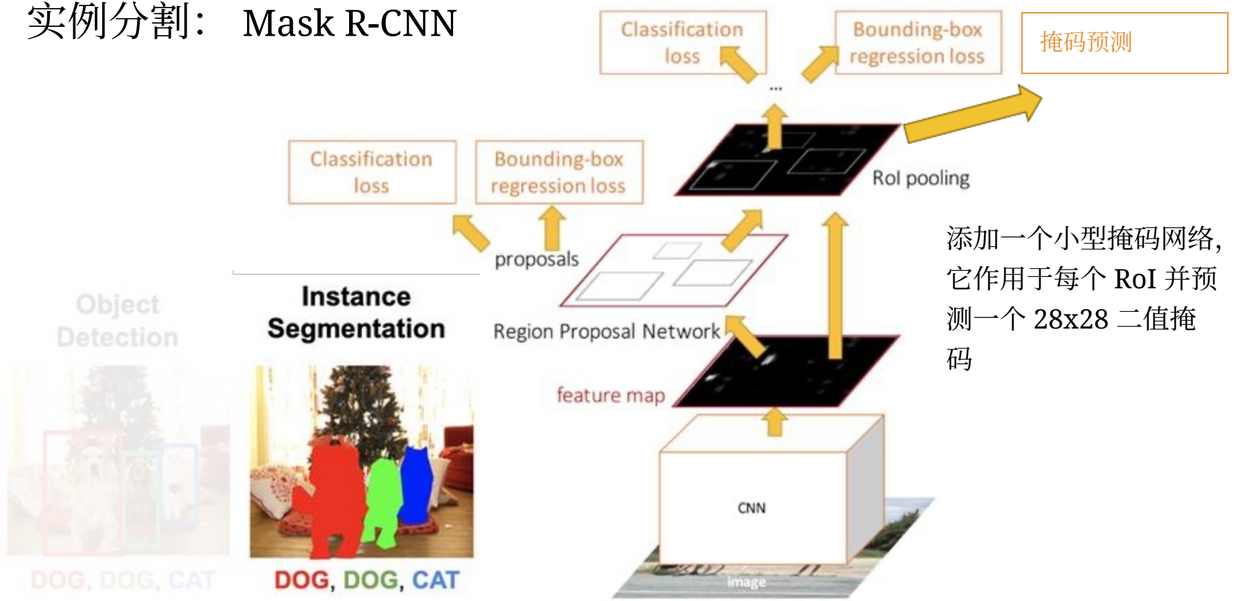


Object Detection:

Faster R-CNN



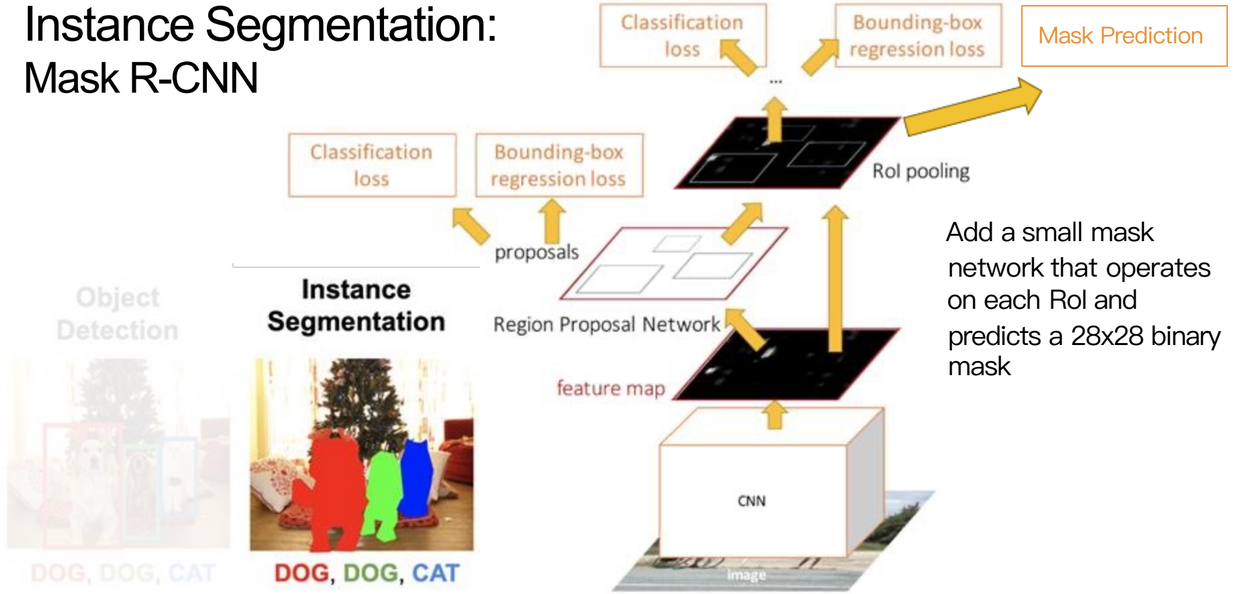
实例分割： Mask R-CNN



添加一个小型掩码网络，它作用于每个 RoI 并预测一个 28x28 二值掩码

He 等人，“Mask R-CNN”，ICCV 2017

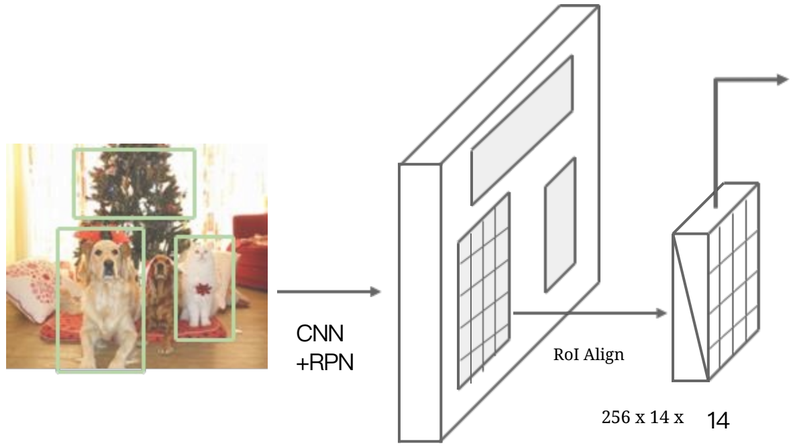
Instance Segmentation: Mask R-CNN



Add a small mask network that operates on each RoI and predicts a 28x28 binary mask

He et al，“Mask R-CNN”，ICCV 2017

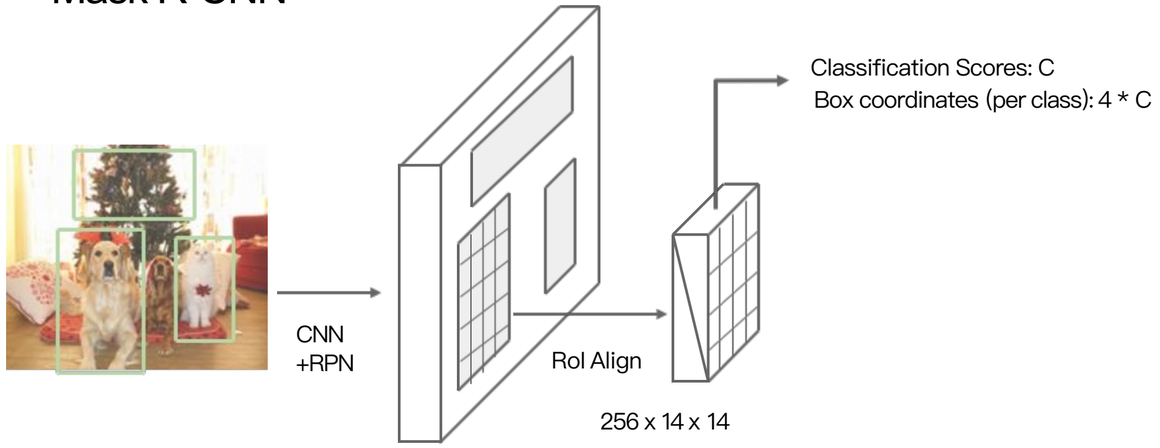
Mask R-CNN



分类得分：C 边界框坐标（每类）：
4 * C

He 等人，“Mask R-CNN”，ICCV 2017

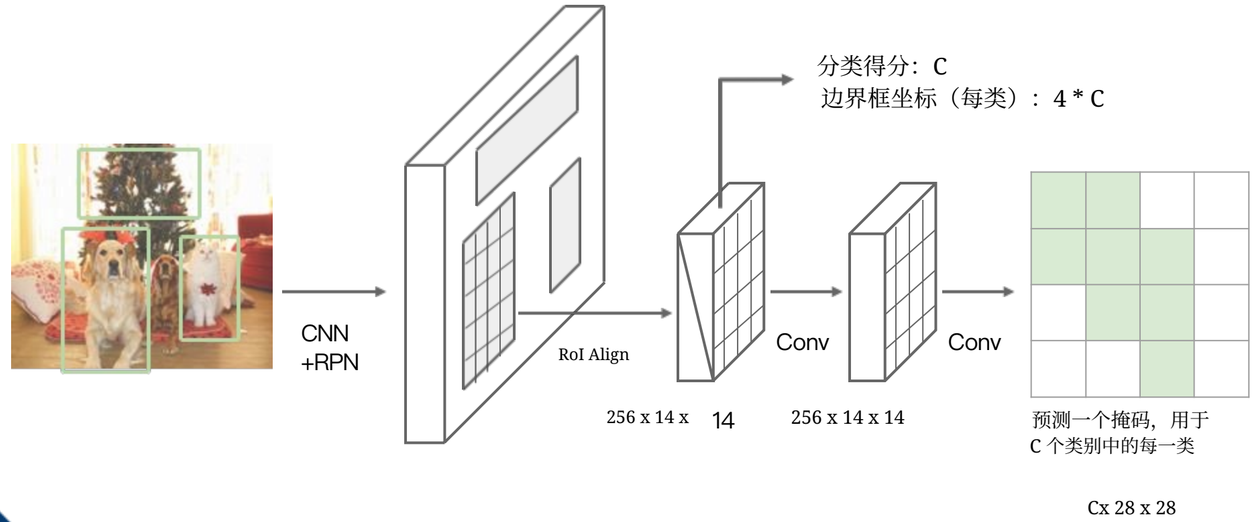
Mask R-CNN



Classification Scores: C
Box coordinates (per class): 4 * C

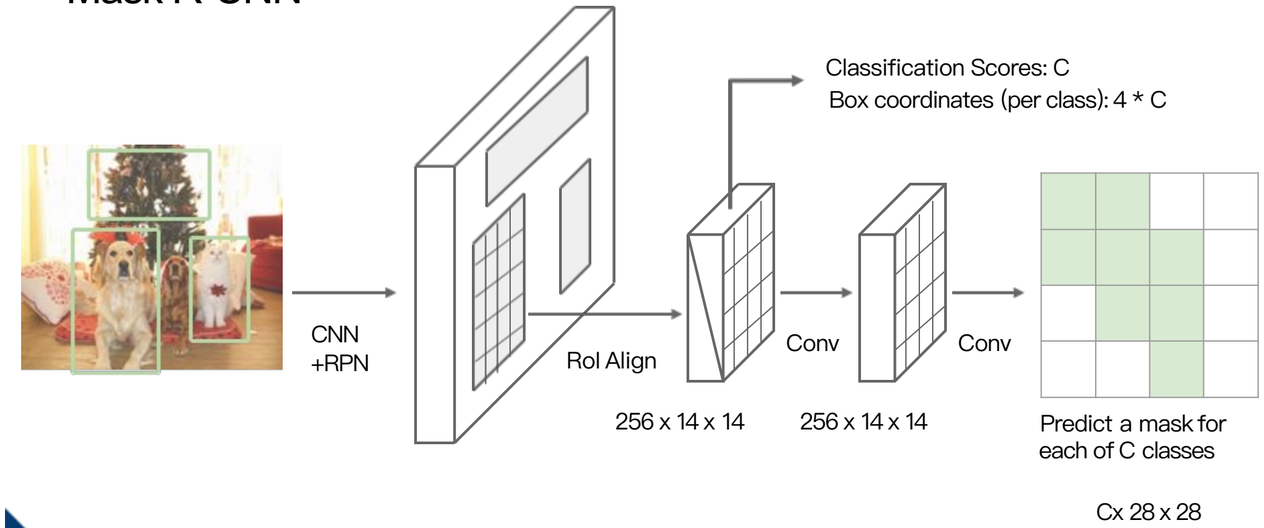
He et al，“Mask R-CNN”，ICCV 2017

Mask R-CNN



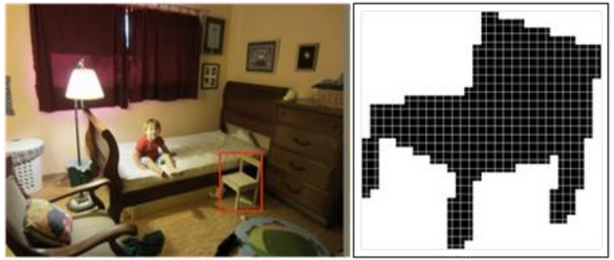
He 等人，“Mask R-CNN”，ICCV 2017

Mask R-CNN

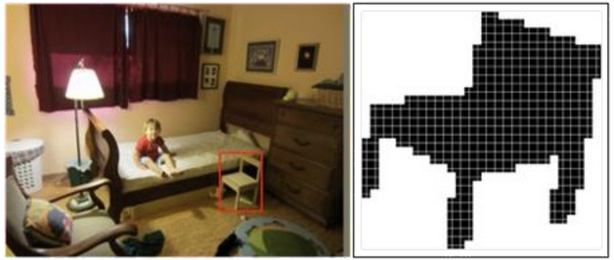


He et al，“Mask R-CNN”，ICCV 2017

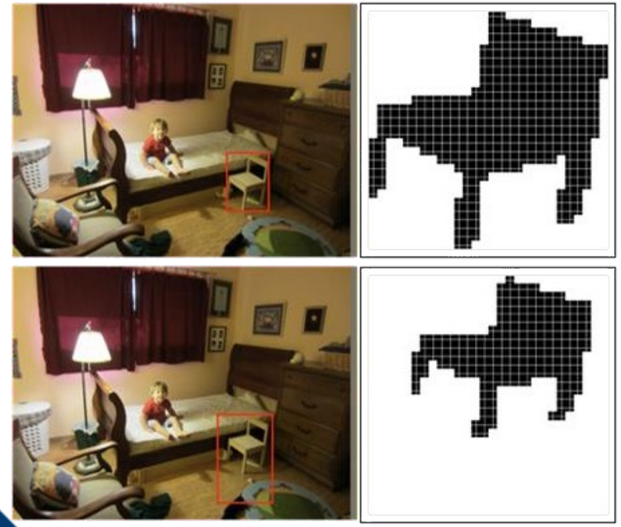
Mask R-CNN：掩码训练目标示例



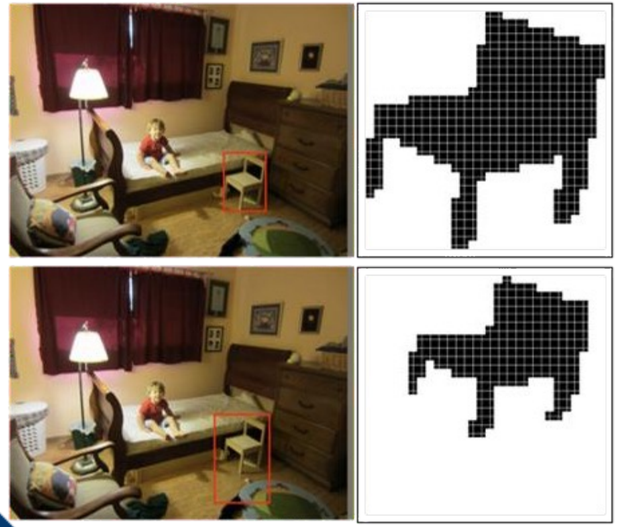
Mask R-CNN: Example Mask Training Targets



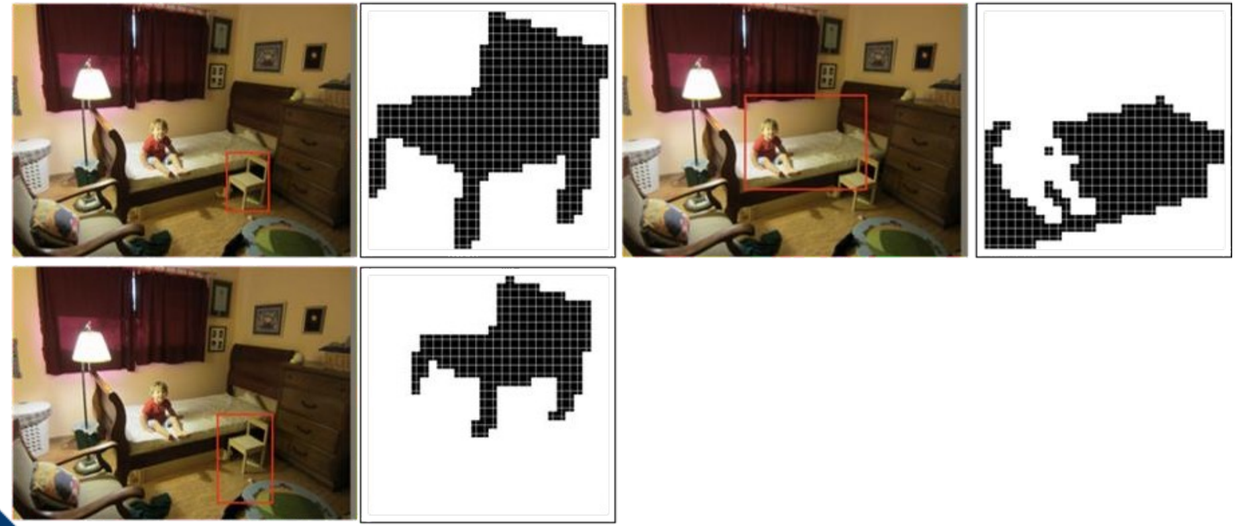
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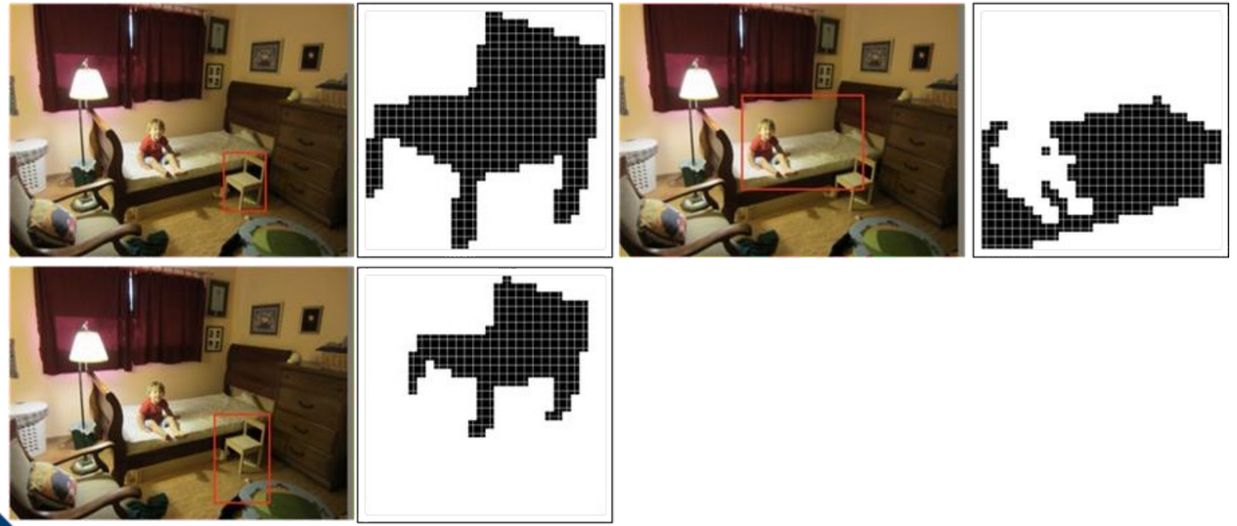
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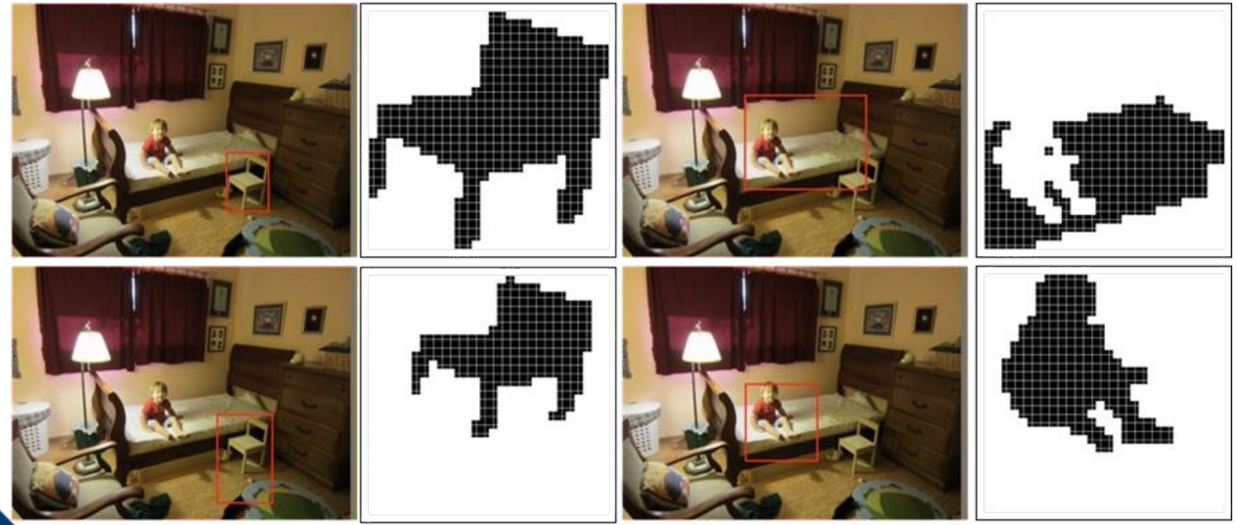
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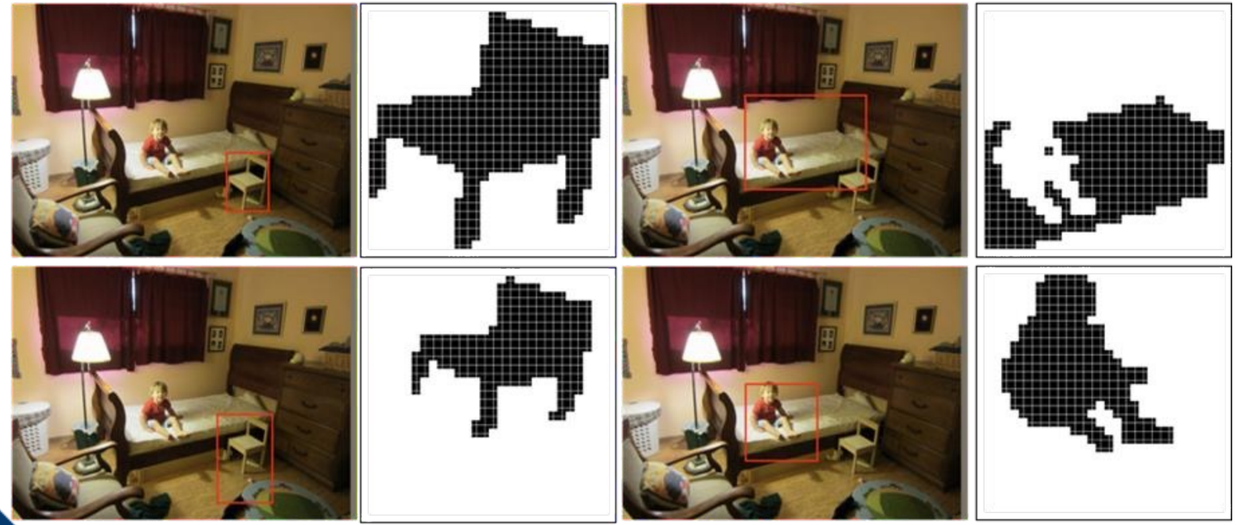
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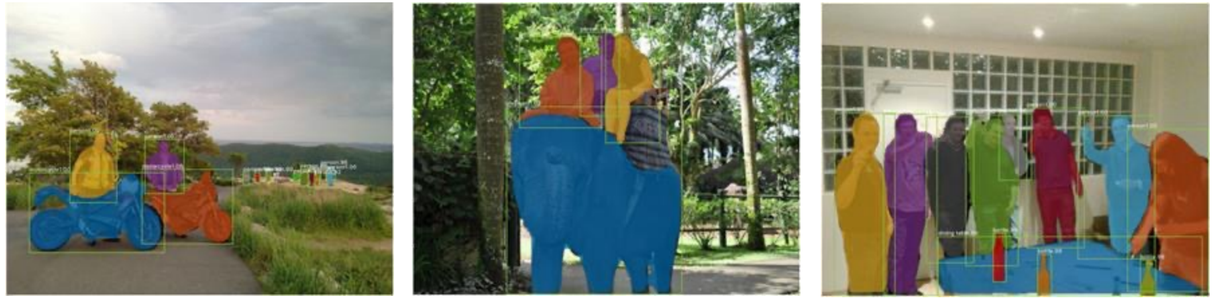
Mask R-CNN：掩码训练目标示例



Mask R-CNN: Example Mask Training Targets



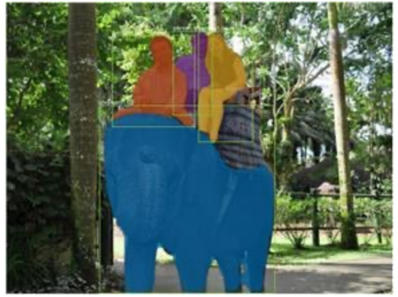
Mask R-CNN：非常好的结果！



He et al, “Mask R-CNN”, ICCV 2017

Mask R-CNN: Very Good Results!







He et al, “Mask R-CNN”, ICCV 2017

Rk□y b-Ca a还可进行姿态估计



He 等人，“Mask R-CNN”，ICCV 2017

Mask R-CNN

Also does pose



He et al，“Mask R-CNN”，ICCV 2017

回顾：许多计算机视觉任务！

分类



CAT

无空间范围

语义分割



草地、猫、树，
S□□

没有物体，只有像素

目标检测



DOG, DOG, CAT

实例分割



狗、狗、猫

多个物体

此图像 是CC0 公有领域

Recap: Lots of computer vision tasks!

Classification



CAT

No spatial extent

Semantic Segmentation



GRASS, CAT, TREE, SKY

No objects, just pixels

Object Detection



DOG, DOG, CAT

Instance Segmentation



DOG, DOG, CAT

Multiple Object

This image is CC0 public domain

今天

- 计算机视觉任务
  - 语义分割
  - 目标检测
  - 实例分割

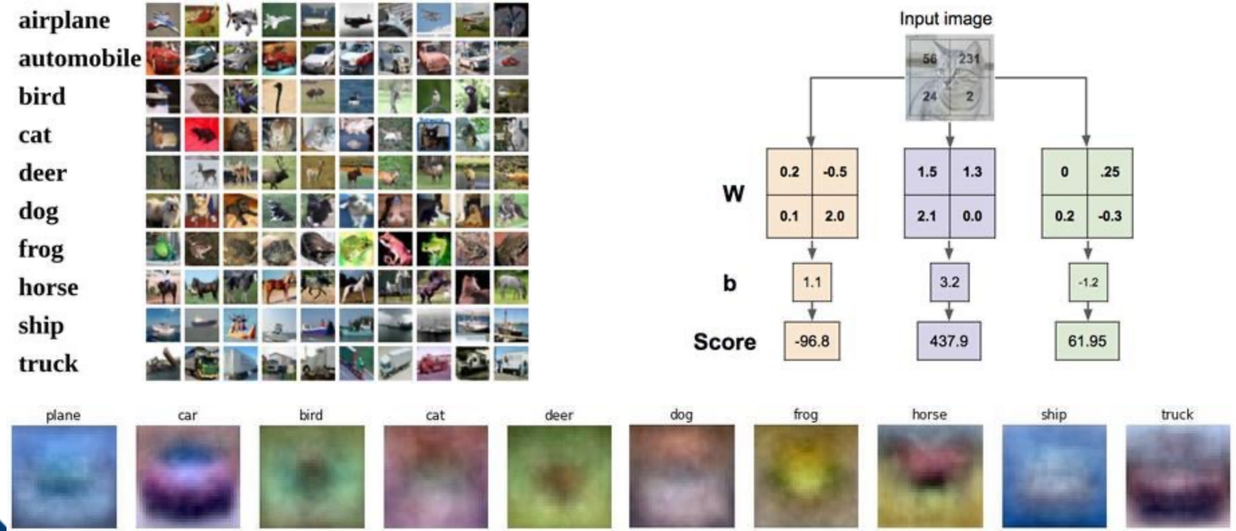
- **可视化与理解**
  - **模型层可视化**
  - 显著性图
  - CAM 与 Grad-CAM

Today

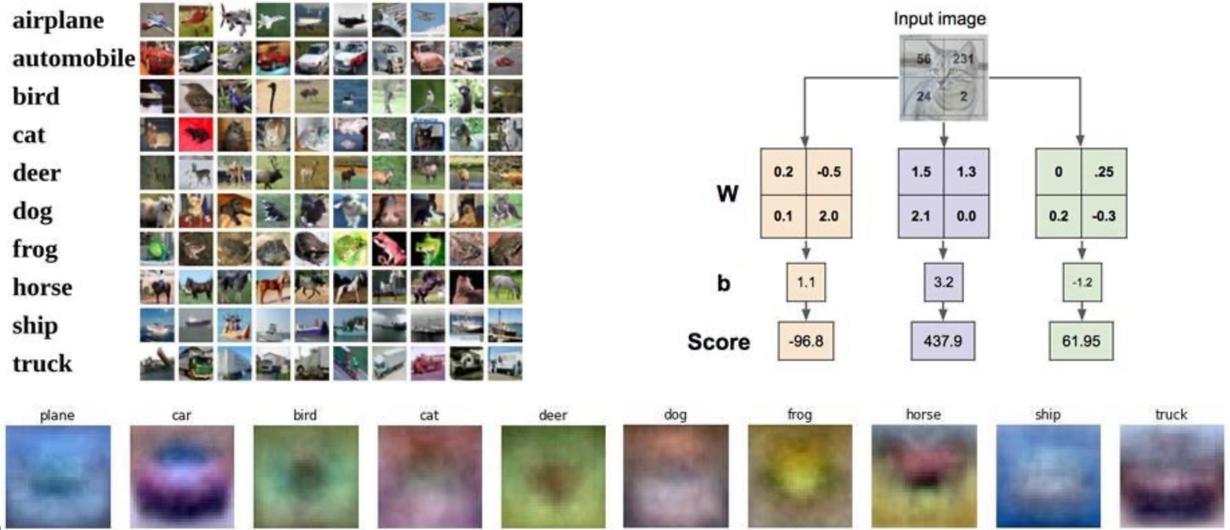
- Computer Vision Tasks
  - Semantic Segmentation
  - Object Detection
  - Instance Segmentation

- **Visualization & Understanding**
  - **Model Layers Visualization**
  - Saliency Maps
  - CAM & Grad-CAM

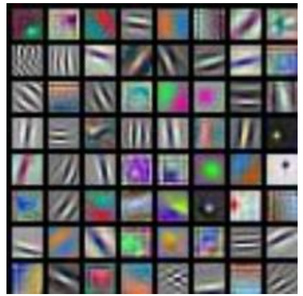
解读线性分类器：视觉视角



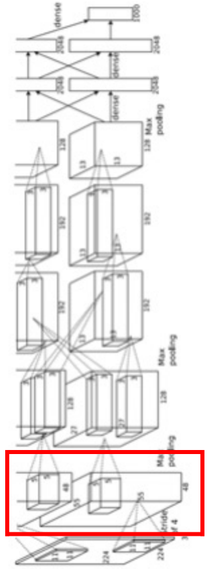
Interpreting a Linear Classifier: Visual Viewpoint



第一层：可视化滤波器

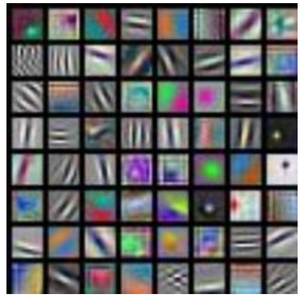


AlexNet:
64 x 3 x 11 x 11

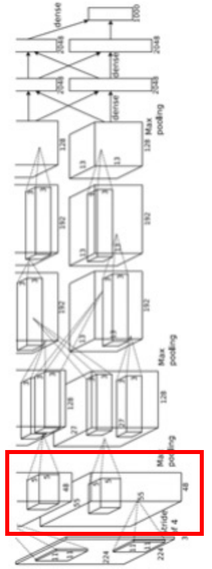


Krizhevsky，“并行化卷积神经网络的一种奇特技巧”，arXiv 2014 He 等，“用于图像识别的深度残差学习”，CVPR 2016 Huang 等，“密集连接卷积网络”，CVPR 2017

First Layer: Visualize Filters

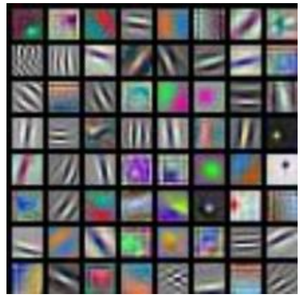


AlexNet:
64 x 3 x 11 x 11



Krizhevsky , “One weird trick for parallelizing convolutional neural networks”, arXiv 2014 He et al, “Deep Residual Learning for Image Recognition”, CVPR 2016
Huang et al, “Densely Connected Convolutional Networks”, CVPR 2017

第一层：可视化滤波器



AlexNet:
64 x 3 x 11 x 11



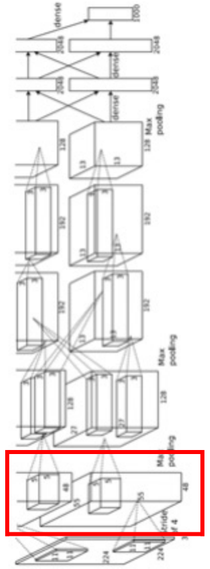
esNet15C:64
x 3 x 7 x 7



esNet1545:64
x 3 x 7 x 7

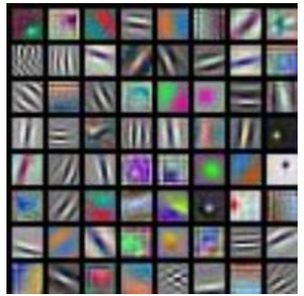


DenseNet-121:
64 x 3 x 7 x 7



Krizhevsky，“一种用于并行化卷积神经网络的奇特技巧”，arXiv 2014 He 等人，“用于图像识别的深度残差学习”，CVPR 2016 Huang 等人，“密集连接卷积网络”，CVPR 2017

First Layer: Visualize Filters



AlexNet:
64 x 3 x 11 x 11



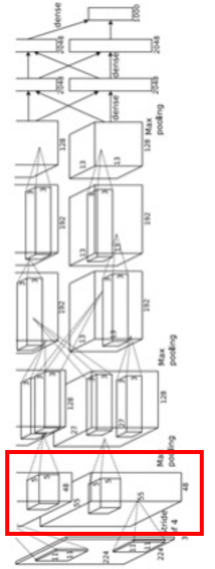
ResNet-18:
64 x 3 x 7 x 7



ResNet-101:
64 x 3 x 7 x 7



DenseNet-121:
64 x 3 x 7 x 7



Krizhevsky , “One weird trick for parallelizing convolutional neural networks”, arXiv 2014 He et al, “Deep Residual Learning for Image Recognition”, CVPR 2016
Huang et al, “Densely Connected Convolutional Networks”, CVPR 2017

今天

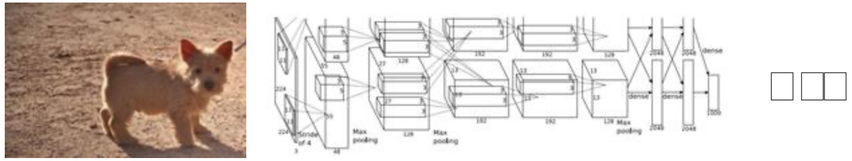
- 计算机视觉任务
  - 语义分割 - 目标检测 - 实例分割
- **可视化与理解**
  - 模型层可视化 - **显著性图** - CAM & Grad-CAM

Today

- Computer Vision Tasks
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- **Visualization & Understanding**
  - Model Layers Visualization
  - **Saliency Maps**
  - CAM & Grad-CAM

哪些像素重要：通过反向传播的显著性

前向传播：计算概率

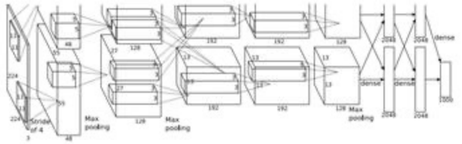


Simonyan, Vedaldi, and Zisserman, “Deep Inside Convolutional Networks: Visualising Image Classification Models and Saliency Maps”, ICLR Workshop 2014. 图片版权归 Karen Simonyan、Andrea Vedaldi 和 Andrew Zisserman 所有，2014；经许可转载。

Which pixels matter: Saliency via Backprop

Forward pass: Compute probabilities



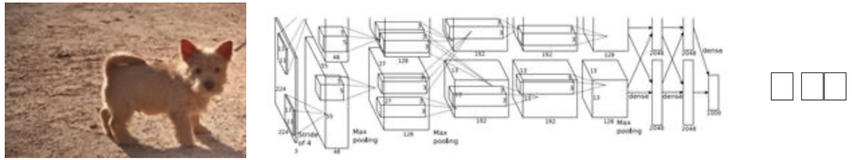


Dog

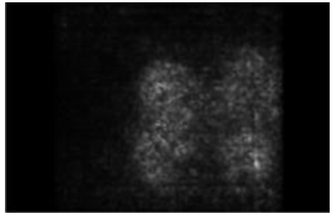
Simonyan, Vedaldi, and Zisserman, “Deep Inside Convolutional Networks: Visualising Image Classification Models and Saliency Maps”, ICLR Workshop 2014. Figures copyright Karen Simonyan, Andrea Vedaldi, and Andrew Zisserman, 2014; reproduced with permission.

哪些像素重要：通过反向传播的显著性

前向传播：计算概率



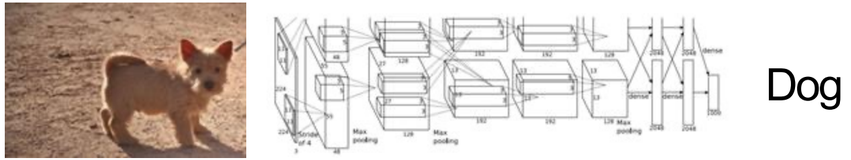
计算（未归一化的）类别分数相对于图像像素的梯度，取绝对值并在 RGB 通道上取最大值



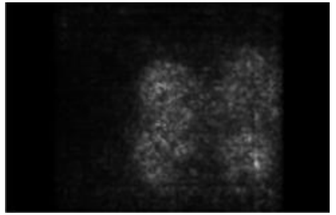
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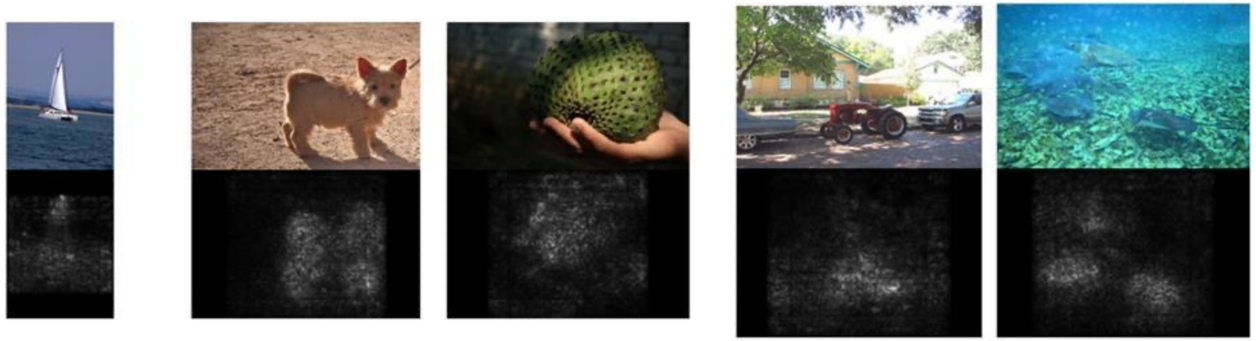


Compute gradient of (unnormalized) class score with respect to image pixels, take absolute value and max over RGB channels



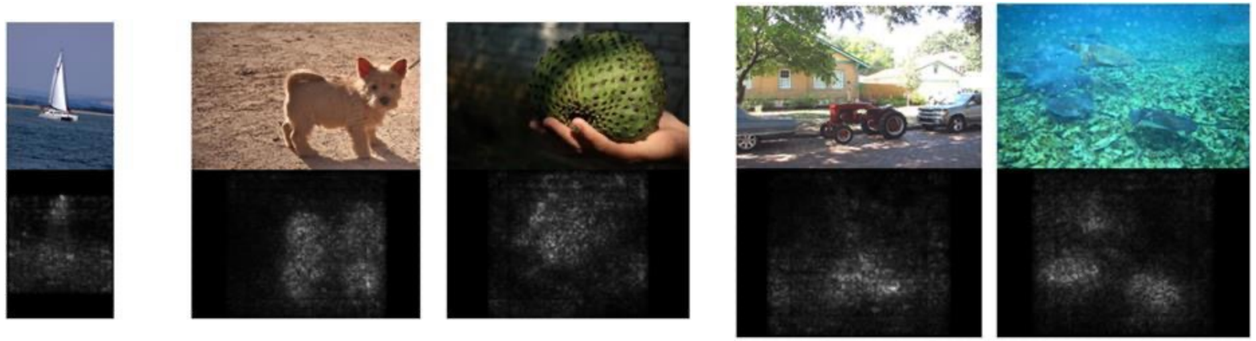
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显著性图



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Saliency Maps



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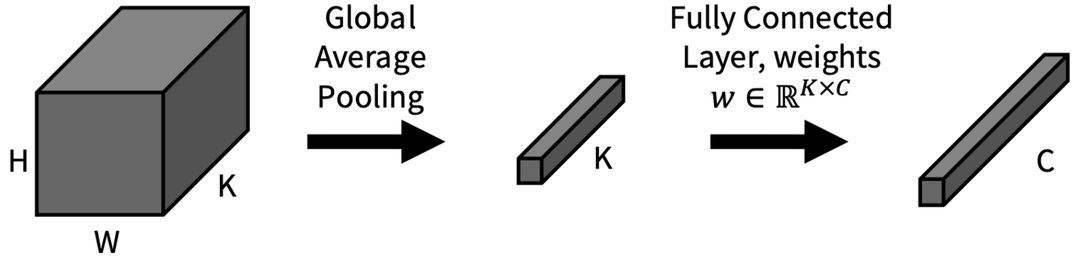
今天

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- 基础模型与提示分割

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- Foundation Models and Prompt Segmentation

Class Activation Mapping (CAM)



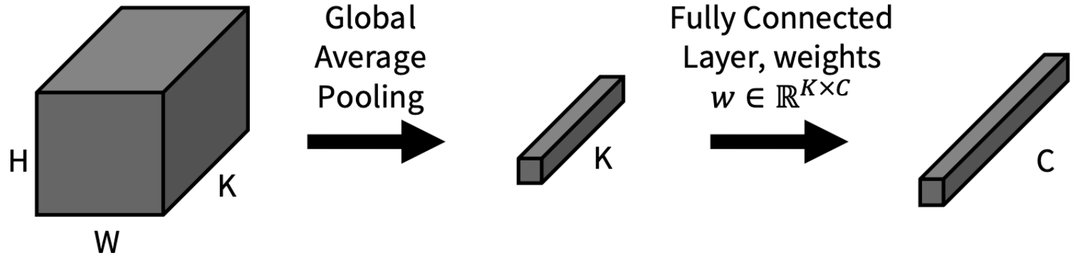
Last layer CNN features:
$f \in \mathbb{R}^{H \times W \times K}$

Pooled features:
$F \in \mathbb{R}^{K}$

Class Scores:
$S \in \mathbb{R}^{C}$

Zhou 等人，“用于判别性定位的深度特征学习”，CVPR 2016

Class Activation Mapping (CAM)



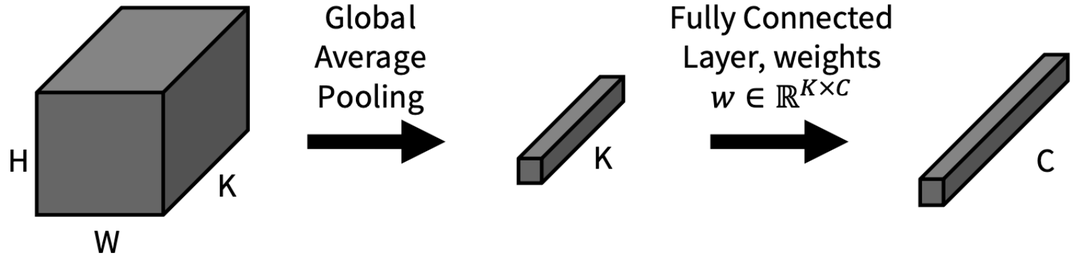
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Zhou et al，“Learning Deep Features for Discriminative Localization”，CVPR 2016

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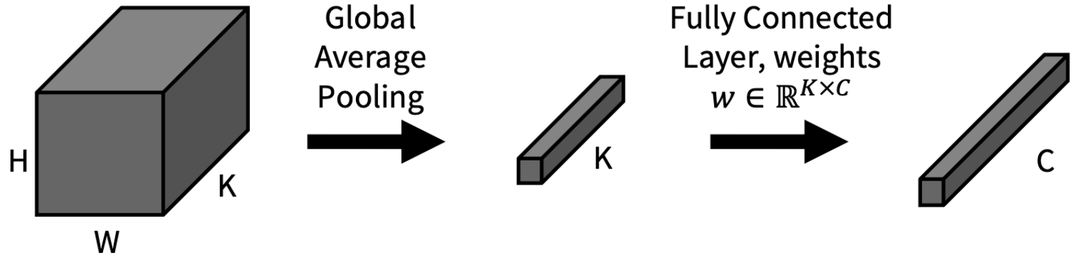
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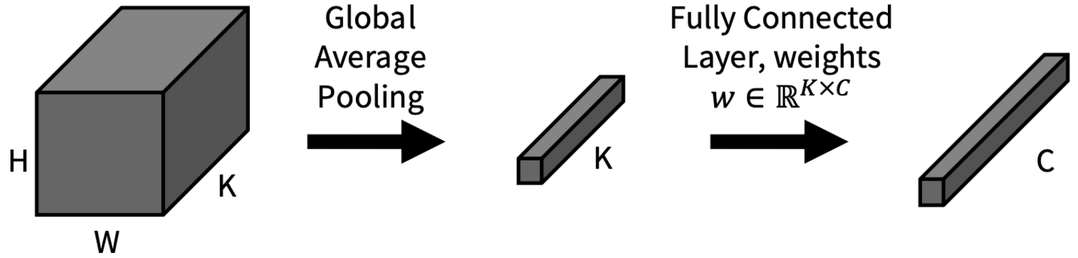
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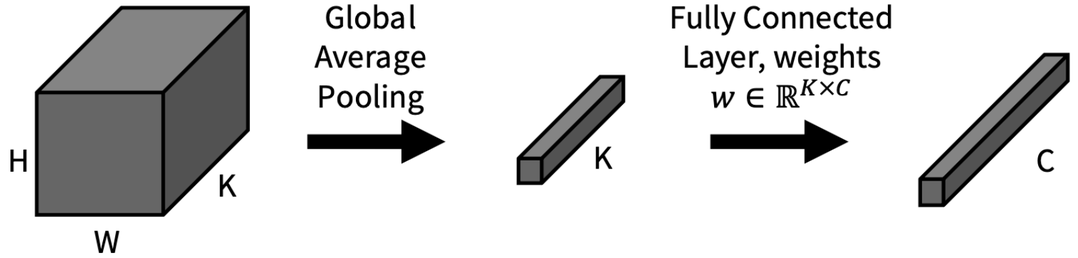
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Zhou et al,"Learning Deep Features for Discriminative Localization",CVPR 2016

Class Activation Mapping (CAM)



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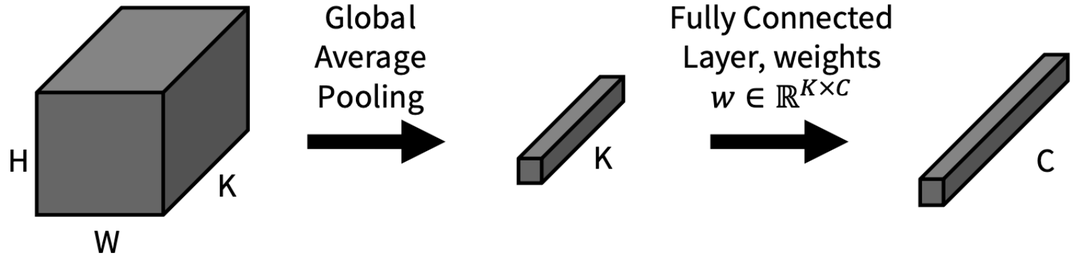
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Zhou et al, "Learning Deep Features for Discriminative Localization", CVPR 2016

Class Activation Mapping (CAM)



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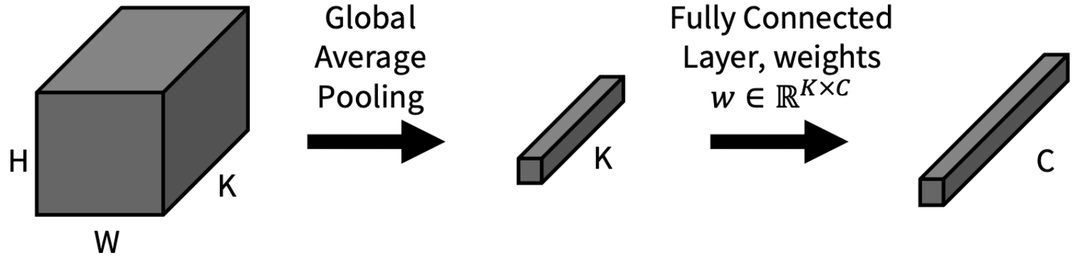
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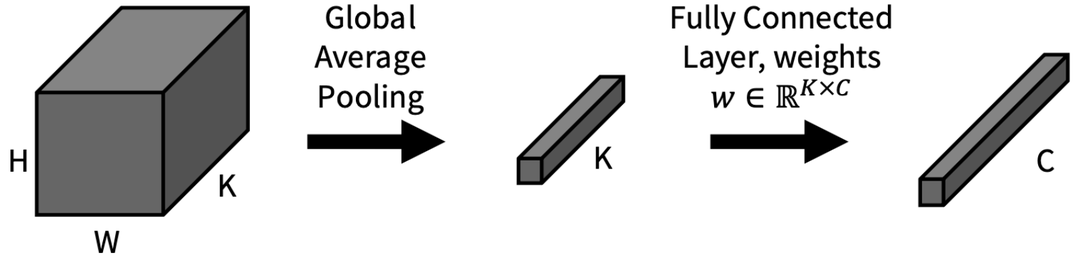
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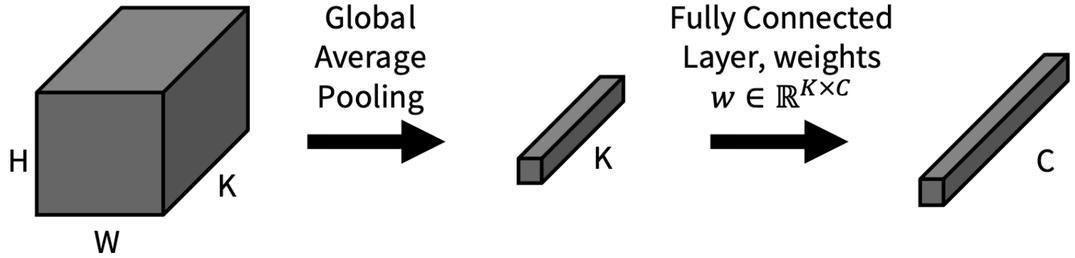
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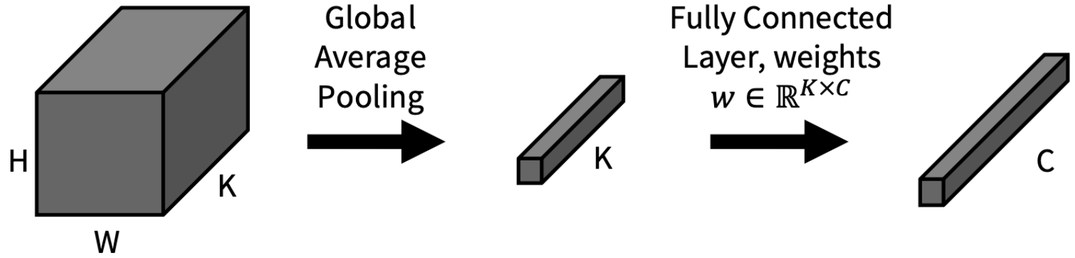
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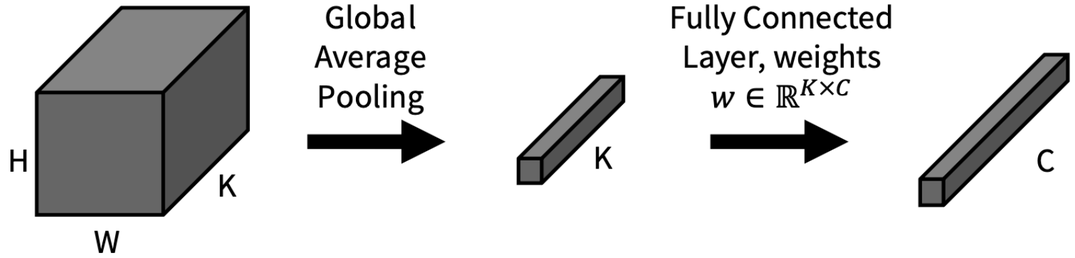
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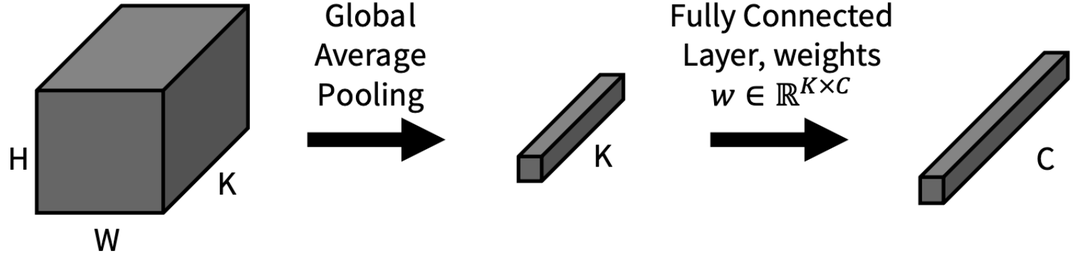
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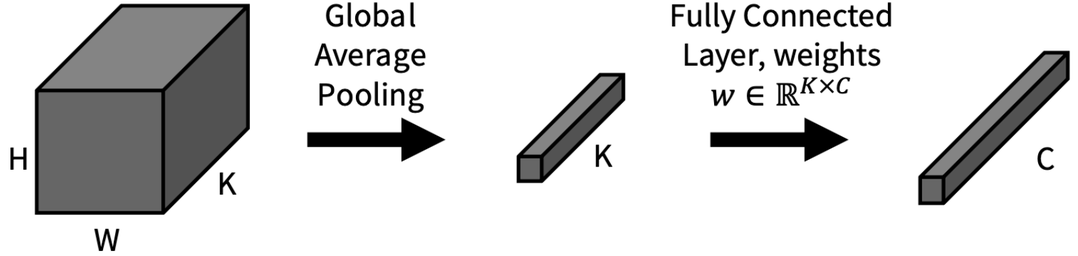
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Class Activation Maps:
$\boldsymbol{M} \in \mathbb{R}^{C,H,W}$

$$\boldsymbol{M_{c,h,w}} = \sum_k \boldsymbol{w_{k,c} f_{h,w,k}}$$

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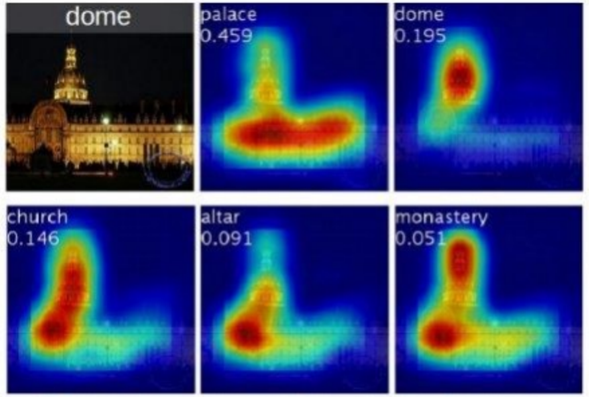
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类激活映射（CAM）



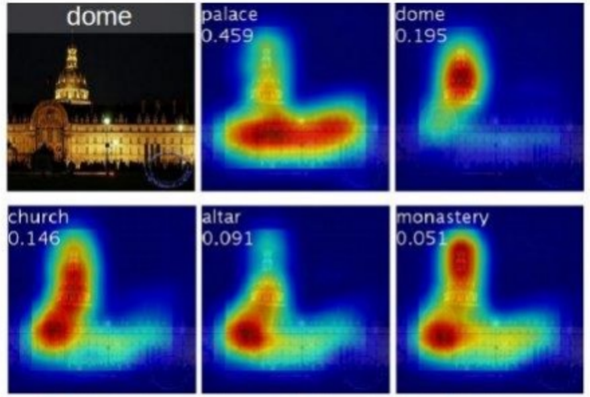
Class activation maps of top 5 predictions



Class activation maps for one object class

Zhou 等人，“Learning Deep Features for Discriminative Localization”，CVPR 2016

Class Activation Mapping (CAM)



Class activation maps of top 5 predictions

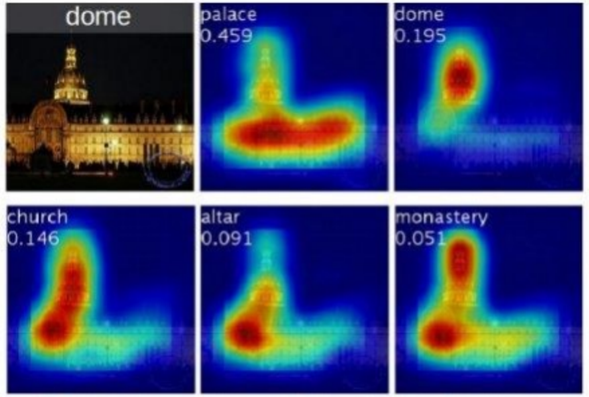


Class activation maps for one object class

Zhou et al，“Learning Deep Features for Discriminative Localization”，CVPR 2016

类激活映射 (CAM)

问题：只能应用于最后一个卷积层



Class activation maps of top 5 predictions

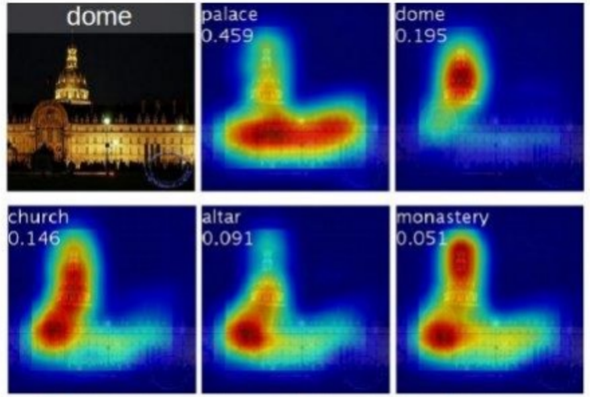


Class activation maps for one object class

Zhou 等，“用于判别性定位的深度特征学习”，CVPR 2016

Class Activation Mapping (CAM)

Problem: Can only apply to last conv layer



Class activation maps of top 5 predictions



Class activation maps for one object class

Zhou et al，“Learning Deep Features for Discriminative Localization”，CVPR 2016

梯度加权类激活映射（Grad-CAM）

1. 选择任意层，其激活为 A ∈ $\mathbb{R}^{□×□×□}$

Gradient-Weighted Class Activation Mapping (Grad-CAM)

1. Pick any layer, with activations $A \in \mathbb{R}^{H×w×k}$

梯度加权类激活映射（Grad-CAM）

1. 选择任意层，其激活为 A ∈ $\mathbb{R}^{H\times W\times K}$ 2. 计算类别分数 S 的梯度c关于A：

$$\frac{\partial S_c}{\partial A} \in \mathbb{R}^{H\times W\times K}$$

Gradient-Weighted Class Activation Mapping (Grad-CAM)

1. Pick any layer, with activations $A \in \mathbb{R}^{H\times W\times K}$
2. Compute gradient of class score S_c with respect to A:

$$\frac{\partial S_c}{\partial A} \in \mathbb{R}^{H\times W\times K}$$

梯度加权类激活映射（Grad-CAM）

1. 任选一层，其激活为 A ∈ $\mathbb{R}^{□×□×□}$
2. 计算类别得分 S_c 关于 A 的梯度：

$$\frac{\partial S_c}{\partial A} \in \mathbb{R}^{□×□×□}$$

3. 对梯度进行全局平均池化，以得到权重 □ ∈ R □

$$\alpha_k = \frac{1}{HW}\sum_{h,w}\frac{\partial S_c}{\partial A_{h,w,k}}$$

Selvaraju 等，“Grad-CAM：通过基于梯度的定位得到深度网络的视觉解释”，CVPR 2017

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2. Compute gradient of class score S_c with respect to A:

$$\frac{\partial S_c}{\partial A} \in \mathbb{R}^{H×W×k}$$

3. Global Average Pool the gradients to get weights α ∈ $\mathbb{R}^{k}$:

$$\alpha_k = \frac{1}{HW}\sum_{h,w}\frac{\partial S_c}{\partial A_{h,w,k}}$$

Selvaraju et al，“Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization”，CVPR 2017

梯度加权类激活映射（Grad-CAM）

1. 选择任意层，其激活为 A ∈ $\mathbb{R}^{H \times W \times k}$
2. 计算类别得分 S_c 关于 A 的梯度：

$$\frac{\partial S_c}{\partial A} \in \mathbb{R}^{H \times W \times k}$$

3. 对梯度进行全局平均池化以获得权重 $\alpha \in \mathbb{R}^k$：

$$\alpha_k = \frac{1}{HW} \sum_{h,w} \frac{\partial S_c}{\partial A_{h,w,k}}$$

4. 计算激活图 $M^c \in \mathbb{R}^{H,W}$：

$$M^c_{h,w} = ReLU\left(\sum_k \alpha_k A_{h,w,k}\right)$$



Gradient-Weighted Class Activation Mapping (Grad-CAM)

1. Pick any layer, with activations A ∈ $\mathbb{R}^{H \times W \times k}$
2. Compute gradient of class score S_c with respect to A:

$$\frac{\partial S_c}{\partial A} \in \mathbb{R}^{H \times W \times k}$$

3. Global Average Pool the gradients to get weights α ∈ $\mathbb{R}^k$:

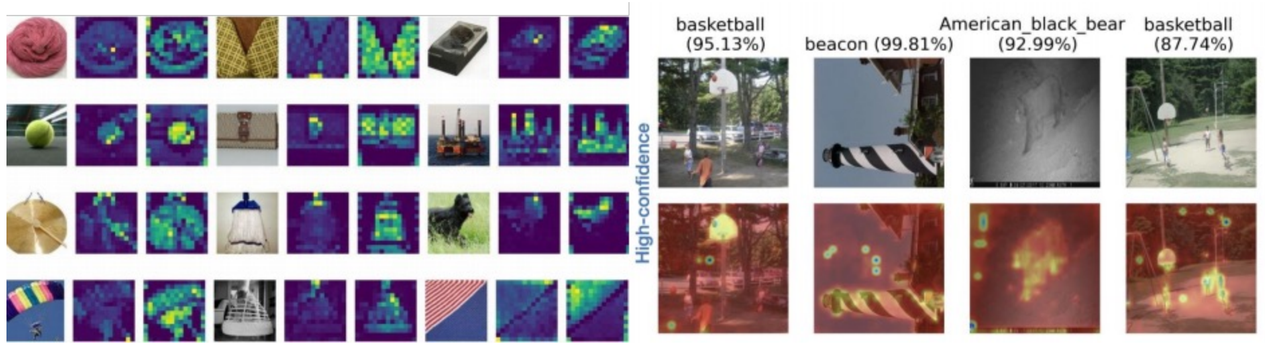
$$\alpha_k = \frac{1}{HW} \sum_{h,w} \frac{\partial S_c}{\partial A_{h,w,k}}$$

4. Compute activation map $M^c \in \mathbb{R}^{H,W}$:

$$M^c_{h,w} = ReLU\left(\sum_k \alpha_k A_{h,w,k}\right)$$

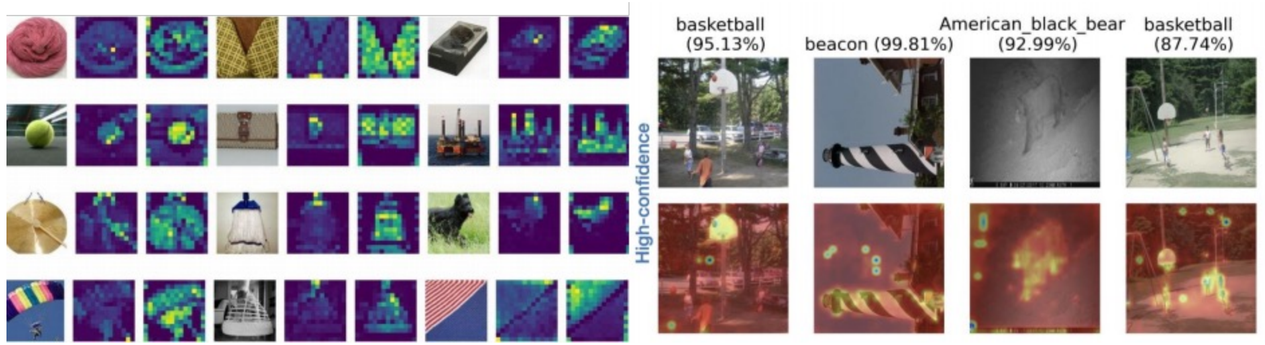


可视化 ViT 特征



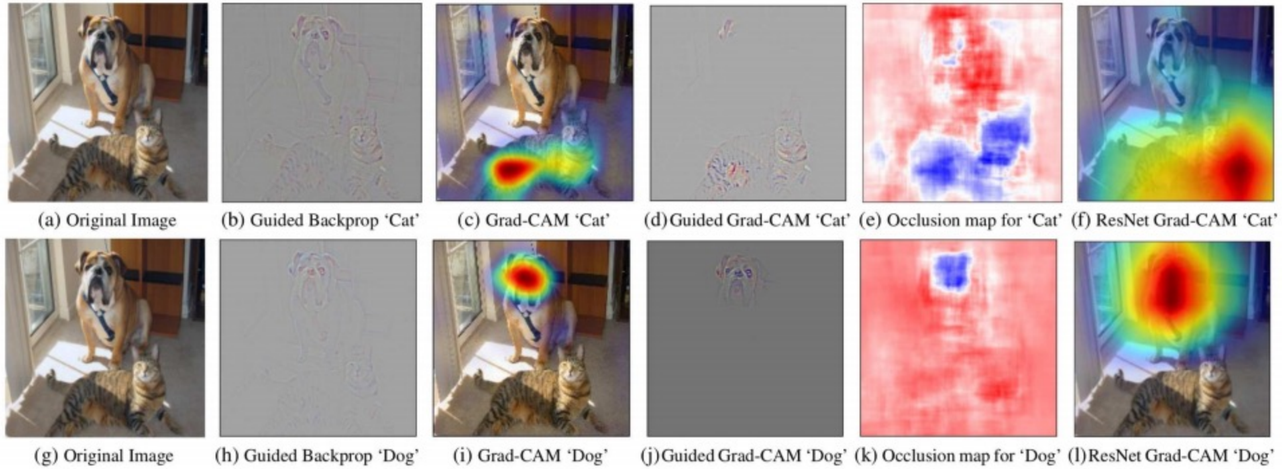
Chen 等人，When Vision Transformers OutperformResnets Without Pre-training Or Strong Data Augmentations, ICLR 2022；Paul 和 Chen，Vision Transformers are Robust Learners，AAAI 2022。为教育目的复现。

Visualizing ViT features

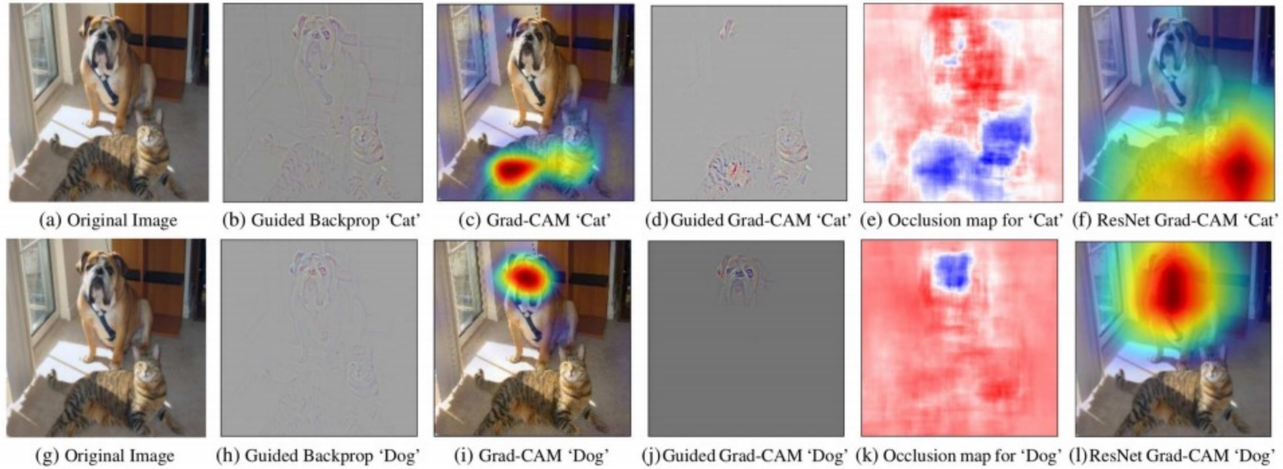


Chen et al., When Vision Transformers OutperformResnets Without Pre-training Or Strong Data Augmentations, ICLR 2022; Paul and Chen, Vision Transformers are Robust Learners, AAAI 2022. Reproduced for educational purposes.

梯度加权类激活映射（Grad-CAM）



Gradient-Weighted Class Activation Mapping (Grad-CAM)



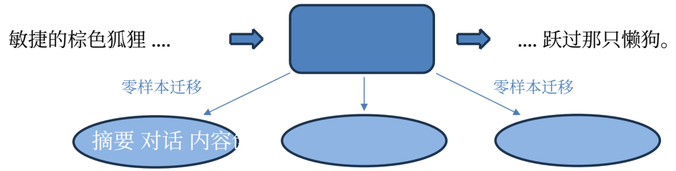
今天

- 计算机视觉任务
  - 语义分割 目标检测 实例分割
- 可视化与理解
  - 模型层可视化 显著性图 CAM 与 Grad-CAM
- **基础模型与提示分割**

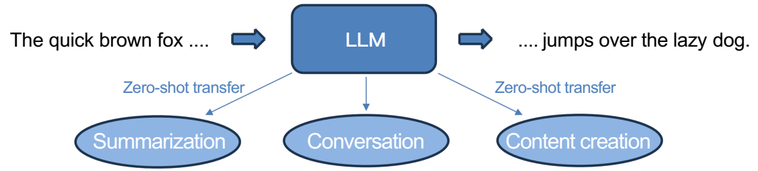
Today

- Computer Vision Tasks
  - Semantic Segmentation
  - Object Detection
  - Instance Segmentation
- Visualization & Understanding
  - Model Layers Visualization
  - Saliency Maps
  - CAM & Grad-CAM
- **Foundation Models and Prompt Segmentation**

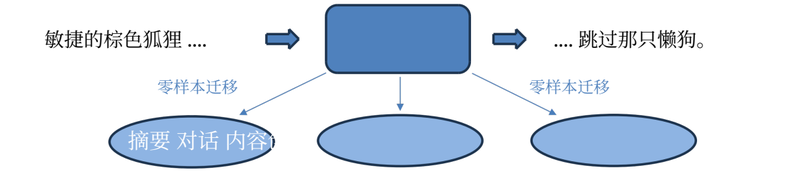
NLP中的零样本迁移提示

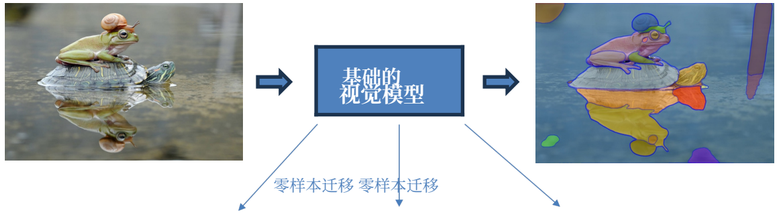


Prompting for Zero-Shot Transfer in NLP

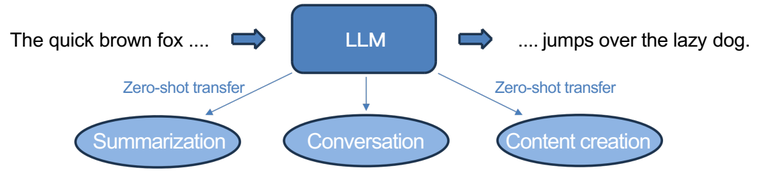


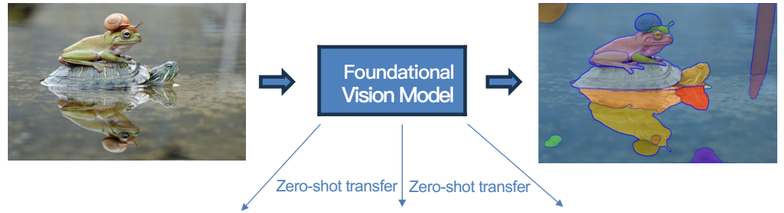
基础视觉模型



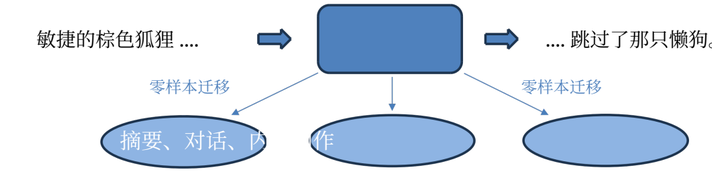


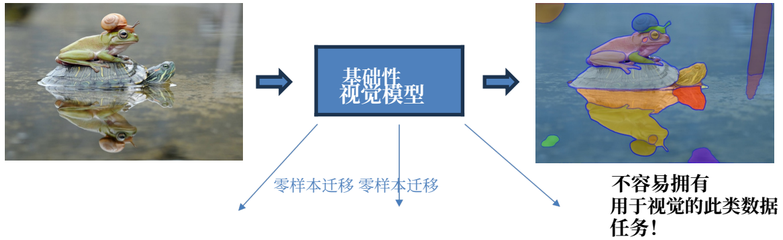
Foundational Vision Model



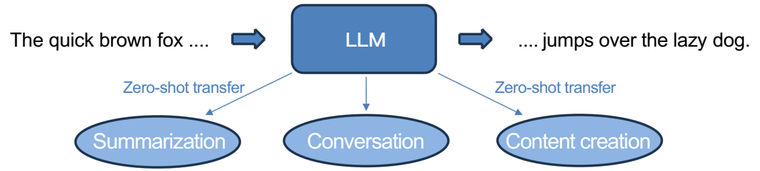


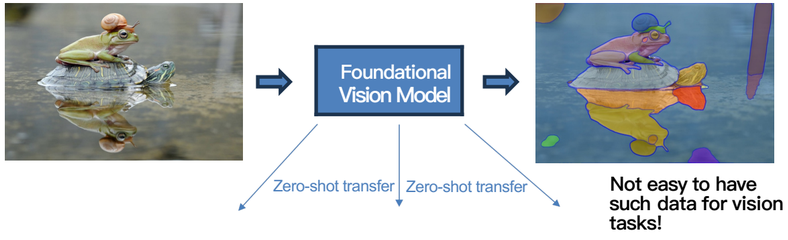
基础视觉模型：数据匮乏



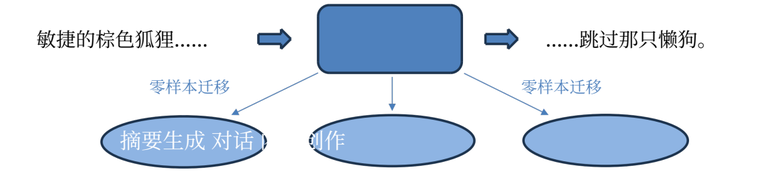


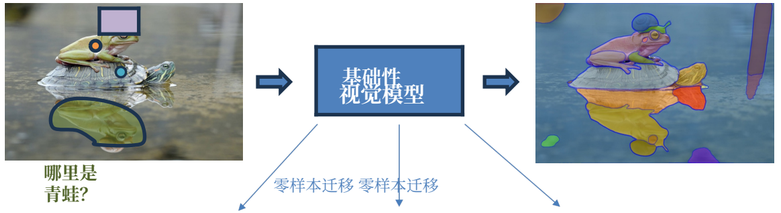
Foundational Vision Model: Data Paucity



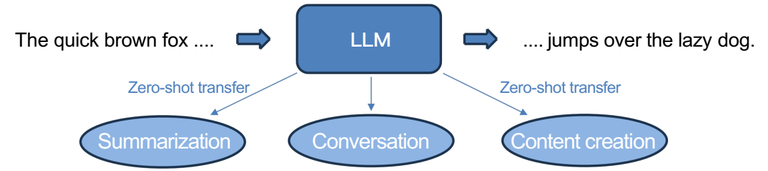


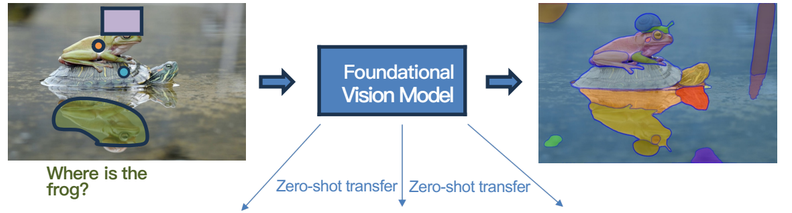
基础视觉模型：可提示分割



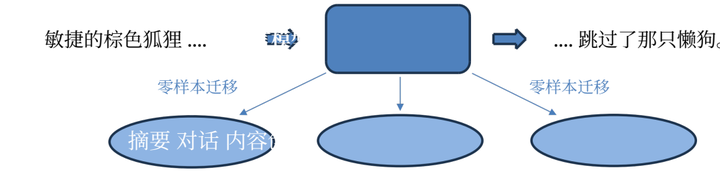


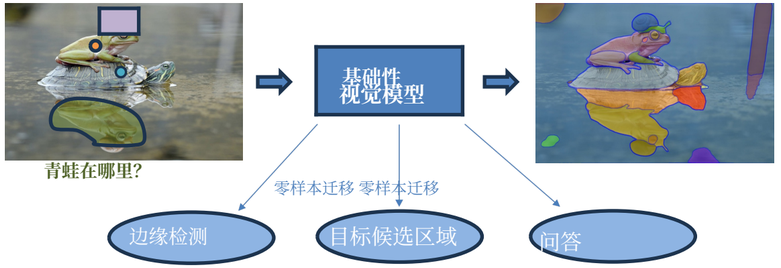
Foundational Vision Model: Promptable Segmentation



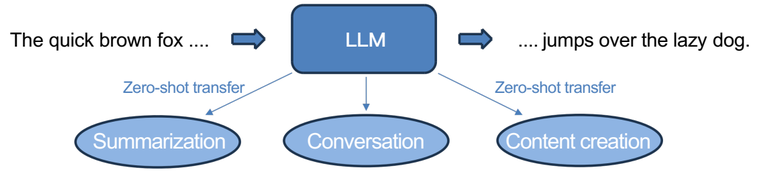


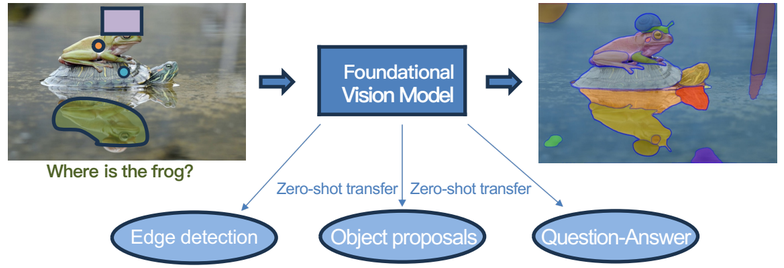
基础视觉模型：零样本迁移





Foundational Vision Model: Zero-Shot Transfer



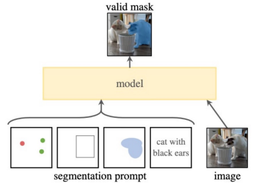


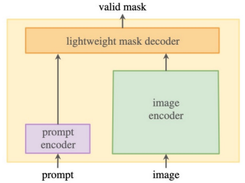
Segment Anything（基于 ViT）

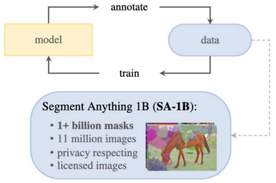
任务： 支持交互式和自动使用，并具备零样本能力泛化与复用

模型： 允许灵活的提示，支持实时使用，能够感知歧义

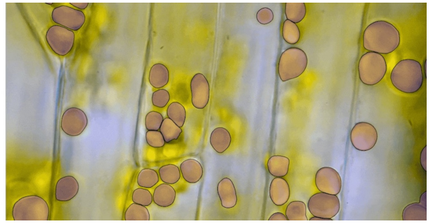
数据引擎： 模型在环数据集标注，兼顾公平性及隐私方面的考量









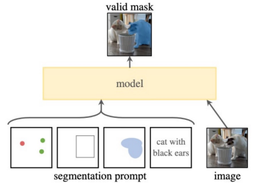


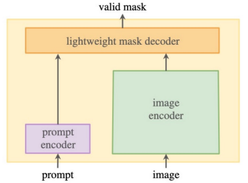
Segment Anything (ViT Based)

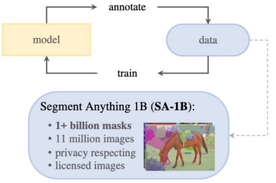
Task: Allow for interactive and automatic use, with zero-shot generalization and re-use

Model: Allow flexible prompts, support real-time usage, be ambiguity-aware

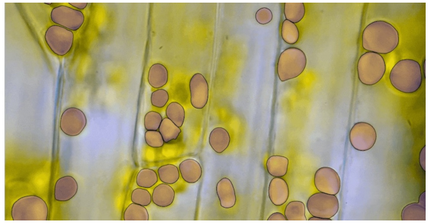
Data engine: Model-in-the-loop dataset annotation, with equity and privacy considerations











Segment Anything：任务

可提示分割

- 为给定项生成有效掩码提示作为输入
- 提示可能存在歧义，但始终返回有效掩码

支持大规模模型在环数据标注

支持零样本迁移类似于提示工程



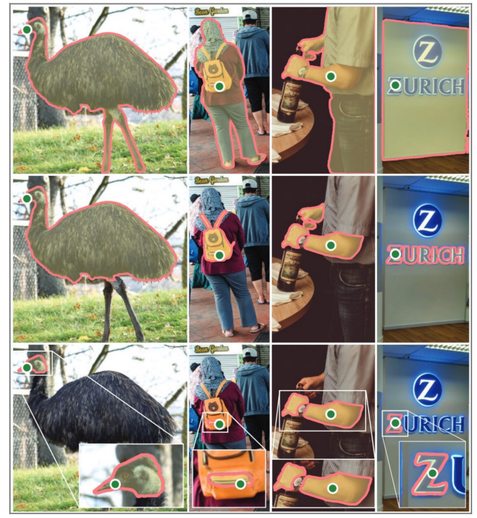
Segment Anything: Task

Promptable segmentation

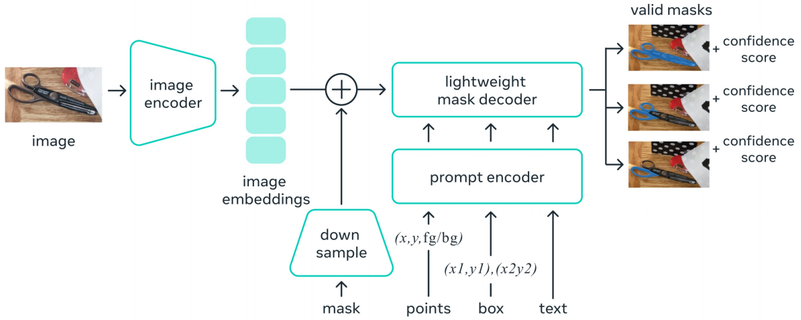
- Generate a *valid* mask for a given *prompt* as input
- Prompts might be *ambiguous*, but always return a valid mask

Allows large-scale model-in-the-loop data annotation

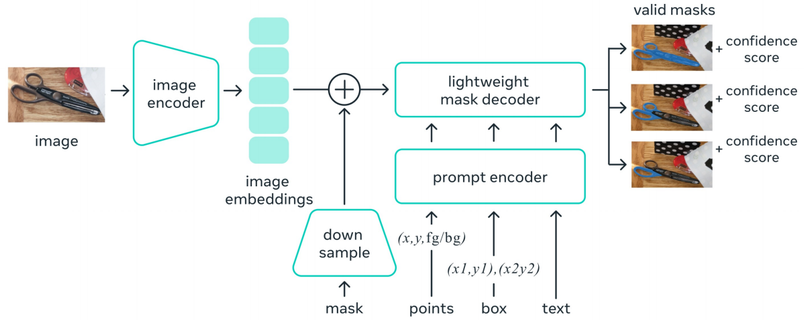
Allows zero-shot transfer similar to prompt engineering



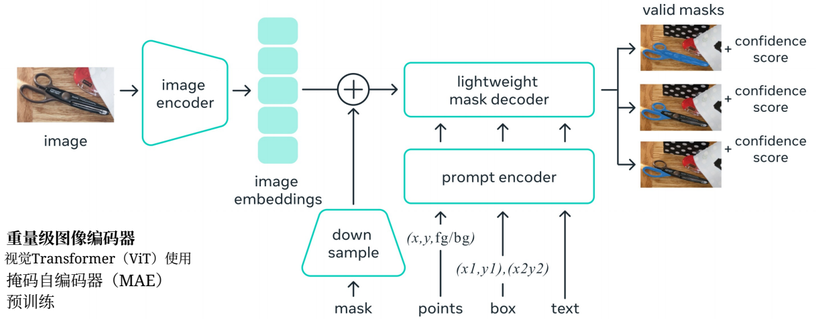
Segment Anything：通用分割模型



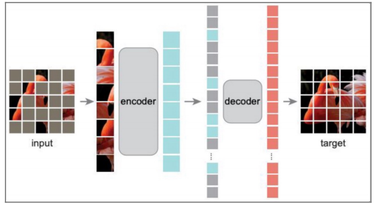
Segment Anything: Universal Segmentation Model



Segment Anything：通用分割模型

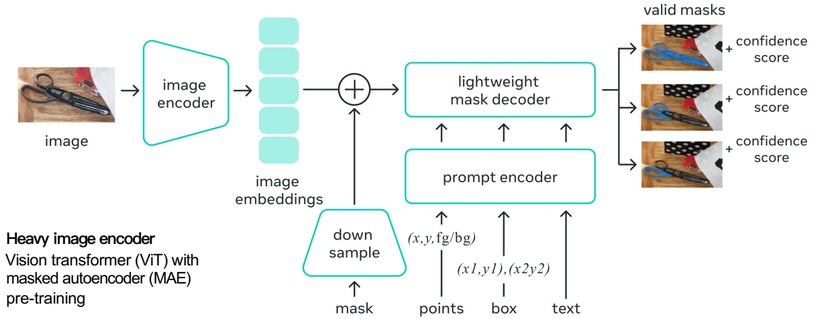


重量级图像编码器
视觉Transformer（ViT）使用
掩码自编码器（MAE）
预训练

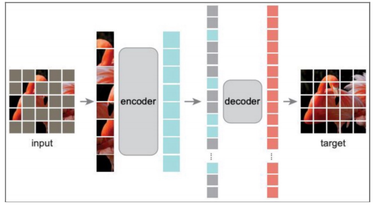


- 在NLP中取得成功，但在视觉领域需特殊考量
- **信息：**相邻图像块更容易重建
- 编码器仅处理25%的图像块（75%掩码）
- **语义层面：**重建像素不同于词语 强解码器（文本仅需MLP），但比编码器更轻
- 处理已编码的图像块和掩码标记以预测像素
- 高容量网络中的可扩展特征学习

Segment Anything: Universal Segmentation Model

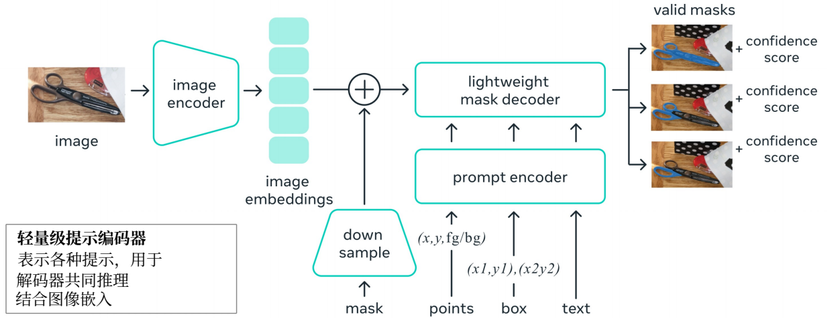


Heavy image encoder
Vision transformer (ViT) with
masked autoencoder (MAE)
pre-training



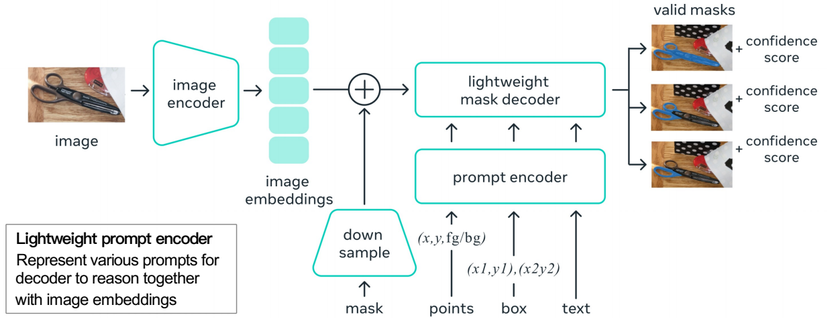
- Successful in NLP, but special considerations for vision
- **Information:** neighbor patches reconstruct more easily
- *Encoder* processes only 25% patches (75% masking)
- **Semantic level:** reconstructing pixels different from words
- Strong *decoder* (just MLP for text), but lighter than encoder
- Process encoded patches and mask tokens to predict pixels
- Scalable feature learning in high capacity networks

Segment Anything：通用分割模型



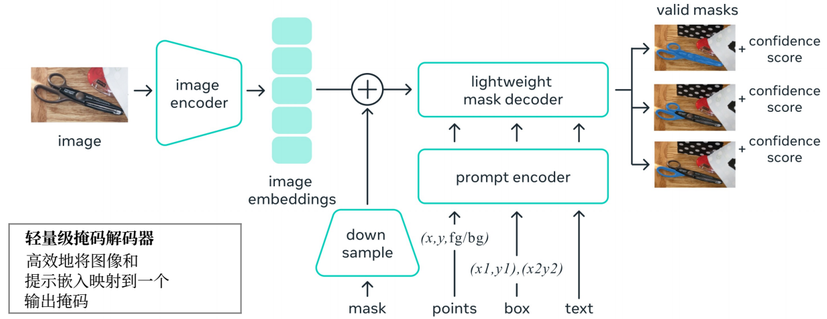
- **点：** 点位置的位置编码 + 用于前景或背景的可学习编码
- **框：** 角点的位置编码 + 哪个角点的可学习编码
- **文本：** CLIP 文本编码器
- **掩码：** 与图像嵌入进行卷积和逐元素求和

Segment Anything: Universal Segmentation Model

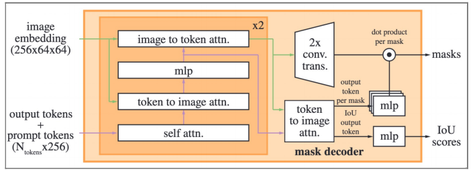


- **Point:** Positional encoding of point location + learnable encoding for foreground or background
- **Box:** Position encoding of corner + learnable encoding of which corner
- **Text:** CLIP text encoder
- **Mask:** Convolutions and element-wise sum with image embeddings

Segment Anything：通用分割模型



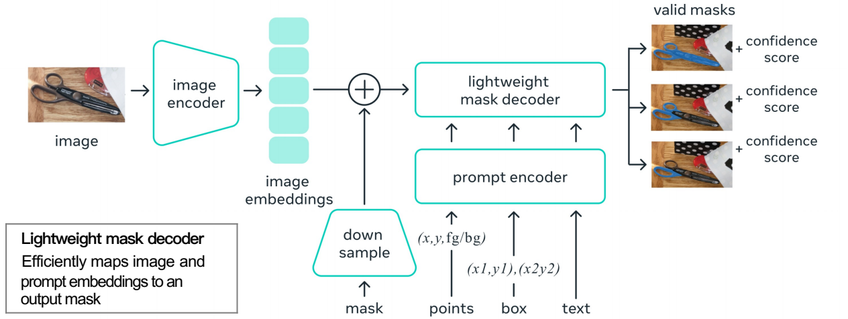
轻量级掩码解码器
高效地将图像和
提示嵌入映射到一个
输出掩码



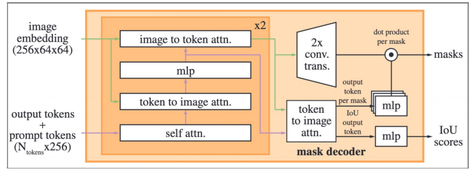
两层 Transformer 解码器
- 提示词上的自注意力
- 从查询提示到图像的交叉注意力
- 逐点 MLP 用于非线性 □ 从查询图像到提示的交叉注意力

转置卷积上采样至掩码

Segment Anything: Universal Segmentation Model



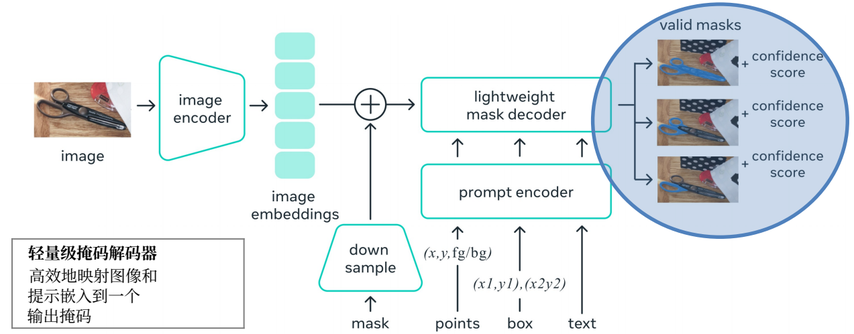
Lightweight mask decoder
Efficiently maps image and prompt embeddings to an output mask



Two layers of transformer decoders
- Self-attention on the prompts
- Cross-attention from query prompts to image
- Pointwise MLP for non-linearity
- Cross-attention from query image to prompts

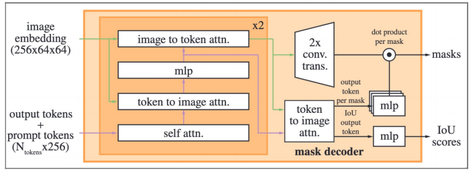
Transposed convolution to upsample to mask

Segment Anything：通用分割模型



轻量级掩码解码器

高效地映射图像和
提示嵌入到一个
输出掩码

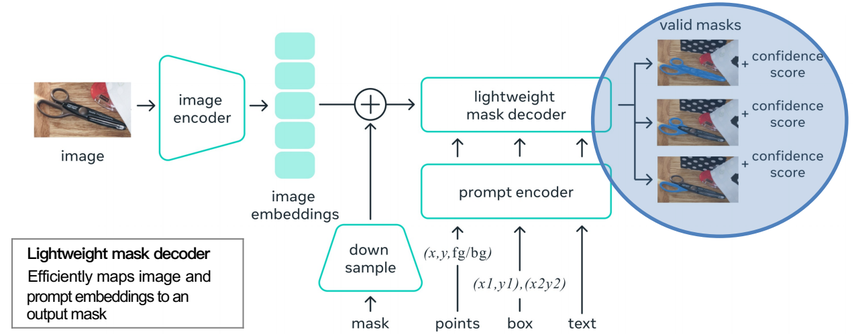


预测三个掩码输出

- 足以涵盖整体、部分、子部分概念
- 为每个预测的掩码生成一个输出标记
- 仅对多个掩码中的最小损失进行反向传播

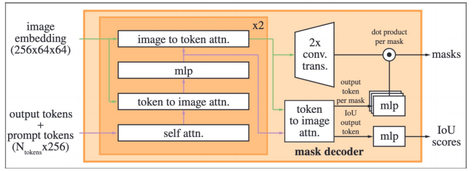
- 用于 IoU 预测以对掩码进行排序的 Token

Segment Anything: Universal Segmentation Model



Lightweight mask decoder

Efficiently maps image and prompt embeddings to an output mask



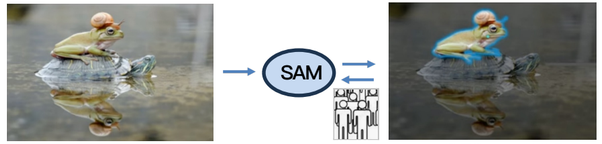
Predict three mask outputs

- Suffices for whole, part, sub-part concepts
- An output token for each mask predicted
- Backpropagate only minimum loss over masks
- Token for IoU prediction to rank masks

Segment Anything：模型在环数据引擎

辅助手动阶段

- 从数据集开始
- 通过多轮人工校正迭代重新训练模型

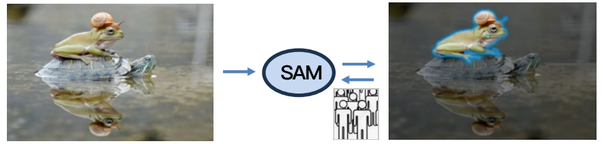


12万张图像，430万个掩码

Segment Anything: Model-in-the-Loop Data Engine

Assisted manual stage

- Start from datasets
- Iteratively retrain model with several rounds of human correction

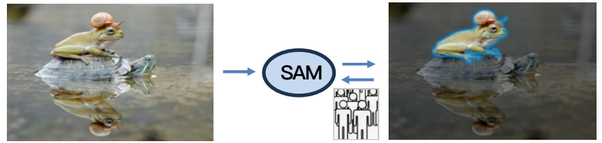


120k images, 4.3M masks

Segment Anything：模型在环数据引擎

辅助手动阶段

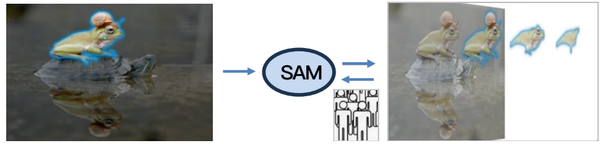
- 从数据集开始
- 通过多轮人工校正迭代重新训练模型



12 万张图像，430 万个掩码

半自动阶段

- 通过人工对当前输出进行精炼来提升多样性
- 定期重新训练模型

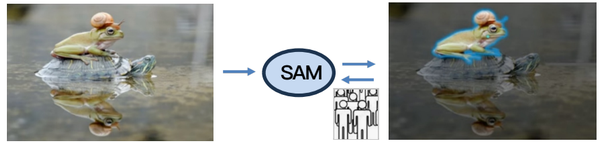


18 万张图像，590 万个掩码

Segment Anything: Model-in-the-Loop Data Engine

Assisted manual stage

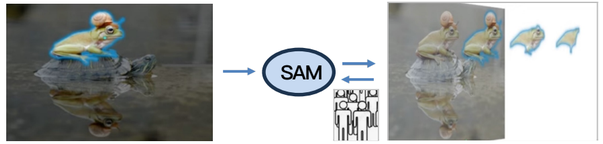
- Start from datasets
- Iteratively retrain model with several rounds of human correction



120k images, 4.3M masks

Semi–automatic stage

- Improve diversity with human refinement of current outputs
- Periodically retrain model

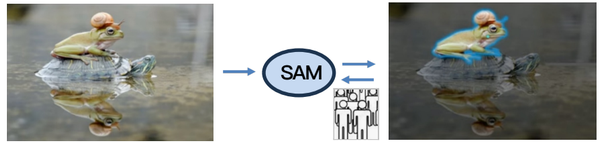


180k images, 5.9M masks

Segment Anything：模型在环数据引擎

辅助人工阶段

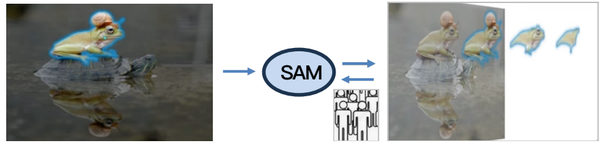
- 从数据集开始
- 通过多轮人工校正迭代重新训练模型



12万张图像，430万个掩码

半自动阶段

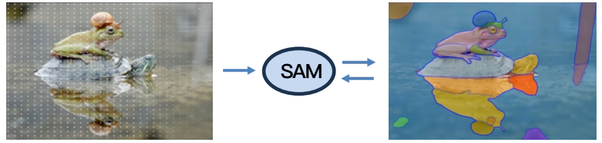
- 通过人工对当前输出进行精炼来提升多样性
- 定期重新训练模型



18万张图像，590万个掩码

全自动阶段

- 已训练好一个歧义感知模型
- 引入采用 32 x 32 点网格的提示
- 保留高置信度掩码（IoU 预测），执行 NMS

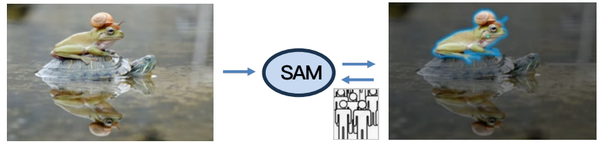


1100万张图像，11亿个掩码

Segment Anything: Model-in-the-Loop Data Engine

Assisted manual stage

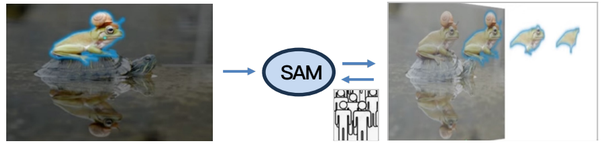
- Start from datasets
- Iteratively retrain model with several rounds of human correction



120k images, 4.3M masks

Semi–automatic stage

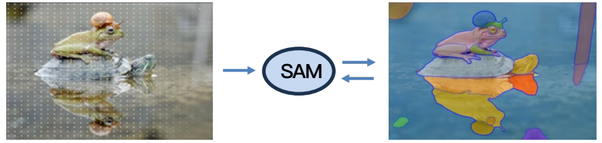
- Improve diversity with human refinement of current outputs
- Periodically retrain model



180k images, 5.9M masks

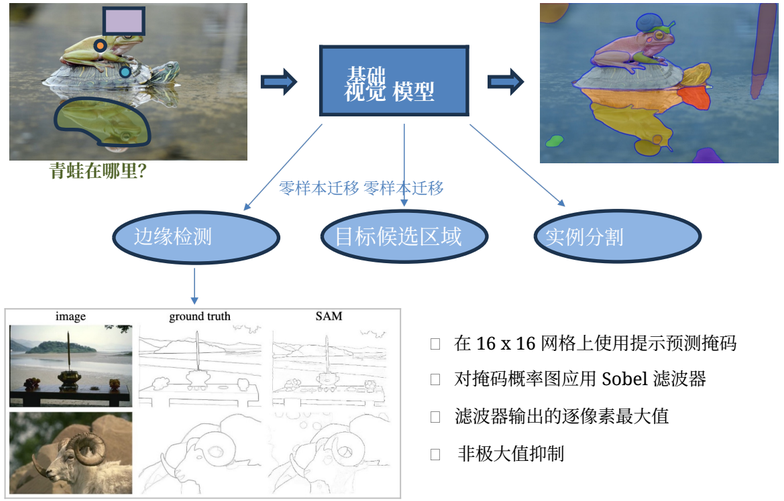
Fully automatic stage

- Already have trained an ambiguity-aware model
- Introduce prompting with 32 x 32 grid of points
- Keep high confidence masks (IoU prediction), do NMS



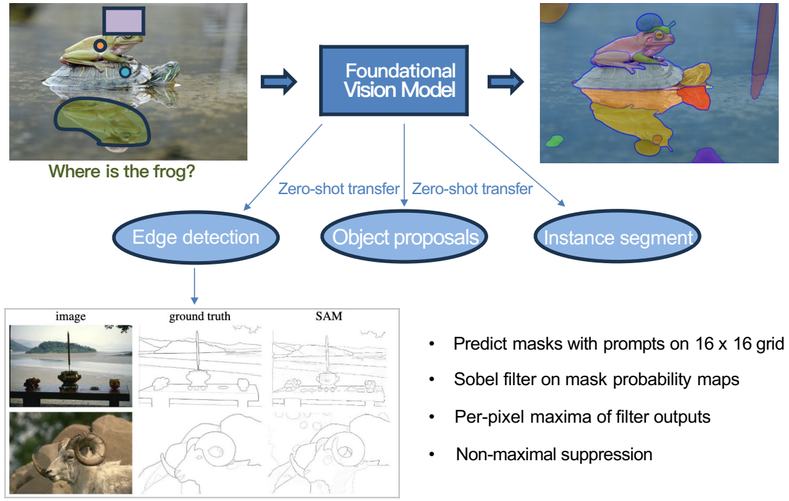
11M images, 1.1B masks

Segment Anything：通过提示实现零样本迁移



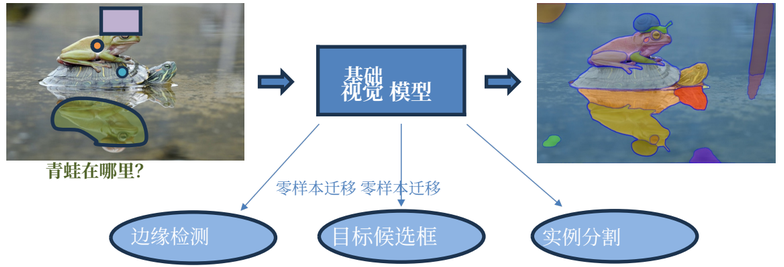
- 在 16 x 16 网格上使用提示预测掩码
- 对掩码概率图应用 Sobel 滤波器
- 滤波器输出的逐像素最大值
- 非极大值抑制

Segment Anything: Prompting for Zero-Shot Transfer

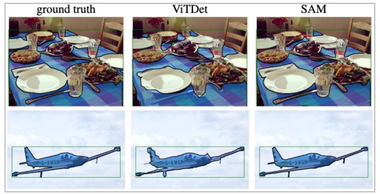


- Predict masks with prompts on 16 x 16 grid
- Sobel filter on mask probability maps
- Per-pixel maxima of filter outputs
- Non-maximal suppression

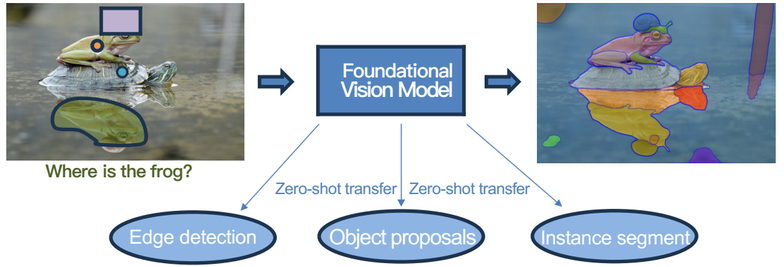
Segment Anything：通过提示实现零样本迁移



- 来自目标检测器的边界框
- 将其用作 SAM 的提示以获取掩码
- 通过馈入来细化置信度最高的掩码带框提示返回解码器



Segment Anything: Prompting for Zero-Shot Transfer



- Bounding boxes from object detector
- Use as prompts for SAM to get mask
- Refine most confident mask by feeding back to decoder with box prompt

